



Predicting Electric Vehicle Purchases ⚡



Overview

- ****Background****

The adoption of ****electric vehicles (EVs)**** is becoming increasingly important as consumers and governments move toward more sustainable transportation. However, EV adoption decisions can be influenced by several factors, including ****income****, ****daily commuting needs****, ****charging infrastructure****, ****environmental concerns****, ****government subsidies****, and ****range anxiety****. Understanding these factors can help identify the characteristics and conditions that influence a consumer's decision to purchase an electric vehicle.

- ****Goal of the Project****

Using demographic information, financial characteristics, driving behavior, charging infrastructure, environmental attitudes, and range anxiety indicators, this project aims to build a predictive model that determines whether a consumer is likely to ****purchase an electric vehicle****. The analysis also helps uncover the key factors associated with EV adoption and provides insights into consumer preferences and barriers to EV ownership.

Key Features

****Target Variable****

- `Will_Buy_EV` : Whether the consumer will purchase an electric vehicle (`Yes` / `No`)

****Buyer Demographics****

- `Age` : Age of the buyer
- `Gender` : Gender of the buyer

****Financial & Geographic Characteristics****

- `Annual_Income_USD` : Annual income of the buyer in USD
- `City_Type` : Type of city where the buyer lives

****Driving & Vehicle Ownership****

- `Daily_Commute_km` : Average daily commuting distance in kilometers
- `Number_of_Cars_Owned` : Number of cars currently owned

- `Current_Car_Type` : Type of vehicle currently owned

****Charging Infrastructure****

- `Charging_Stations_Near_Home` : Number of charging stations available near the buyer's home
- `Charging_Stations_Near_Work` : Number of charging stations available near the buyer's workplace
- `Home_Charging_Possible` : Whether home charging is possible (`Yes` / `No`)

****Environmental & Policy Factors****

- `Environmental_Concern_Level` : Buyer's level of concern about environmental issues
- `Subsidy_Available` : Whether an EV purchase subsidy is available (`Yes` / `No`)

****EV Adoption Barriers****

- `Range_Anxiety_Level` : Buyer's level of concern about EV driving range

****Files Provided****

- `EV_Adoption_and_Range_Anxiety_Dataset.csv` : Dataset containing buyer characteristics, driving behavior, charging infrastructure, environmental concerns, range anxiety, and EV purchase decisions

****Project Objective****

The primary objective of this project is to ****perform exploratory data analysis (EDA)**** and develop machine learning models capable of predicting whether a consumer will purchase an electric vehicle.

Through this analysis, we aim to:

- Understand ****EV adoption patterns**** across different consumer groups
- Identify the ****key factors influencing EV purchase decisions****
- Analyze how ****income, commuting distance, and vehicle ownership**** relate to EV adoption
- Evaluate the impact of ****charging infrastructure and home charging availability**** on purchase intention
- Investigate the relationship between ****environmental concern, subsidies, and range anxiety**** and EV adoption
- Build an accurate **classification model** for predicting EV purchase decisions.

Import Libraries

```
# =====  
# Clone Project Source from GitHub  
# =====  
  
import os  
  
# Reset the current working directory before removing any old clone.  
os.chdir("/kaggle/working")  
  
# Remove any previous repository clone.  
!rm -rf /kaggle/working/kaggle-portfolio  
  
# Clone the repository using a partial clone and sparse checkout.  
!git clone --filter=blob:none --sparse https://github.com/daominhthuan42/kaggle-portfolio.git  
  
# Move into the cloned repository.  
%cd /kaggle/working/kaggle-portfolio  
  
# Checkout only the EV project.  
!git sparse-checkout set 50-predicting-electric-vehicle-purchases  
  
# Verify the selected project.  
!git sparse-checkout list  
  
# Install statistical analysis dependencies.  
!pip install statsmodels > pip_log_statsmodels.txt 2>&1  
!pip install scikit_posthocs > pip_log_scikit_posthocs.txt 2>&1  
!pip install pingouin==0.5.5 > pip_log_pingouin.txt 2>&1  
  
# Install the required Pandas version.  
!pip install -q pandas==3.0.0 > pip_log_pandas.txt 2>&1
```

Cloning into 'kaggle-portfolio'...

remote: Enumerating objects: 591, done.

remote: Counting objects: 100% (16/16), done.

remote: Compressing objects: 100% (13/13), done.

remote: Total 591 (delta 5), reused 12 (delta 2), pack-reused 575 (from 3)

Receiving objects: 100% (591/591), 84.69 KiB | 1.49 MiB/s, done.

Resolving deltas: 100% (249/249), done.

remote: Enumerating objects: 3, done.

remote: Counting objects: 100% (1/1), done.

remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 2 (from 1)

Receiving objects: 100% (3/3), 1.43 KiB | 1.43 MiB/s, done.

/kaggle/working/kaggle-portfolio

remote: Enumerating objects: 21, done.

remote: Counting objects: 100% (10/10), done.

remote: Compressing objects: 100% (10/10), done.

remote: Total 21 (delta 0), reused 2 (delta 0), pack-reused 11 (from 3)

Receiving objects: 100% (21/21), 19.90 MiB | 20.67 MiB/s, done.

Resolving deltas: 100% (1/1), done.

Updating files: 100% (24/24), done.

50-predicting-electric-vehicle-purchases

```
import sys
from pathlib import Path

# Define the root directory of the current project.
PROJECT_ROOT = Path("/kaggle/working/kaggle-portfolio/50-predicting-electric-vehicle-
purchases")

# Add the project root to Python's module search path.
sys.path.insert(0, str(PROJECT_ROOT))

# Core Libraries
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from visualization.visualization_plots import *

# Project Utilities
from utils.logger import LoggerConfig
from utils.config_loader import *
```

```

from utils.utils import Utils
from utils.stats_utils import StatUtils
from preprocessing.validation import DataValidator

# Modeling Utilities
from models.benchmark import *
from models.oof import *

# Hyperparameter Tuning
from models.catboost_tuning import *
from models.lgbm_tunning import *
from models.xgboost_tuning import *

# Notebook Display
from IPython.display import display, HTML

# Optimization
import optuna
optuna.logging.set_verbosity(optuna.logging.WARNING)

# Environment Settings
import warnings
warnings.filterwarnings("ignore")

pd.set_option("display.max_columns", 500)
pd.set_option("display.max_colwidth", None)

```

Load Data

```

# Configure a consistent dark theme for Matplotlib and Seaborn.
setup_plot_style()

```

```

# Initialize the project logger.
logger = LoggerConfig.setup_logger()

```

```

# Load the global project configuration from the YAML file.
config = load_config(logger=logger)

```

```

2026-09-19 04:36:38 | INFO      | Logging configured: level=DEBUG, format=colored
2026-09-19 04:36:38 | INFO      | Loading configuration from config.yaml
2026-09-19 04:36:38 | INFO      | Configuration loaded successfully

```

```

# Load datasets safely
df_train = Utils.read_csv_safe(path=config["dataset"]["paths"]["train"], logger=logger,
                               encodings=config["dataset"]
["default_encoding"]).pipe(Utils.clean_columns)
df_original = Utils.read_csv_safe(path=config["dataset"]["paths"]["original"], logger=logger,
                                  encodings=config["dataset"]
["default_encoding"]).pipe(Utils.clean_columns)
df_test = Utils.read_csv_safe(path=config["dataset"]["paths"]["test"], logger=logger,

```

```
encodings=config["dataset"]
["default_encoding"].pipe(Utils.clean_columns)
```

```
2026-09-19 04:36:39 | INFO      | Loaded file with encoding: utf-8 | Shape: (668665, 15)
2026-09-19 04:36:39 | INFO      | Loaded file with encoding: utf-8 | Shape: (10000, 15)
2026-09-19 04:36:40 | INFO      | Loaded file with encoding: utf-8 | Shape: (286571, 14)
```

Data Preview and Info

```
# Preview Datasets
```

```
# Display the first five rows of each dataset for a quick
# inspection of structure, columns, and sample values.
```

```
datasets = {"Train": df_train, "Test": df_test, "Original": df_original}
```

```
for name, df in datasets.items():
    # Log the dataset currently being previewed.
    logger.info(f"{name} Data Preview")
```

```
# Display the first five rows.
display(df.head())
```

```
2026-09-19 04:36:40 | INFO      | Train Data Preview
```

	id	Age	Annual_Income_USD	Daily_Commute_km	Number_of_Cars_Owned	Charging_Stations_Near_Hoi
0	0	66	92887.0	23.4	2	
1	1	38	30000.0	5.0	1	
2	2	26	94389.0	36.8	1	
3	3	66	73580.0	23.7	2	
4	4	54	57898.0	50.8	1	

```
2026-09-19 04:36:40 | INFO      | Test Data Preview
```

	id	Age	Annual_Income_USD	Daily_Commute_km	Number_of_Cars_Owned	Charging_Stations_Nea
0	668665	61	67725.0	16.9	2	
1	668666	42	152835.0	41.9	2	
2	668667	68	86877.0	53.3	1	
3	668668	39	46794.0	34.1	2	
4	668669	55	112172.0	57.2	1	

```
2026-09-19 04:36:40 | INFO      | Original Data Preview
```

	Buyer_ID	Age	Gender	Annual_Income_USD	City_Type	Daily_Commute_km	Number_of_Cars_Owned
0	EV00001	56	Male	44906.0	Rural	18.1	1
1	EV00002	36	Male	88754.0	Rural	5.0	1
2	EV00003	42	Male	155958.0	Urban	32.2	2
3	EV00004	31	Male	37059.0	Rural	89.9	1
4	EV00005	48	Female	48552.0	Rural	71.0	1

Display information about the DataFrames

```
for name, df in datasets.items():  
    logger.info(f"{name} Data Info")  
    df.info()
```

2026-09-19 04:36:40 | INFO | Train Data Info

<class 'pandas.DataFrame'>

RangeIndex: 668665 entries, 0 to 668664

Data columns (total 15 columns):

#	Column	Non-Null Count	Dtype
0	id	668665 non-null	int64
1	Age	668665 non-null	int64
2	Annual_Income_USD	668665 non-null	float64
3	Daily_Commute_km	668665 non-null	float64
4	Number_of_Cars_Owned	668665 non-null	int64
5	Charging_Stations_Near_Home	668665 non-null	int64
6	Charging_Stations_Near_Work	668665 non-null	int64
7	Environmental_Concern_Level	668665 non-null	float64
8	Gender	668665 non-null	str
9	City_Type	668665 non-null	str
10	Current_Car_Type	668665 non-null	str
11	Home_Charging_Possible	668665 non-null	str
12	Subsidy_Available	668665 non-null	str
13	Range_Anxiety_Level	668665 non-null	str
14	Will_Buy_EV	668665 non-null	str

dtypes: float64(3), int64(5), str(7)

memory usage: 93.5 MB

2026-09-19 04:36:40 | INFO | Test Data Info

<class 'pandas.DataFrame'>

RangeIndex: 286571 entries, 0 to 286570

Data columns (total 14 columns):

#	Column	Non-Null Count	Dtype
0	id	286571 non-null	int64
1	Age	286571 non-null	int64
2	Annual_Income_USD	286571 non-null	float64
3	Daily_Commute_km	286571 non-null	float64
4	Number_of_Cars_Owned	286571 non-null	int64
5	Charging_Stations_Near_Home	286571 non-null	int64
6	Charging_Stations_Near_Work	286571 non-null	int64
7	Environmental_Concern_Level	286571 non-null	float64
8	Gender	286571 non-null	str
9	City_Type	286571 non-null	str
10	Current_Car_Type	286571 non-null	str
11	Home_Charging_Possible	286571 non-null	str
12	Subsidy_Available	286571 non-null	str
13	Range_Anxiety_Level	286571 non-null	str

dtypes: float64(3), int64(5), str(6)

memory usage: 37.3 MB

2026-09-19 04:36:40 | INFO | Original Data Info

<class 'pandas.DataFrame'>

RangeIndex: 10000 entries, 0 to 9999

Data columns (total 15 columns):

#	Column	Non-Null Count	Dtype
0	Buyer_ID	10000 non-null	str
1	Age	10000 non-null	int64
2	Gender	10000 non-null	str
3	Annual_Income_USD	9822 non-null	float64
4	City_Type	10000 non-null	str
5	Daily_Commute_km	9819 non-null	float64
6	Number_of_Cars_Owned	10000 non-null	int64
7	Current_Car_Type	10000 non-null	str
8	Charging_Stations_Near_Home	10000 non-null	int64

```
9   Charging_Stations_Near_Work  10000 non-null  int64
10  Home_Charging_Possible        10000 non-null  str
11  Environmental_Concern_Level   9816 non-null   float64
12  Subsidy_Available            10000 non-null  str
13  Range_Anxiety_Level          10000 non-null  str
14  Will_Buy_EV                  10000 non-null  str
dtypes: float64(3), int64(4), str(8)
memory usage: 1.5 MB
```



Insights from Initial Data Exploration

Dataset Overview

- **Training dataset:** 668,665 samples with 15 columns, including the target `Will_Buy_EV`.
- **Test dataset:** 286,571 samples with 14 columns, with the target variable excluded.
- **Original dataset:** 10,000 observations with 15 columns, using `Buyer_ID` as the identifier and containing the original target `Will_Buy_EV`.
- The competition datasets contain a combined **955,236 observations**, substantially larger than the original 10,000-row dataset.

Feature Overview

- **Numerical features (8):** `Age`, `Annual_Income_USD`, `Daily_Commute_km`, `Number_of_Cars_Owned`, `Charging_Stations_Near_Home`, `Charging_Stations_Near_Work`, and `Environmental_Concern_Level`, together with the numerical identifier `id`.
- **Categorical features (6):** `Gender`, `City_Type`, `Current_Car_Type`, `Home_Charging_Possible`, `Subsidy_Available`, and `Range_Anxiety_Level`.
- `Will_Buy_EV` is the binary classification target indicating whether a consumer is likely to purchase an electric vehicle.

Data Quality

- The **competition train and test datasets are complete**, with no missing values across any of the predictor variables.
- In contrast, the **original dataset contains missing values** in `Annual_Income_USD`, `Daily_Commute_km`, and `Environmental_Concern_Level`.
- The competition version therefore appears to have undergone **data preprocessing or transformation** compared with the original dataset, resulting in a much larger dataset with no missing values.
- Numerical variables are stored as `int64` or `float64`, while categorical variables are stored as `object` and will require appropriate encoding for machine learning models.

Memory and Efficiency

- The training dataset occupies approximately **76.5 MB** of memory, while the test dataset occupies approximately **30.6 MB**.
- Given the combined dataset size of more than **955,000 observations**, memory-efficient data processing becomes important during EDA and model development.
- Converting categorical variables to the `category` dtype can help reduce memory usage and improve computational efficiency.

Initial Observations

- The dataset combines **demographic**, **financial**, **driving**, **charging infrastructure**, **environmental**, and **behavioral** factors, providing multiple perspectives for predicting EV purchase decisions.
- Features such as `Annual_Income_USD`, `Daily_Commute_km`, and `Number_of_Cars_Owned` may capture the consumer's financial capacity and transportation needs.
- Charging-related variables, particularly `Charging_Stations_Near_Home`, `Charging_Stations_Near_Work`, and `Home_Charging_Possible`, may provide important signals about the practicality of EV ownership.
- `Environmental_Concern_Level`, `Subsidy_Available` and `Range_Anxiety_Level` represent important behavioral and policy-related factors that may influence EV adoption.
- The identifier `id`, `Buyer_ID` do not represent a meaningful consumer characteristic and should be **excluded from model training** to avoid introducing non-predictive information.

```
# Preserve test IDs for the final submission file.
list_test_id = df_test[config["columns"]["train_test"]][0].copy().to_list()

# Remove identifier columns from the Train and Test datasets.
# These columns do not provide predictive information.
df_train.drop(columns=config["columns"]["train_test"], inplace=True)
df_test.drop(columns=config["columns"]["train_test"], inplace=True)

# Remove the original dataset identifier column.
df_original.drop(columns=config["columns"]["original"], inplace=True)

# Numerical features
num_features = ["Age", "Annual_Income_USD", "Daily_Commute_km", "Charging_Stations_Near_Home",
"Charging_Stations_Near_Work"]

# Categorical features
cat_features = ["Gender", "City_Type", "Current_Car_Type", "Home_Charging_Possible",
"Subsidy_Available",
"Environmental_Concern_Level", "Range_Anxiety_Level", "Number_of_Cars_Owned"]

# Reduce memory and convert to category
df_train = Utils.reduce_mem_usage(cols=num_features, df=df_train)
df_test = Utils.reduce_mem_usage(cols=num_features, df=df_test)
df_original = Utils.reduce_mem_usage(cols=num_features, df=df_original)

df_train = Utils.convert_cat(features=cat_features, df=df_train)
df_test = Utils.convert_cat(features=cat_features, df=df_test)
df_original = Utils.convert_cat(features=cat_features, df=df_original)

# Display information about the DataFrames
for name, df in datasets.items():
    logger.info(f"{name} Data Info After Reducing")
    df.info()
```

2026-09-19 04:36:40 | INFO | Train Data Info After Reducing

<class 'pandas.DataFrame'>

RangeIndex: 668665 entries, 0 to 668664

Data columns (total 14 columns):

#	Column	Non-Null Count	Dtype
0	Age	668665 non-null	int8
1	Annual_Income_USD	668665 non-null	float32
2	Daily_Commute_km	668665 non-null	float32
3	Number_of_Cars_Owned	668665 non-null	category
4	Charging_Stations_Near_Home	668665 non-null	int8
5	Charging_Stations_Near_Work	668665 non-null	int8
6	Environmental_Concern_Level	668665 non-null	category
7	Gender	668665 non-null	category
8	City_Type	668665 non-null	category
9	Current_Car_Type	668665 non-null	category
10	Home_Charging_Possible	668665 non-null	category
11	Subsidy_Available	668665 non-null	category
12	Range_Anxiety_Level	668665 non-null	category
13	Will_Buy_EV	668665 non-null	str

dtypes: category(8), float32(2), int8(3), str(1)

memory usage: 18.6 MB

2026-09-19 04:36:40 | INFO | Test Data Info After Reducing

<class 'pandas.DataFrame'>

RangeIndex: 286571 entries, 0 to 286570

Data columns (total 13 columns):

#	Column	Non-Null Count	Dtype
0	Age	286571 non-null	int8
1	Annual_Income_USD	286571 non-null	float32
2	Daily_Commute_km	286571 non-null	float32
3	Number_of_Cars_Owned	286571 non-null	category
4	Charging_Stations_Near_Home	286571 non-null	int8
5	Charging_Stations_Near_Work	286571 non-null	int8
6	Environmental_Concern_Level	286571 non-null	category
7	Gender	286571 non-null	category
8	City_Type	286571 non-null	category
9	Current_Car_Type	286571 non-null	category
10	Home_Charging_Possible	286571 non-null	category
11	Subsidy_Available	286571 non-null	category
12	Range_Anxiety_Level	286571 non-null	category

dtypes: category(8), float32(2), int8(3)

memory usage: 5.2 MB

2026-09-19 04:36:40 | INFO | Original Data Info After Reducing

<class 'pandas.DataFrame'>

RangeIndex: 10000 entries, 0 to 9999

Data columns (total 14 columns):

#	Column	Non-Null Count	Dtype
0	Age	10000 non-null	int8
1	Gender	10000 non-null	category
2	Annual_Income_USD	9822 non-null	float32
3	City_Type	10000 non-null	category
4	Daily_Commute_km	9819 non-null	float32
5	Number_of_Cars_Owned	10000 non-null	category
6	Current_Car_Type	10000 non-null	category
7	Charging_Stations_Near_Home	10000 non-null	int8
8	Charging_Stations_Near_Work	10000 non-null	int8
9	Home_Charging_Possible	10000 non-null	category
10	Environmental_Concern_Level	9816 non-null	category

11 Subsidy_Available 10000 non-null category
12 Range_Anxiety_Level 10000 non-null category
13 Will_Buy_EV 10000 non-null str
dtypes: category(8), float32(2), int8(3), str(1)
memory usage: 285.3 KB

Numerical Feature Summary — Data Analyst Perspective

```
# Create a light blue colormap for highlighting summary statistics.  
cm = sns.light_palette("blue", as_cmap=True)  
for name, df in datasets.items():  
    # Log the dataset currently being analyzed.  
    logger.info(f"{name} Data Describe:")  
  
    # Display descriptive statistics for numerical features.  
    display(df[num_features].describe().T.style.background_gradient(cmap=cm))
```

2026-09-19 04:36:40 | INFO | Train Data Describe:

	count	mean	std	min	25%
Age	668665.000000	47.039171	12.875448	25.000000	36.000000
Annual_Income_USD	668665.000000	84769.289062	28637.109375	30000.000000	67376.000000
Daily_Commute_km	668665.000000	32.158298	18.725174	5.000000	17.200001
Charging_Stations_Near_Home	668665.000000	4.960408	3.926843	0.000000	2.000000
Charging_Stations_Near_Work	668665.000000	7.176314	5.186627	0.000000	3.000000

2026-09-19 04:36:41 | INFO | Test Data Describe:

	count	mean	std	min	25%
Age	286571.000000	47.070824	12.864850	25.000000	36.000000
Annual_Income_USD	286571.000000	84799.515625	28625.003906	30000.000000	67380.000000
Daily_Commute_km	286571.000000	32.159302	18.717730	5.000000	17.200001
Charging_Stations_Near_Home	286571.000000	4.950693	3.928626	0.000000	2.000000
Charging_Stations_Near_Work	286571.000000	7.155626	5.186880	0.000000	3.000000

2026-09-19 04:36:41 | INFO | Original Data Describe:

	count	mean	std	min	25%
Age	10000.000000	46.936200	12.945968	25.000000	36.000000
Annual_Income_USD	9822.000000	85378.492188	32838.671875	30000.000000	61369.500000
Daily_Commute_km	9819.000000	41.105415	23.351192	5.000000	23.250000
Charging_Stations_Near_Home	10000.000000	5.346900	4.049074	0.000000	2.000000
Charging_Stations_Near_Work	10000.000000	7.458200	5.259374	0.000000	3.000000



Descriptive Statistics

- **Age**: Consumers have an average age of approximately **47 years**, with the middle 50% ranging from **36 to 58 years**. The distribution is highly consistent across the Train, Test, and Original datasets, indicating a stable consumer age profile.
- **Annual_Income_USD**: Average annual income is approximately **\$84,769** in the Train dataset and **\$84,800** in the Test dataset, with median income around **84,880–84,926**. The middle 50% of consumers earn roughly between **67,400 and 102,753**, suggesting a relatively broad income range across the population.
- **Daily_Commute_km**: Consumers travel approximately **32.2 km per day** on average in the Train dataset, with the middle 50% commuting between **17.0 and 47.4 km**. The Test dataset shows a very similar distribution, while the Original dataset has a noticeably higher average commute of approximately **41.1 km**, indicating some distribution differences between the original and competition data.
- **Charging_Stations_Near_Home**: Consumers have approximately **5 charging stations** near their homes on average, with a median of **4 stations**. Around 75% of observations have no more than **7 stations** nearby, indicating relatively moderate access to home-area charging infrastructure.
- **Charging_Stations_Near_Work**: The average number of charging stations near the workplace is approximately **7.2**, with a median of **6 stations**. The higher availability compared with home charging suggests that workplace areas may provide broader access to charging infrastructure.
- **Train vs Test**: The Train and Test datasets show **highly consistent descriptive statistics** across age, income, commuting distance, and charging infrastructure. The close alignment of means, medians, quartiles, and standard deviations suggests **no major distribution shift** between the competition training and test sets.
- **Train/Test vs Original**: The competition datasets generally preserve the overall characteristics of the Original dataset, particularly for **Age** and **Annual_Income_USD**. However, **Daily_Commute_km** shows a noticeable difference, with the Original dataset having a higher average and wider upper range than the competition data.
- **Overall**: The numerical variables capture several important dimensions of EV adoption, including **consumer demographics, financial capacity, daily transportation needs, and charging infrastructure availability**. These characteristics provide a strong foundation for analyzing and predicting electric vehicle purchase decisions.

Categorical Feature Summary — Data Analyst Perspective

```
for name, df in datasets.items():
    # Log the dataset currently being analyzed.
    logger.info(f"{name} Data Describe:")

    # Display descriptive statistics for categorical features.

display(df[cat_features].astype("category").describe().T.style.background_gradient(cmap="Blues", subset=["unique", "freq"]))
```

2026-09-19 04:36:41 | INFO | Train Data Describe:

	count	unique	top	freq
Gender	668665	3	Male	367954
City_Type	668665	3	Urban	289305
Current_Car_Type	668665	4	Sedan	303459
Home_Charging_Possible	668665	2	Yes	462677
Subsidy_Available	668665	2	Yes	419909
Environmental_Concern_Level	668665	5	1	147476
Range_Anxiety_Level	668665	3	Low	603972
Number_of_Cars_Owned	668665	4	2	298287

2026-09-19 04:36:41 | INFO | Test Data Describe:

	count	unique	top	freq
Gender	286571	3	Male	157647
City_Type	286571	3	Urban	123602
Current_Car_Type	286571	4	Sedan	130185
Home_Charging_Possible	286571	2	Yes	198480
Subsidy_Available	286571	2	Yes	180378
Environmental_Concern_Level	286571	5	1	63125
Range_Anxiety_Level	286571	3	Low	259247
Number_of_Cars_Owned	286571	4	2	127869

2026-09-19 04:36:41 | INFO | Original Data Describe:

	count	unique	top	freq
Gender	10000	3	Male	5201
City_Type	10000	3	Urban	4941
Current_Car_Type	10000	4	Sedan	4000
Home_Charging_Possible	10000	2	Yes	6163
Subsidy_Available	10000	2	Yes	6008
Environmental_Concern_Level	9816	5	1	2013
Range_Anxiety_Level	10000	3	Low	8310
Number_of_Cars_Owned	10000	4	2	4020

Categorical Feature Insights

- **Gender:** All datasets contain **three gender categories**, with **Male** being the most frequently observed group. In the Train and Test datasets, males account for approximately **55%** of observations, while the Original dataset shows a slightly more balanced distribution with males representing **52%**.
- **City_Type:** Three city categories are represented, with **Urban** being the dominant category. It accounts for approximately **43%** of the Train and Test datasets and nearly **49%** of the Original dataset, suggesting that urban consumers form an important segment of the population.
- **Current_Car_Type:** Four vehicle categories are present, with **Sedan** being the most common current vehicle type. Sedans represent approximately **45%** of both the Train and Test datasets, compared with **40%** in the Original dataset.
- **Home_Charging_Possible:** This binary feature is moderately concentrated toward **Yes**, which represents approximately **69%** of the Train and Test observations. The Original dataset has a lower proportion of consumers with home charging availability at approximately **62%**.
- **Subsidy_Available:** Approximately **63%** of consumers in the Train and Test datasets have access to an available subsidy, compared with approximately **60%** in the Original dataset. This indicates that government incentives are available to a substantial proportion of the observed consumers.
- **Environmental_Concern_Level:** This ordinal feature contains **five levels**. Level **1** is the most frequently observed category, accounting for approximately **22%** of the Train and Test observations. The distribution across the five levels should be examined further because the feature represents ordered levels of environmental concern rather than purely nominal categories.
- **Range_Anxiety_Level:** The distribution is strongly concentrated in the **Low** category, representing approximately **90%** of both the Train and Test datasets. The Original dataset has a lower proportion of **Low** range anxiety at approximately **83%**, indicating a noticeable distribution difference between the original and competition datasets.
- **Number_of_Cars_Owned:** Four ownership levels are represented, with consumers owning **2** cars being the most common group. This category accounts for approximately **45%** of the Train and Test datasets and **40%** of the Original dataset. Although this variable is numerical in storage, its low number of discrete values makes categorical treatment a reasonable modeling approach to evaluate.
- **Low-cardinality features:** Most categorical variables contain only **2–4 categories**, making them straightforward to encode using One-Hot Encoding or other appropriate categorical encoding techniques. The ordinal variables **Environmental_Concern_Level** and **Range_Anxiety_Level** should preserve their inherent ordering during preprocessing.
- **Distribution consistency:** The Train and Test datasets show **highly consistent categorical distributions** across all examined variables. However, several differences can be observed when

compared with the Original dataset, particularly for `Range_Anxiety_Level`, `Home_Charging_Possible`, and `Current_Car_Type`.

- **Overall:** These categorical and ordinal variables capture important aspects of **consumer demographics**, **vehicle ownership**, **charging accessibility**, **government incentives**, **environmental attitudes**, and **range anxiety**. Together, they provide important behavioral and contextual signals for predicting electric vehicle purchase decisions.

Data Quality Checks

Missing Value

```
logger.info("Start checking NULL values")
logger.info("-" * 80)
logger.info(f"Checking missing values for {config['dataset']['name']}...")
for name, df in datasets.items():
    logger.info(f"{name} Data Checking:")
    DataValidator.checkNULL(df=df, logger=logger)
logger.info("-" * 80)
logger.info("Completed checking for NULL values across datasets.")
```

```
2026-09-19 04:36:41 | INFO      | Start checking NULL values
2026-09-19 04:36:41 | INFO      | -----
2026-09-19 04:36:41 | INFO      | Checking missing values for EV Adoption Prediction Dataset...
2026-09-19 04:36:41 | INFO      | Train Data Checking:
2026-09-19 04:36:46 | INFO      | No missing values detected in 668,665 rows.
2026-09-19 04:36:46 | INFO      | Test Data Checking:
2026-09-19 04:36:48 | INFO      | No missing values detected in 286,571 rows.
2026-09-19 04:36:48 | INFO      | Original Data Checking:
2026-09-19 04:36:48 | WARNING   |
+-----+-----+-----+
| Feature                | Missing_Count | Missing_% |
+-----+-----+-----+
| Environmental_Concern_Level | 184           | 1.84      |
| Daily_Commute_km        | 181           | 1.81      |
| Annual_Income_USD       | 178           | 1.78      |
+-----+-----+-----+
2026-09-19 04:36:48 | WARNING   | Total missing values: 543 out of 10,000 rows.
2026-09-19 04:36:48 | INFO      | -----
2026-09-19 04:36:48 | INFO      | Completed checking for NULL values across datasets.
```

Checking duplicate Value

```
logger.info("Start checking duplicate values")
logger.info("-" * 80)
logger.info(f"Data duplicate Analysis...")
for name, df in datasets.items():
    logger.info(f"{name} Data Checking:")
    DataValidator.checkDuplicate(df=df, logger=logger)
logger.info("-" * 80)
logger.info("Duplicate value inspection successfully completed across datasets.")
```

```
2026-09-19 04:36:48 | INFO      | Start checking duplicate values
2026-09-19 04:36:48 | INFO      | -----
-----
2026-09-19 04:36:48 | INFO      | Data duplicate Analysis...
2026-09-19 04:36:48 | INFO      | Train Data Checking:
2026-09-19 04:36:48 | INFO      | No duplicate rows found out of 668,665 total rows.
2026-09-19 04:36:48 | INFO      | Test Data Checking:
2026-09-19 04:36:48 | INFO      | No duplicate rows found out of 286,571 total rows.
2026-09-19 04:36:48 | INFO      | Original Data Checking:
2026-09-19 04:36:48 | INFO      | No duplicate rows found out of 10,000 total rows.
2026-09-19 04:36:48 | INFO      | -----
-----
2026-09-19 04:36:48 | INFO      | Duplicate value inspection successfully completed across
datasets.
```

Checking Outlier Value

```
logger.info("-" * 80)
outlier_data = []
for name, df in datasets.items():
    logger.info(f"{name} Data Checking:")
    outlier_data.append(DataValidator.checkOutlier(df=df, dataset_name=config["dataset"]
["name"],
                                                    list_feature=num_features, logger=logger))
if outlier_data:
    for outlier in outlier_data:
        logger.warning("\n" + outlier.to_string())

logger.info("-" * 80)
logger.info("Outlier checking successfully completed across datasets.")
```

```
2026-09-19 04:36:48 | INFO      | -----
-----
2026-09-19 04:36:48 | INFO      | Train Data Checking:
2026-09-19 04:36:48 | INFO      | Checking outliers for EV Adoption Prediction Dataset...
2026-09-19 04:36:48 | INFO      | -----
-----
2026-09-19 04:36:48 | DEBUG     | [EV Adoption Prediction Dataset] Age: No outliers detected
2026-09-19 04:36:48 | WARNING   | [EV Adoption Prediction Dataset] Annual_Income_USD: 3,678
outliers detected (lower=14310.500, upper=155818.500)
2026-09-19 04:36:48 | WARNING   | [EV Adoption Prediction Dataset] Daily_Commute_km: 41 outliers
detected (lower=-28.100, upper=92.700)
2026-09-19 04:36:48 | DEBUG     | [EV Adoption Prediction Dataset] Charging_Stations_Near_Home:
No outliers detected
2026-09-19 04:36:48 | DEBUG     | [EV Adoption Prediction Dataset] Charging_Stations_Near_Work:
No outliers detected
2026-09-19 04:36:48 | INFO      | Test Data Checking:
2026-09-19 04:36:48 | INFO      | Checking outliers for EV Adoption Prediction Dataset...
2026-09-19 04:36:48 | INFO      | -----
-----
2026-09-19 04:36:48 | DEBUG     | [EV Adoption Prediction Dataset] Age: No outliers detected
2026-09-19 04:36:48 | WARNING   | [EV Adoption Prediction Dataset] Annual_Income_USD: 1,538
outliers detected (lower=14323.500, upper=155807.500)
2026-09-19 04:36:48 | WARNING   | [EV Adoption Prediction Dataset] Daily_Commute_km: 16 outliers
detected (lower=-27.950, upper=92.450)
2026-09-19 04:36:48 | DEBUG     | [EV Adoption Prediction Dataset] Charging_Stations_Near_Home:
No outliers detected
2026-09-19 04:36:48 | DEBUG     | [EV Adoption Prediction Dataset] Charging_Stations_Near_Work:
No outliers detected
2026-09-19 04:36:48 | INFO      | Original Data Checking:
2026-09-19 04:36:48 | INFO      | Checking outliers for EV Adoption Prediction Dataset...
2026-09-19 04:36:48 | INFO      | -----
-----
2026-09-19 04:36:48 | DEBUG     | [EV Adoption Prediction Dataset] Age: No outliers detected
2026-09-19 04:36:48 | WARNING   | [EV Adoption Prediction Dataset] Annual_Income_USD: 55 outliers
detected (lower=-7651.125, upper=176403.875)
2026-09-19 04:36:48 | WARNING   | [EV Adoption Prediction Dataset] Daily_Commute_km: 39 outliers
detected (lower=-27.375, upper=107.625)
2026-09-19 04:36:48 | DEBUG     | [EV Adoption Prediction Dataset] Charging_Stations_Near_Home:
No outliers detected
2026-09-19 04:36:48 | DEBUG     | [EV Adoption Prediction Dataset] Charging_Stations_Near_Work:
No outliers detected
2026-09-19 04:36:48 | WARNING   |
Feature      Outlier_Count
0  Annual_Income_USD      3678
1  Daily_Commute_km       41
2026-09-19 04:36:48 | WARNING   |
Feature      Outlier_Count
0  Annual_Income_USD      1538
1  Daily_Commute_km       16
2026-09-19 04:36:48 | WARNING   |
Feature      Outlier_Count
0  Annual_Income_USD       55
1  Daily_Commute_km       39
2026-09-19 04:36:48 | INFO      | -----
-----
2026-09-19 04:36:48 | INFO      | Outlier checking successfully completed across datasets.
```

Exploratory Data Analysis

Class Distribution

```
# Target Distribution Comparison
order = ["No", "Yes"]
colors = ["#16A34A", "#DC2626"]
datasets = {"Train Dataset": df_train, "Original Dataset": df_original}

fig, axes = plt.subplots(1, 2, figsize=(16, 7), facecolor="#181818")
distribution_summary = {}
for ax, (dataset_name, df) in zip(axes, datasets.items()):
    # Copy dataframe
    df_plot = df.copy()

    # Target distribution
    target_distribution = (
        df_plot[config["target"]["feature"]]
        .value_counts()
        .reindex(order, fill_value=0)
    )

    total = len(df_plot)

    distribution_summary[dataset_name] = {
        "No": target_distribution["No"] / total * 100,
        "Yes": target_distribution["Yes"] / total * 100
    }

    # Count Plot
    sns.countplot(
        data=df_plot,
        x=config["target"]["feature"],
        order=order,
        hue=config["target"]["feature"],
        palette=colors,
        legend=False,
        ax=ax
    )

    ax.set_title(
        dataset_name,
        fontsize=12,
        fontweight="bold",
        pad=15
    )

    ax.set_xlabel("")
```

```

ax.set_ylabel(
    "Number of Consumers",
    fontsize=11
)

sns.despine(ax=ax)

# Count + Percentage Labels
for container in ax.containers:
    labels = [
        f"{int(bar.get_height()):,}\n"
        f"({bar.get_height() / total:.1%})"
        for bar in container
    ]
    ax.bar_label(container, labels=labels, fontsize=10, fontweight="bold", padding=3)

# Disable grid
for ax in axes:
    ax.grid(False)

# Overall Title
plt.suptitle(
    "Target Distribution Comparison",
    fontsize=18,
    fontweight="bold"
)

# Overall Insight
train_no = distribution_summary["Train Dataset"]["No"]
train_yes = distribution_summary["Train Dataset"]["Yes"]

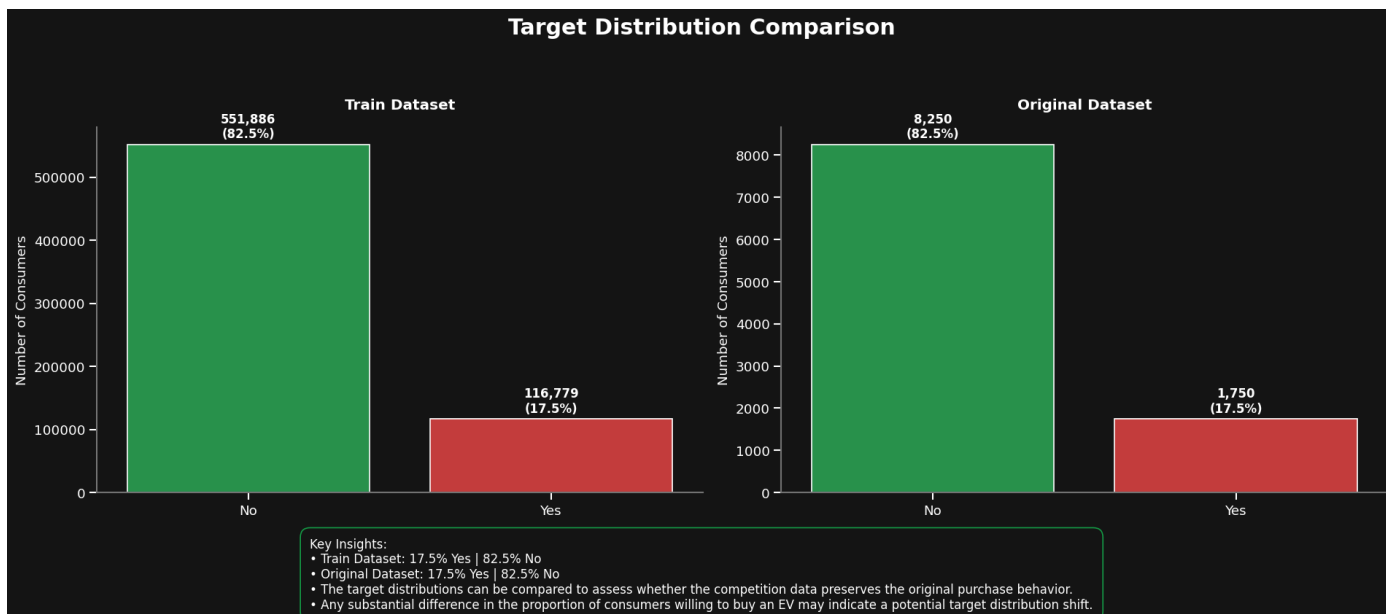
original_no = distribution_summary["Original Dataset"]["No"]
original_yes = distribution_summary["Original Dataset"]["Yes"]

insight = (
    "Key Insights:\n"
    f"• Train Dataset: {train_yes:.1f}% Yes | {train_no:.1f}% No\n"
    f"• Original Dataset: {original_yes:.1f}% Yes | {original_no:.1f}% No\n"
    "• The target distributions can be compared to assess whether "
    "the competition data preserves the original purchase behavior.\n"
    "• Any substantial difference in the proportion of consumers "
    "willing to buy an EV may indicate a potential target distribution shift."
)

fig.text(
    0.5,
    0.01,
    insight,
    ha="center",
    va="bottom",
    multialignment="left",
    fontsize=10,
    bbox=dict(
        boxstyle="round,pad=0.8",
        facecolor="#181818",
        edgecolor="#16A34A",
        alpha=0.95
    )
)

```

```
# Leave space for insight box
plt.tight_layout(
    rect=[0, 0.14, 1, 0.93]
)
plt.show()
```

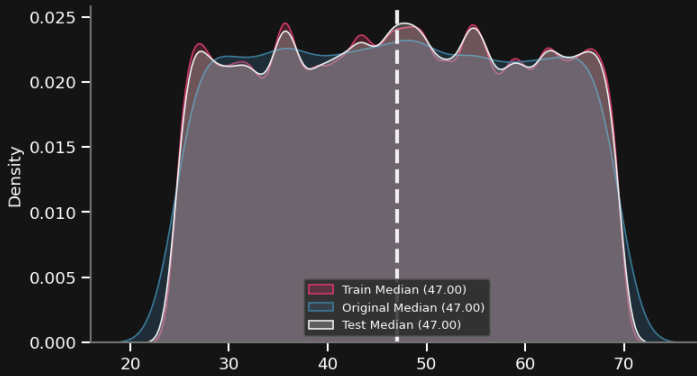


Numerical Feature Distributions

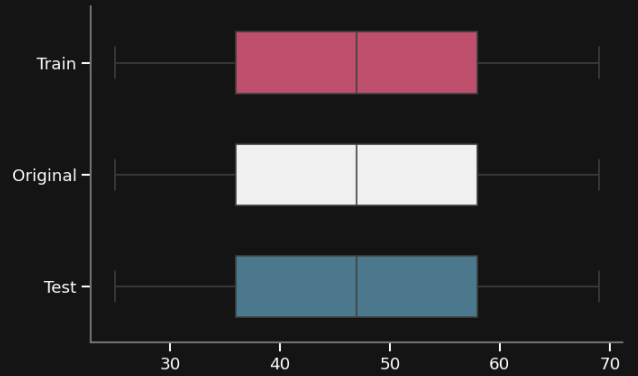
```
plot_numerical_features_v3(df_train=df_train, df_test=df_test, df_original=df_original,
    num_features=num_features)
```

Numerical Feature Distributions

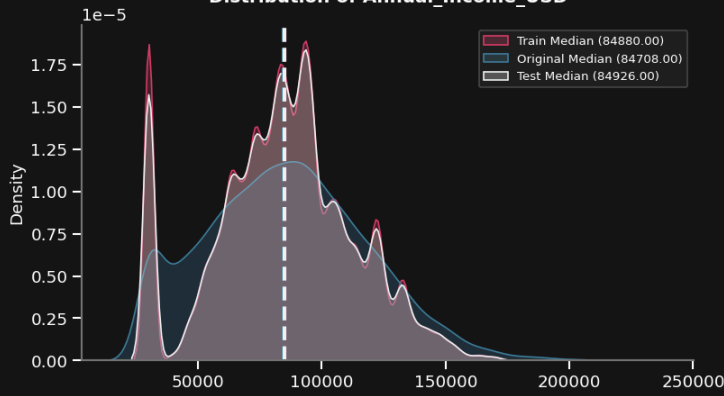
Distribution of Age



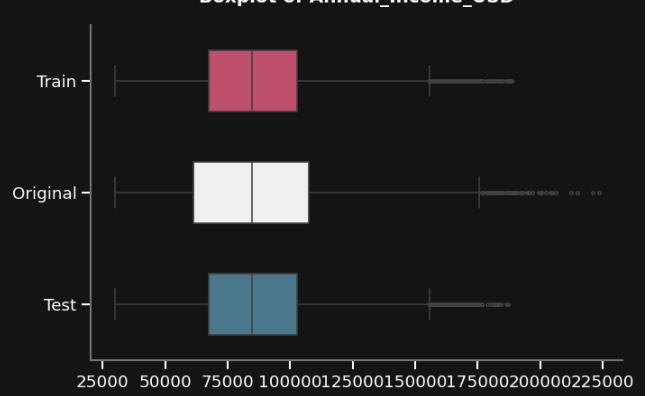
Boxplot of Age



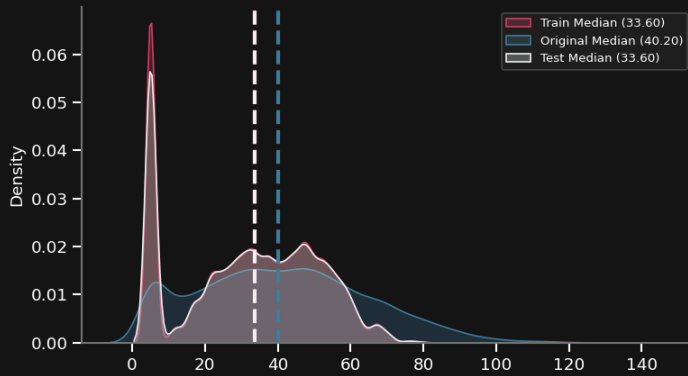
Distribution of Annual_Income_USD



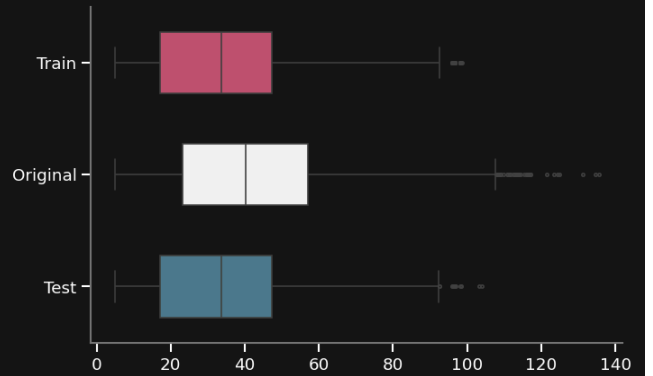
Boxplot of Annual_Income_USD



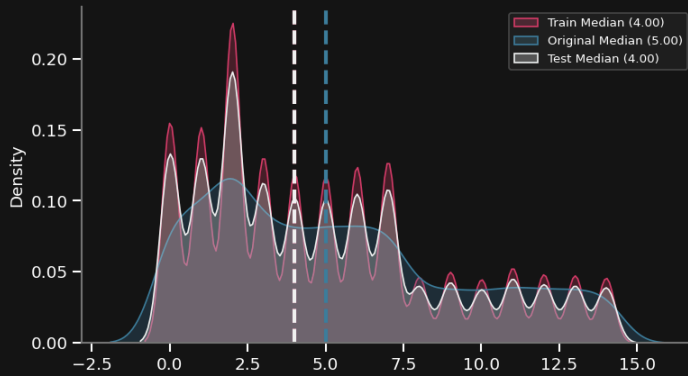
Distribution of Daily_Commute_km



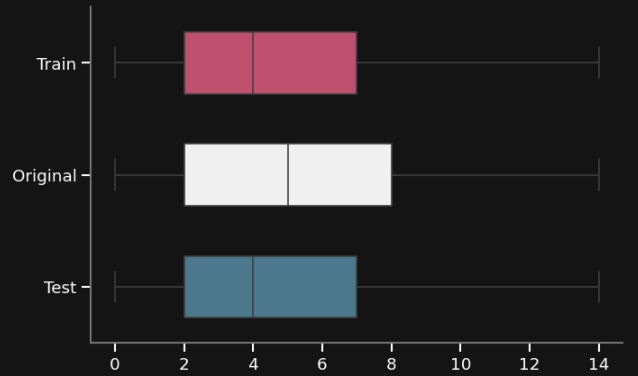
Boxplot of Daily_Commute_km



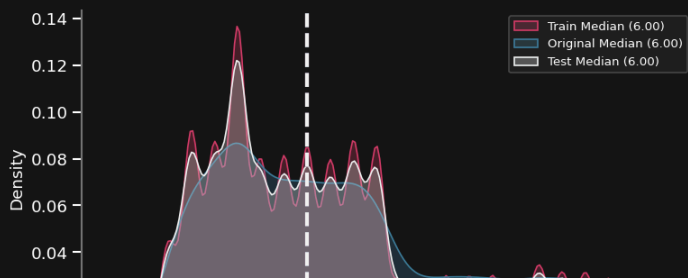
Distribution of Charging_Stations_Near_Home



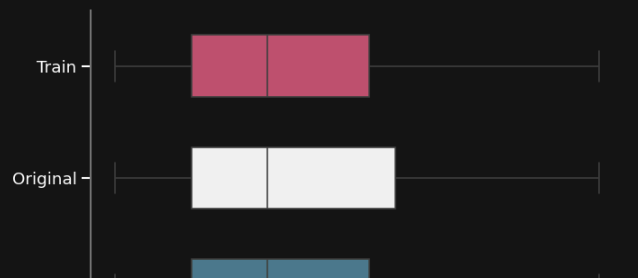
Boxplot of Charging_Stations_Near_Home

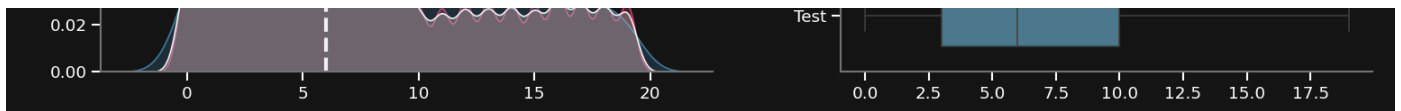


Distribution of Charging_Stations_Near_Work



Boxplot of Charging_Stations_Near_Work





Analyze skewness

```
logger.info("-" * 80)
skew_feature = []
for name, df in datasets.items():
    logger.info(f"{name} Data Checking Skewness:")
    skew_feature.append(DataValidator.check_skewness(data=df, dataset_name= config["dataset"]
["name"],
                                                    numerical_features=num_features,
logger=logger))
logger.info("-" * 80)
logger.info("Checking skewness successfully completed the dataset.")
```

```
2026-09-19 04:37:16 | INFO | -----
-----
2026-09-19 04:37:16 | INFO | Train Dataset Data Checking Skewness:
2026-09-19 04:37:16 | INFO | Skewness analysis for EV Adoption Prediction Dataset
2026-09-19 04:37:16 | INFO | Total numerical features checked: 5
2026-09-19 04:37:16 | DEBUG | -----
-----
2026-09-19 04:37:16 | DEBUG | Feature | Skewness | Remark
2026-09-19 04:37:16 | DEBUG | -----
-----
2026-09-19 04:37:16 | DEBUG | Charging_Stations_Near_Work | +0.7002 | Moderately skewed
2026-09-19 04:37:16 | DEBUG | Charging_Stations_Near_Home | +0.6987 | Moderately skewed
2026-09-19 04:37:16 | DEBUG | Daily_Commute_km | -0.0846 | Approximately
symmetric
2026-09-19 04:37:16 | DEBUG | Annual_Income_USD | -0.0131 | Approximately
symmetric
2026-09-19 04:37:16 | DEBUG | Age | -0.0038 | Approximately
symmetric
2026-09-19 04:37:16 | DEBUG | -----
-----
2026-09-19 04:37:16 | INFO | 2 features are skewed (|skew| > 0.5).
2026-09-19 04:37:16 | INFO | Recommendation: Consider log / Box-Cox / Yeo-Johnson
transformation.
2026-09-19 04:37:16 | INFO | Original Dataset Data Checking Skewness:
2026-09-19 04:37:16 | INFO | Skewness analysis for EV Adoption Prediction Dataset
2026-09-19 04:37:16 | INFO | Total numerical features checked: 5
2026-09-19 04:37:16 | DEBUG | -----
-----
2026-09-19 04:37:16 | DEBUG | Feature | Skewness | Remark
2026-09-19 04:37:16 | DEBUG | -----
-----
2026-09-19 04:37:16 | DEBUG | Charging_Stations_Near_Work | +0.6080 | Moderately skewed
2026-09-19 04:37:16 | DEBUG | Charging_Stations_Near_Home | +0.5712 | Moderately skewed
2026-09-19 04:37:16 | DEBUG | Daily_Commute_km | +0.3361 | Approximately
symmetric
2026-09-19 04:37:16 | DEBUG | Annual_Income_USD | +0.3052 | Approximately
symmetric
2026-09-19 04:37:16 | DEBUG | Age | +0.0090 | Approximately
symmetric
2026-09-19 04:37:16 | DEBUG | -----
-----
2026-09-19 04:37:16 | INFO | 2 features are skewed (|skew| > 0.5).
2026-09-19 04:37:16 | INFO | Recommendation: Consider log / Box-Cox / Yeo-Johnson
transformation.
2026-09-19 04:37:16 | INFO | -----
-----
2026-09-19 04:37:16 | INFO | Checking skewness successfully completed the dataset.
```



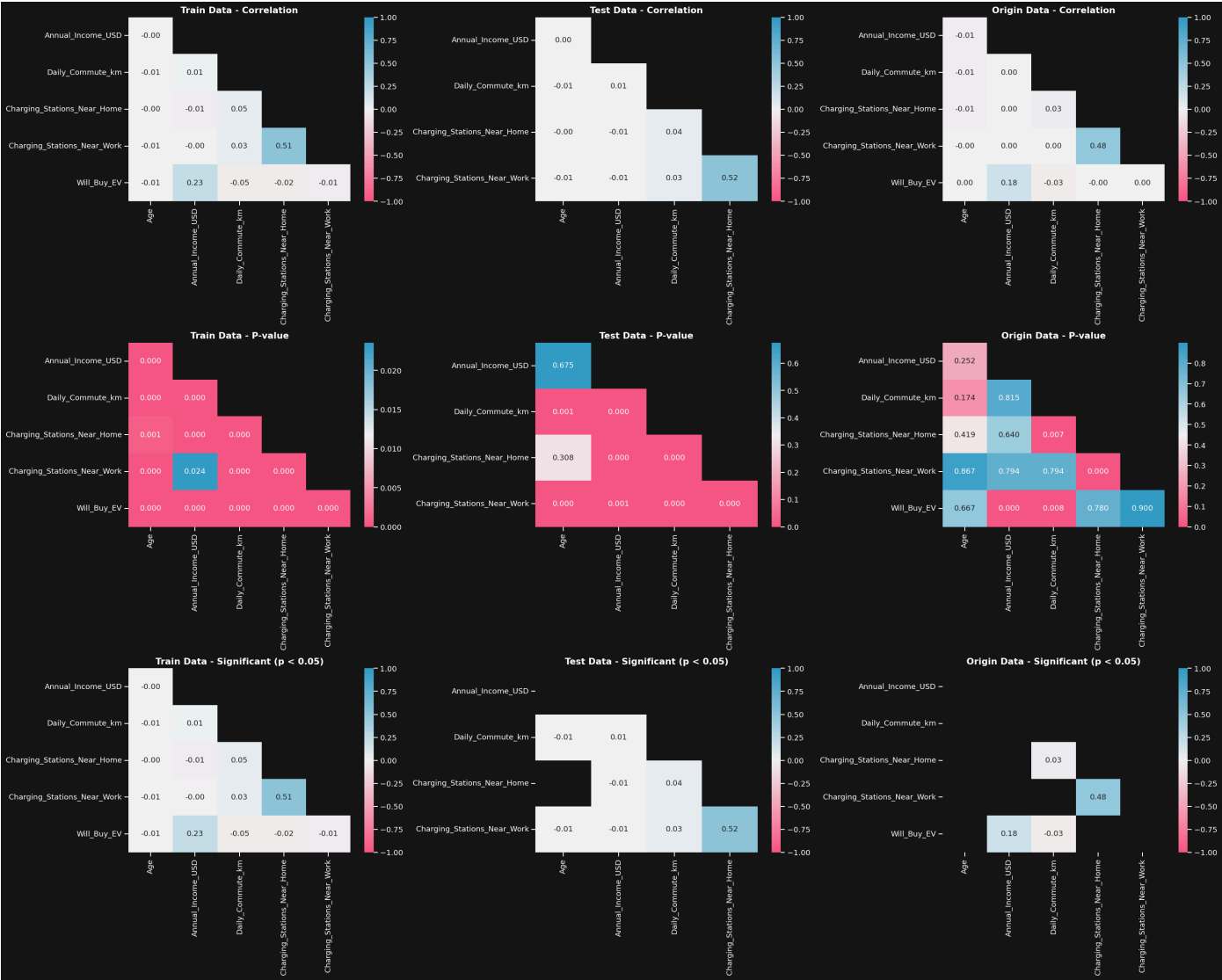
Distribution & Skewness Insights

- The **training and test datasets** exhibit highly consistent numerical distributions across most features, with nearly identical medians, quartiles, and value ranges. This suggests **no meaningful distribution shift** between the competition training and test sets.
- **Age** is approximately symmetric in the training dataset ($\text{skew} \approx -0.004$), with a median of **47 years**. The distributions across Train, Test, and Original datasets are highly similar.
- **Annual_Income_USD** is also approximately symmetric in the training dataset ($\text{skew} \approx -0.013$). The Train and Test datasets show very similar income distributions, while the Original dataset exhibits a slightly more right-skewed distribution ($\text{skew} \approx 0.305$) with a broader upper range.
- **Daily_Commute_km** is approximately symmetric in the training dataset ($\text{skew} \approx -0.085$). However, the Original dataset has a higher median of approximately **40.2 km**, compared with approximately **33.6 km** in Train and Test, indicating a noticeable distribution difference in commuting distance.
- **Charging_Stations_Near_Home** and **Charging_Stations_Near_Work** are **moderately right-skewed** in the training dataset, with skewness values of approximately **0.70** and **0.70**, respectively. This indicates that most consumers have relatively moderate charging availability, while a smaller group has access to substantially more charging stations.
- The same charging-related variables remain moderately right-skewed in the Original dataset, with skewness of approximately **0.57** for **Charging_Stations_Near_Home** and **0.61** for **Charging_Stations_Near_Work**. The overall distribution patterns remain comparable despite differences in skewness magnitude.
- **No highly skewed numerical features** were identified, as all observed skewness values remain below **1**. Therefore, there is no strong statistical requirement for logarithmic or Box-Cox transformation based solely on skewness.
- For the charging-related count variables, the observed skewness is **structurally reasonable** because the number of available charging stations is a discrete count variable bounded within a relatively small range.
- **Overall:** The numerical features show generally stable and well-behaved distributions in the competition datasets. The main distribution difference compared with the Original dataset is observed in **Daily_Commute_km**, while the charging infrastructure variables exhibit moderate positive skewness that should be considered during modeling.

Correlation Analysis of Numerical Features

```
# Encode the binary target variable as 0/1 for correlation analysis.
mapping = {"No": 0, "Yes": 1}
df_train[config["target"]["feature"]] = df_train[config["target"]["feature"]].map(mapping)
df_original[config["target"]["feature"]] = df_original[config["target"]
["feature"]].map(mapping)

# Visualize correlations and statistical significance across datasets.
StatUtils.plot_correlation_with_pvalue(df_train=df_train, df_test=df_test,
df_origin=df_original)
```





Correlation Analysis Insights

- The **training, test, and original datasets** exhibit highly consistent correlation patterns, with similar correlation directions and magnitudes across the numerical features. This suggests **no meaningful structural shift** between the competition datasets and the original data.
- `Charging_Stations_Near_Home` and `Charging_Stations_Near_Work` show the strongest feature-to-feature relationship, with a moderate positive correlation of approximately $r \approx 0.48\text{--}0.52$ across the three datasets. This indicates that consumers living in areas with better charging infrastructure near home also tend to have greater charging availability near their workplace.
- `Will_Buy_EV` has the strongest relationship with `Annual_Income_USD`, with a positive correlation of approximately $r \approx 0.18\text{--}0.23$. This suggests that higher-income consumers tend to show a greater likelihood of purchasing an electric vehicle.
- `Daily_Commute_km` has a **weak negative correlation** with `Will_Buy_EV` ($r \approx -0.03$ to -0.05). Although statistically significant in the larger datasets, the small correlation magnitude indicates that commuting distance alone has **limited linear association** with EV purchase intention.
- The remaining numerical features, including `Age`, `Charging_Stations_Near_Home`, and `Charging_Stations_Near_Work`, show **very weak correlations** with `Will_Buy_EV` (approximately $|r| < 0.03$). These variables may still provide predictive value through nonlinear relationships or interactions with other features.
- Most feature pairs have correlations close to **zero**, indicating generally **low multicollinearity**. The main exception is the relationship between the two charging infrastructure variables, which should be considered when interpreting feature importance or coefficients.
- The **large training and test sample sizes** make even relatively weak correlations statistically significant ($p < 0.05$). Therefore, statistical significance should not be confused with practical importance; the **correlation magnitude** remains important when evaluating predictive relevance.
- Compared with the competition datasets, the **original dataset shows fewer statistically significant weak relationships**, which is expected given its substantially smaller sample size. However, the major correlation patterns remain consistent.
- **Overall:** `Annual_Income_USD` appears to be the strongest numerical indicator associated with EV purchase intention, while charging infrastructure variables show moderate correlation with each other. Overall multicollinearity is limited, providing a relatively clean numerical feature set for subsequent modeling.

Categorical Feature Distributions

[illegible]

```

2026-09-19 04:37:22 | INFO      | Analyzing categorical feature: Gender
2026-09-19 04:37:22 | DEBUG    | Gender: 3 categories detected
2026-09-19 04:37:22 | INFO      | Categorical Feature Distribution: Gender:
  Category  Train_Count  Train_%  Test_Count  Test_%  Original_Count  Original_%
1    Male      367954    55.03    157647    55.01         5201      52.01
0    Female    295427    44.18    126669    44.20         4498      44.98
2    Other      5284      0.79     2255     0.79          301       3.01
2026-09-19 04:37:22 | INFO      | Completed analysis for: Gender
2026-09-19 04:37:22 | INFO      |
*****
2026-09-19 04:37:22 | INFO      | Analyzing categorical feature: City_Type
2026-09-19 04:37:23 | DEBUG    | City_Type: 3 categories detected
2026-09-19 04:37:23 | INFO      | Categorical Feature Distribution: City_Type:
  Category  Train_Count  Train_%  Test_Count  Test_%  Original_Count  Original_%
2    Urban      289305    43.27    123602    43.13         4941      49.41
1  Suburban      255377    38.19    109676    38.27         3563      35.63
0    Rural      123983    18.54     53293    18.60         1496      14.96
2026-09-19 04:37:23 | INFO      | Completed analysis for: City_Type
2026-09-19 04:37:23 | INFO      |
*****
2026-09-19 04:37:23 | INFO      | Analyzing categorical feature: Current_Car_Type
2026-09-19 04:37:23 | DEBUG    | Current_Car_Type: 4 categories detected
2026-09-19 04:37:23 | INFO      | Categorical Feature Distribution: Current_Car_Type:
  Category  Train_Count  Train_%  Test_Count  Test_%  Original_Count  Original_%
2    Sedan      303459    45.38    130185    45.43         4000      40.00
1    SUV        246545    36.87    105687    36.88         3486      34.86
0  Hatchback     79438    11.88     34104    11.90         1523      15.23
3    Truck      39223     5.87     16595     5.79          991       9.91
2026-09-19 04:37:23 | INFO      | Completed analysis for: Current_Car_Type
2026-09-19 04:37:23 | INFO      |
*****
2026-09-19 04:37:23 | INFO      | Analyzing categorical feature: Home_Charging_Possible
2026-09-19 04:37:23 | DEBUG    | Home_Charging_Possible: 2 categories detected
2026-09-19 04:37:23 | INFO      | Categorical Feature Distribution: Home_Charging_Possible:
  Category  Train_Count  Train_%  Test_Count  Test_%  Original_Count  Original_%
1    Yes      462677    69.19    198480    69.26         6163      61.63
0    No       205988    30.81     88091    30.74         3837      38.37
2026-09-19 04:37:23 | INFO      | Completed analysis for: Home_Charging_Possible
2026-09-19 04:37:23 | INFO      |
*****
2026-09-19 04:37:23 | INFO      | Analyzing categorical feature: Subsidy_Available
2026-09-19 04:37:23 | DEBUG    | Subsidy_Available: 2 categories detected
2026-09-19 04:37:23 | INFO      | Categorical Feature Distribution: Subsidy_Available:
  Category  Train_Count  Train_%  Test_Count  Test_%  Original_Count  Original_%
1    Yes      419909    62.8     180378    62.94         6008      60.08
0    No       248756    37.2     106193    37.06         3992      39.92
2026-09-19 04:37:23 | INFO      | Completed analysis for: Subsidy_Available
2026-09-19 04:37:23 | INFO      |
*****
2026-09-19 04:37:23 | INFO      | Analyzing categorical feature: Environmental_Concern_Level
2026-09-19 04:37:23 | DEBUG    | Environmental_Concern_Level: 5 categories detected
2026-09-19 04:37:23 | INFO      | Categorical Feature Distribution: Environmental_Concern_Level:
  Category  Train_Count  Train_%  Test_Count  Test_%  Original_Count  Original_%
0    1      147476    22.06     63125    22.03         2013      20.51
1    2      135133    20.21     58019    20.25         1948      19.85
3    4      130469    19.51     56326    19.66         1989      20.26
4    5      128236    19.18     54594    19.05         1908      19.44
2    3      127351    19.05     54507    19.02         1958      19.95
2026-09-19 04:37:23 | INFO      | Completed analysis for: Environmental_Concern_Level
2026-09-19 04:37:23 | INFO      |

```

```
*****
2026-09-19 04:37:23 | INFO      | Analyzing categorical feature: Range_Anxiety_Level
2026-09-19 04:37:23 | DEBUG    | Range_Anxiety_Level: 3 categories detected
2026-09-19 04:37:23 | INFO      | Categorical Feature Distribution: Range_Anxiety_Level:
  Category  Train_Count  Train_%  Test_Count  Test_%  Original_Count  Original_%
1      Low      603972    90.33      259247    90.47          8310      83.10
2  Medium      62499     9.35       26295     9.18          1523      15.23
0      High      2194      0.33       1029     0.36           167       1.67
2026-09-19 04:37:23 | INFO      | Completed analysis for: Range_Anxiety_Level
2026-09-19 04:37:23 | INFO      |
*****
2026-09-19 04:37:23 | INFO      | Analyzing categorical feature: Number_of_Cars_Owned
2026-09-19 04:37:23 | DEBUG    | Number_of_Cars_Owned: 4 categories detected
2026-09-19 04:37:23 | INFO      | Categorical Feature Distribution: Number_of_Cars_Owned:
  Category  Train_Count  Train_%  Test_Count  Test_%  Original_Count  Original_%
1          2      298287    44.61     127869    44.62         4020      40.20
0          1      288012    43.07     122958    42.91         3946      39.46
2          3       68877    10.30       29889    10.43         1523      15.23
3          4       13489     2.02        5855     2.04          511       5.11
2026-09-19 04:37:23 | INFO      | Completed analysis for: Number_of_Cars_Owned
2026-09-19 04:37:23 | INFO      |
*****
```



Categorical Feature Insights

- **Gender** is moderately dominated by **Male** consumers in the competition datasets, representing approximately **55%** of Train and Test observations, while **Female** accounts for approximately **44%**. The **Other** category is relatively rare at only **0.79%**. The Original dataset shows a similar Male/Female distribution, but a noticeably higher proportion of **Other** consumers (**3.01%**).
- **City_Type** is dominated by **Urban** consumers, accounting for approximately **43%** of both Train and Test datasets, followed by **Suburban** at approximately **38%**. **Rural** consumers represent around **19%**. Compared with the competition datasets, the Original dataset contains a higher proportion of **Urban** consumers (**49.41%**) and a lower proportion of **Rural** consumers (**14.96%**).
- **Current_Car_Type** is dominated by **Sedan** and **SUV** owners, which together account for more than **82%** of Train and Test observations. The Original dataset has a somewhat higher proportion of **Truck** and **Hatchback** owners, indicating a moderate difference in the current vehicle composition.
- **Home_Charging_Possible** is predominantly **Yes** in the competition datasets, representing approximately **69%** of both Train and Test consumers. However, this proportion decreases to **61.63%** in the Original dataset, suggesting a noticeable difference in home-charging availability between the original and competition data.
- **Subsidy_Available** is also dominated by **Yes**, accounting for approximately **63%** of Train and Test observations. The Original dataset shows a very similar distribution at approximately **60%**, indicating only a relatively small shift in subsidy availability.
- **Environmental_Concern_Level** has a **well-balanced distribution across all five levels**. In the Train and Test datasets, each category represents approximately **19–22%** of observations, suggesting that no particular environmental concern level dominates the competition data.
- **Range_Anxiety_Level** is highly concentrated in the **Low** category, representing approximately **90%** of Train and Test observations. The Original dataset has a lower proportion of **Low** range anxiety (**83.10%**) and higher proportions of **Medium** and **High**, indicating a noticeable distribution shift.
- **Number_of_Cars_Owned** is concentrated around **1–2 cars per consumer**, with these two categories accounting for approximately **88%** of Train and Test observations. The Original dataset contains relatively more consumers owning **3–4 cars**, indicating a moderate difference in vehicle ownership patterns.
- The **training and test datasets exhibit highly consistent categorical distributions** across all features, with almost identical category proportions. This is a positive sign for competition modeling because the test set appears to closely represent the training population.

- However, several categorical features show noticeable differences between the **competition datasets and the Original dataset**, particularly `Gender`, `Home_Charging_Possible`, `Range_Anxiety_Level`, and `Number_of_Cars_Owned`. These differences should be considered when interpreting how well the competition data represents the original population.
- **Overall:** The competition datasets provide stable categorical distributions between Train and Test, while the Original dataset reveals several population-level differences. Features related to **home charging, range anxiety, vehicle ownership, and urbanization** may therefore capture important differences in EV adoption behavior.

Multivariate Statistical Analysis

```
# Map the binary target back to readable labels for analysis.
mapping = {0: "No", 1: "Yes"}

df_plot = df_train.copy()
df_plot[config["target"]["feature"]] = df_plot[config["target"]["feature"]].map(mapping)

for feature in num_features:
    # Display a title for the current feature.
    display(
        HTML(
            f"<h2 style='text-align:center; font-size:22px; color:#fde68a;'>"
            f"<b>Distribution of {feature} by EV Purchase Decision</b>"
            f"</h2>"
        )
    )

    # Analyze the feature distribution across target classes.
    StatUtils.analyze_numeric_feature_by_target(
        feature=feature,
        df=df_plot,
        logger=logger,
        target_feature=config["target"]["feature"],
    )
```

Distribution of Age by EV Purchase Decision

- **Overall Mean:** 47.04
- **Overall Median:** 47.0
- **Overall Std:** 12.88

	Will_Buy_EV	Count	Mean	Median	Std
0	No	551886	47.083293	47.000000	12.896488
1	Yes	116779	46.830654	47.000000	12.773543

2026-09-19 04:37:23 | INFO | Checking normality of feature 'Age' by groups of 'Will_Buy_EV'...

2026-09-19 04:37:23 | INFO | Evaluating distribution characteristics (skewness & kurtosis)...

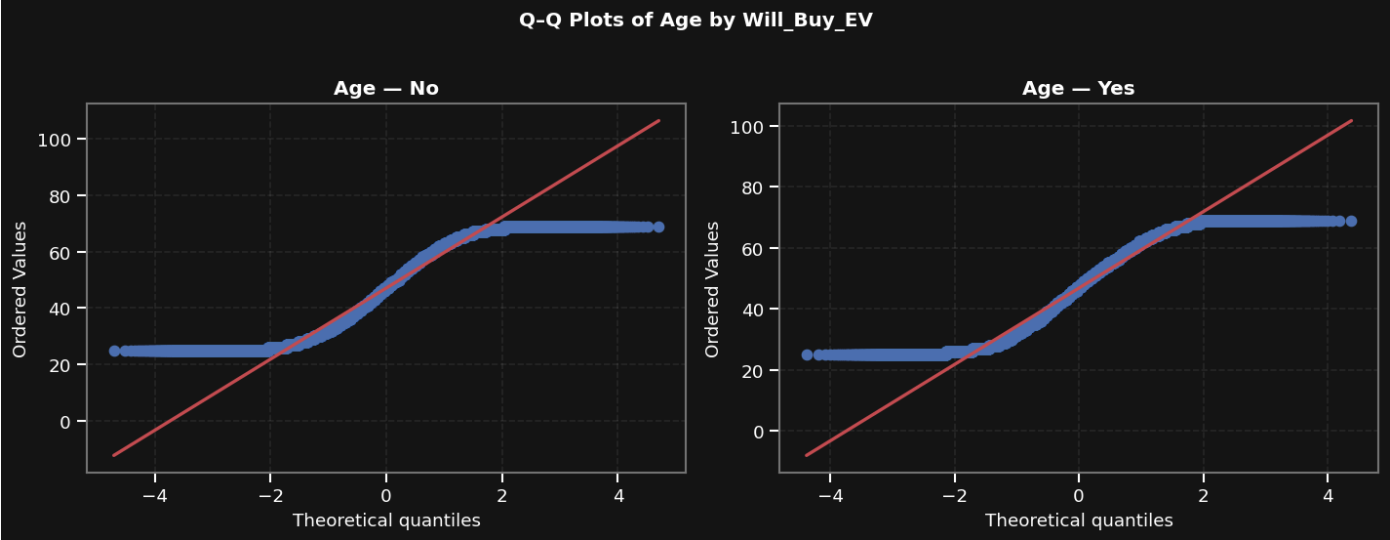
2026-09-19 04:37:23 | DEBUG | Number of groups detected: 2

2026-09-19 04:37:23 | DEBUG | Thresholds → |skew|≤0.5, |kurtosis|≤1.5

2026-09-19 04:37:23 | DEBUG | [Will_Buy_EV=No] skew=-0.0022, kurtosis=-1.1814 → Approximately symmetric, Normal tails

2026-09-19 04:37:23 | DEBUG | [Will_Buy_EV=Yes] skew=-0.0126, kurtosis=-1.1396 → Approximately symmetric, Normal tails

2026-09-19 04:37:23 | INFO | Generating Q-Q plots for visual assessment...



Skewness & Kurtosis of Age by Will_Buy_EV

	Feature	Group	Skewness	Kurtosis	Remark
0	Age	No	-0.0022	-1.1814	Approximately symmetric, Normal tails
1	Age	Yes	-0.0126	-1.1396	Approximately symmetric, Normal tails

2026-09-19 04:37:25 | INFO | All groups approximately follow a normal distribution.

2026-09-19 04:37:25 | INFO | Check homogeneity of variances across groups using Levene's test (median-centered)

2026-09-19 04:37:25 | DEBUG | Number of groups detected: 2

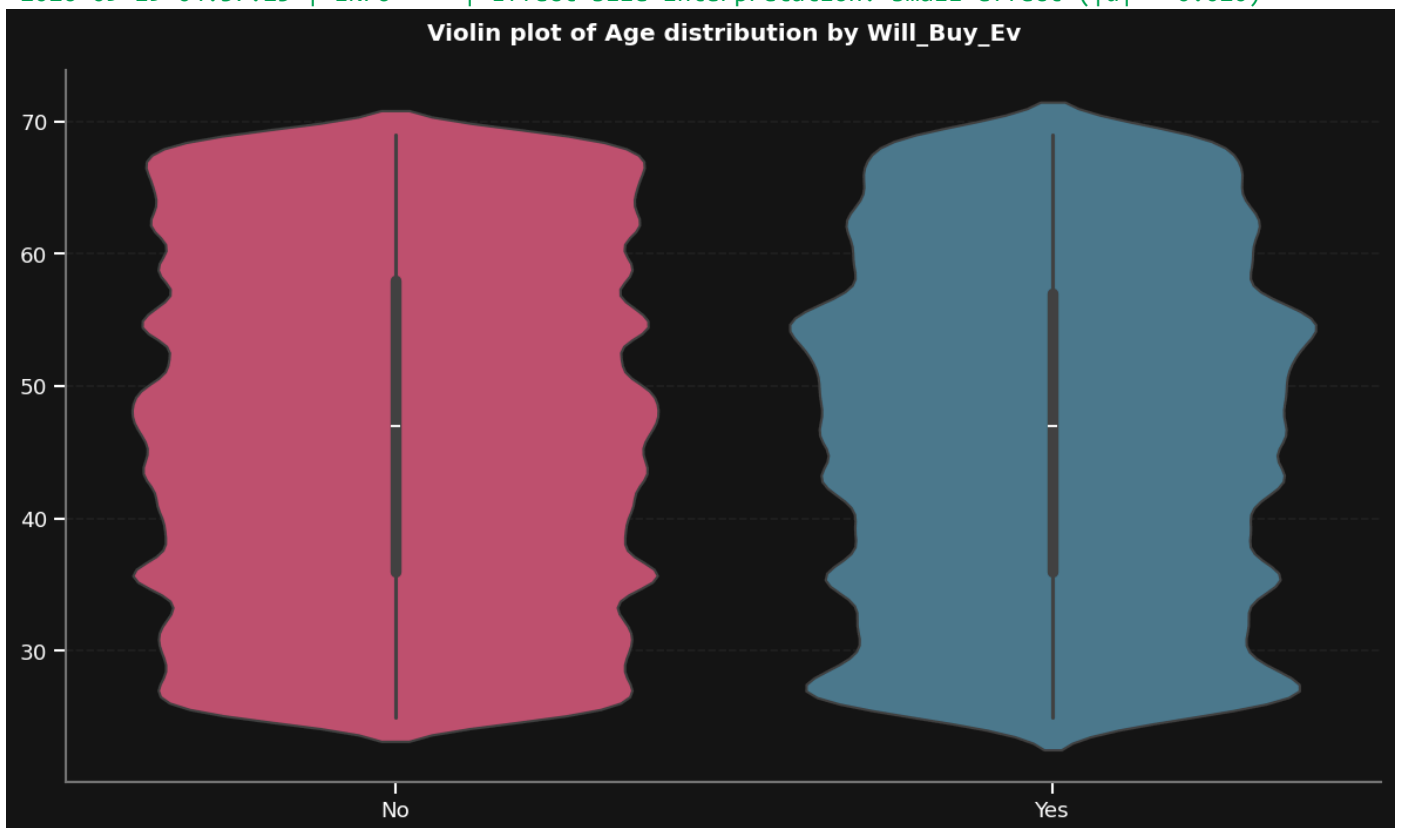
Homogeneity of Variance — Age by Will_Buy_EV

	Metric	Value
0	Levene's Statistic	53.7311
1	p-value	0.000000
2	Max/Min Variance Ratio	1.02

```

2026-09-19 04:37:25 | INFO      | Interpretation:
2026-09-19 04:37:25 | INFO      | - Status: Statistically different variances, but practical
difference is small.
2026-09-19 04:37:25 | INFO      | - Recommendation: ANOVA or T-Test are acceptable, but Welch's
ANOVA or Welch's T-Test is recommended.
2026-09-19 04:37:25 | INFO      | Running Independent Two-Sample T-Test: 'Age' ~ 'Will_Buy_EV'
2026-09-19 04:37:25 | DEBUG     | Number of groups detected: 2
2026-09-19 04:37:25 | DEBUG     | Test Type: Welch's T-Test (unequal variances)
2026-09-19 04:37:25 | INFO      | Comparing groups: 'No' vs 'Yes'
2026-09-19 04:37:25 | INFO      | p-value: 0.000000
2026-09-19 04:37:25 | INFO      | Cohen's d: 0.020
2026-09-19 04:37:25 | DEBUG     | t-statistic: 6.130
2026-09-19 04:37:25 | DEBUG     | Group means → No: 47.083, Yes: 46.831
2026-09-19 04:37:25 | INFO      | Result: p-value ≤ 0.05 → Reject H0
2026-09-19 04:37:25 | INFO      | Conclusion: Significant difference between the two groups.
2026-09-19 04:37:25 | INFO      | Effect size interpretation: small effect (|d| = 0.020)

```



Distribution of Annual_Income_USD by EV Purchase Decision

- **Overall Mean:** 84769.29
- **Overall Median:** 84880.0
- **Overall Std:** 28637.11

	Will_Buy_EV	Count	Mean	Median	Std
0	Yes	116779	98827.304688	96167.000000	26300.675910
1	No	551886	81794.585938	82844.000000	28237.155138

2026-09-19 04:37:28 | INFO | Checking normality of feature 'Annual_Income_USD' by groups of 'Will_Buy_EV'...

2026-09-19 04:37:28 | INFO | Evaluating distribution characteristics (skewness & kurtosis)...

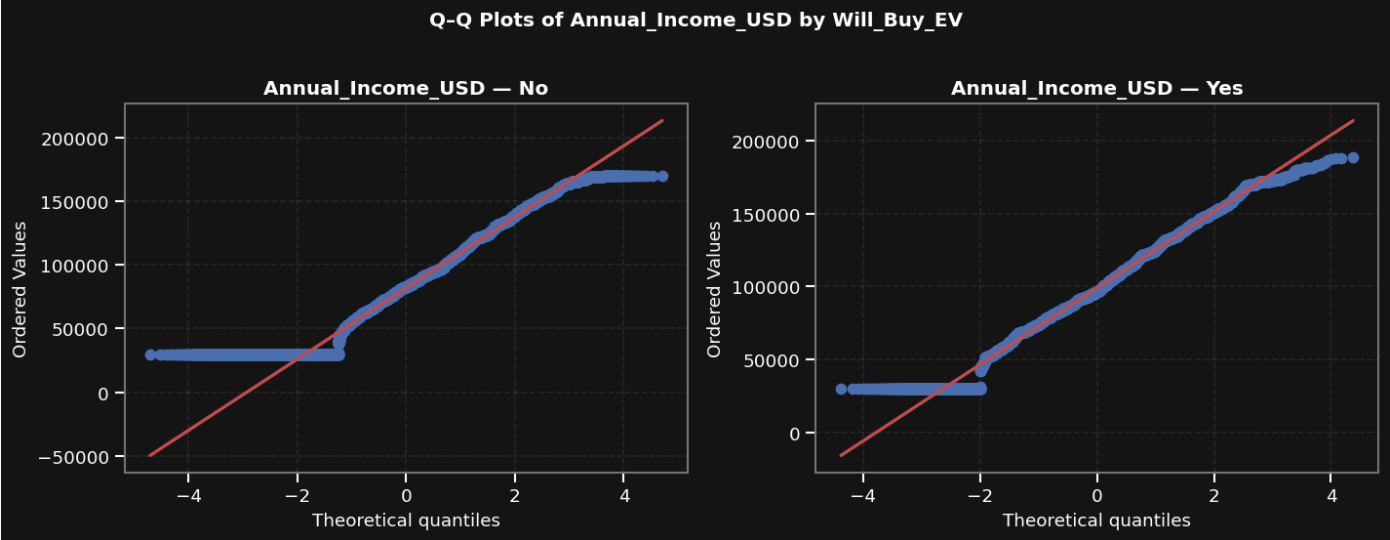
2026-09-19 04:37:28 | DEBUG | Number of groups detected: 2

2026-09-19 04:37:28 | DEBUG | Thresholds → |skew|≤0.5, |kurtosis|≤1.5

2026-09-19 04:37:28 | DEBUG | [Will_Buy_EV=No] skew=0.0042, kurtosis=-0.1797 → Approximately symmetric, Normal tails

2026-09-19 04:37:28 | DEBUG | [Will_Buy_EV=Yes] skew=-0.0226, kurtosis=0.1137 → Approximately symmetric, Normal tails

2026-09-19 04:37:28 | INFO | Generating Q-Q plots for visual assessment...



Skewness & Kurtosis of Annual_Income_USD by Will_Buy_EV

	Feature	Group	Skewness	Kurtosis	Remark
0	Annual_Income_USD	No	0.0042	-0.1797	Approximately symmetric, Normal tails
1	Annual_Income_USD	Yes	-0.0226	0.1137	Approximately symmetric, Normal tails

2026-09-19 04:37:30 | INFO | All groups approximately follow a normal distribution.

2026-09-19 04:37:30 | INFO | Check homogeneity of variances across groups using Levene's test (median-centered)

2026-09-19 04:37:30 | DEBUG | Number of groups detected: 2

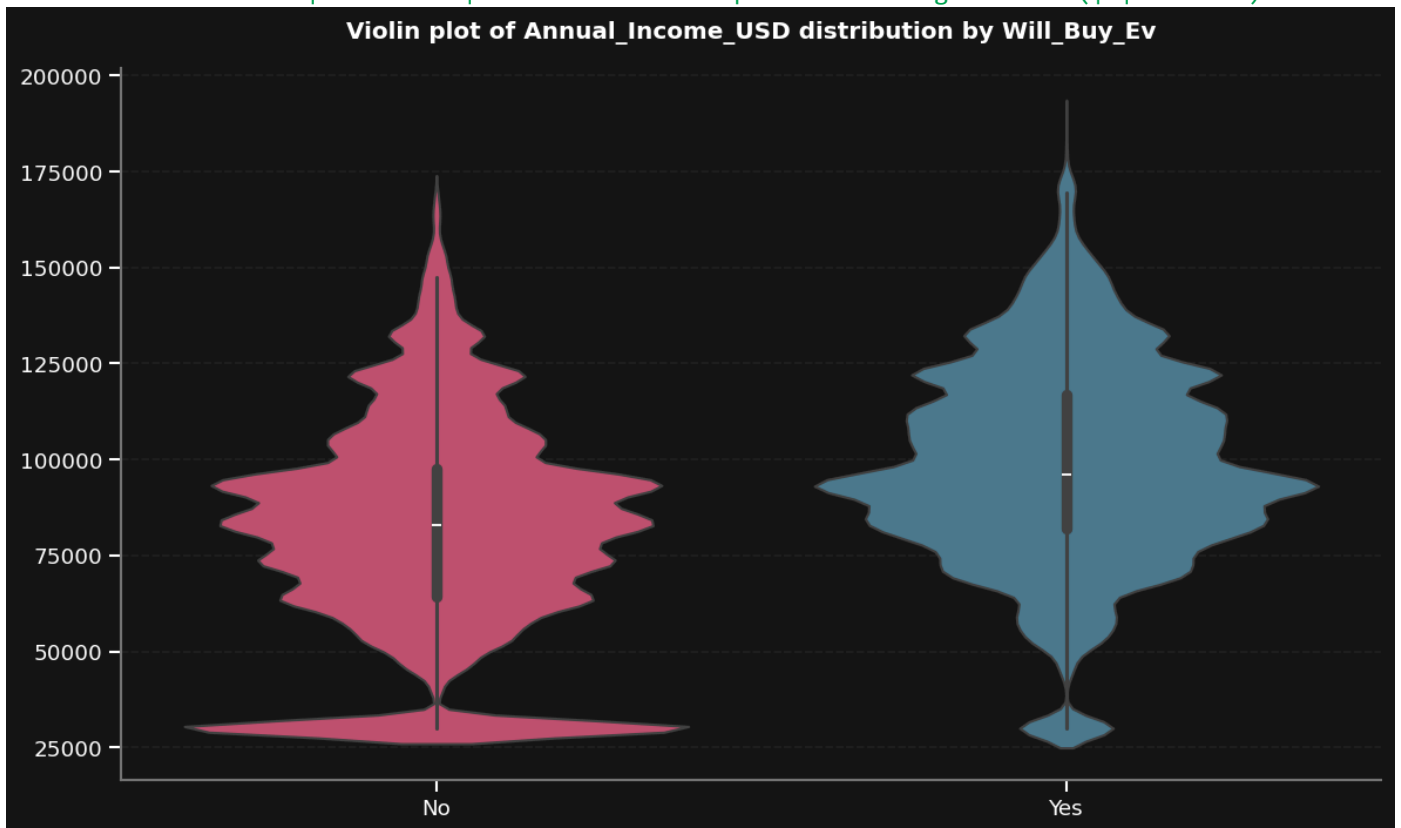
Homogeneity of Variance — Annual_Income_USD by Will_Buy_EV

		Metric	Value
0	Levene's Statistic		506.1255
1	p-value		0.000000
2	Max/Min Variance Ratio		1.15

```

2026-09-19 04:37:30 | INFO      | Interpretation:
2026-09-19 04:37:30 | INFO      | - Status: Statistically different variances, but practical
difference is small.
2026-09-19 04:37:30 | INFO      | - Recommendation: ANOVA or T-Test are acceptable, but Welch's
ANOVA or Welch's T-Test is recommended.
2026-09-19 04:37:30 | INFO      | Running Independent Two-Sample T-Test: 'Annual_Income_USD' ~
'Will_Buy_EV'
2026-09-19 04:37:30 | DEBUG     | Number of groups detected: 2
2026-09-19 04:37:30 | DEBUG     | Test Type: Welch's T-Test (unequal variances)
2026-09-19 04:37:30 | INFO      | Comparing groups: 'No' vs 'Yes'
2026-09-19 04:37:30 | INFO      | p-value: 0.000000
2026-09-19 04:37:30 | INFO      | Cohen's d: -0.624
2026-09-19 04:37:30 | DEBUG     | t-statistic: -198.429
2026-09-19 04:37:30 | DEBUG     | Group means → No: 81794.570, Yes: 98827.305
2026-09-19 04:37:30 | INFO      | Result: p-value ≤ 0.05 → Reject H0
2026-09-19 04:37:30 | INFO      | Conclusion: Significant difference between the two groups.
2026-09-19 04:37:30 | INFO      | Effect size interpretation: large effect (|d| = 0.624)

```



Distribution of Daily_Commute_km by EV Purchase Decision

- **Overall Mean:** 32.16
- **Overall Median:** 33.599998474121094
- **Overall Std:** 18.73

	Will_Buy_EV	Count	Mean	Median	Std
0	No	551886	32.553478	34.000000	18.820772
1	Yes	116779	30.290714	31.700001	18.181942

2026-09-19 04:37:33 | INFO | Checking normality of feature 'Daily_Commute_km' by groups of 'Will_Buy_EV'...

2026-09-19 04:37:33 | INFO | Evaluating distribution characteristics (skewness & kurtosis)...

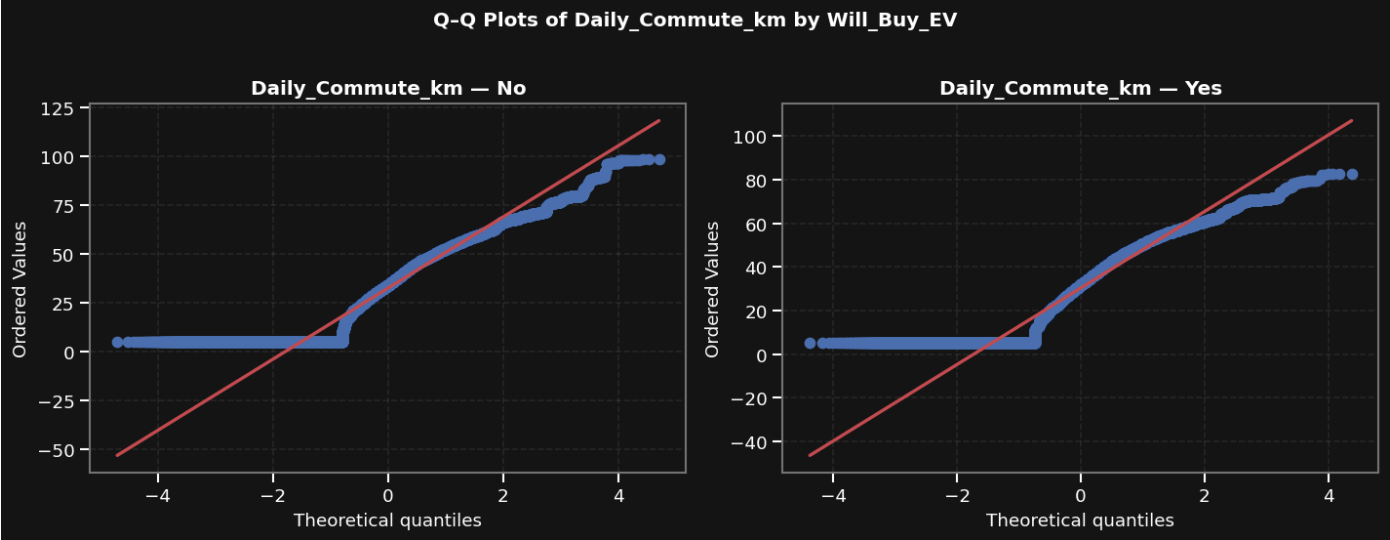
2026-09-19 04:37:33 | DEBUG | Number of groups detected: 2

2026-09-19 04:37:33 | DEBUG | Thresholds → |skew|≤0.5, |kurtosis|≤1.5

2026-09-19 04:37:33 | DEBUG | [Will_Buy_EV=No] skew=-0.0993, kurtosis=-1.0422 → Approximately symmetric, Normal tails

2026-09-19 04:37:33 | DEBUG | [Will_Buy_EV=Yes] skew=-0.0299, kurtosis=-1.1629 → Approximately symmetric, Normal tails

2026-09-19 04:37:33 | INFO | Generating Q-Q plots for visual assessment...



Skewness & Kurtosis of Daily_Commute_km by Will_Buy_EV

	Feature	Group	Skewness	Kurtosis	Remark
0	Daily_Commute_km	No	-0.0993	-1.0422	Approximately symmetric, Normal tails
1	Daily_Commute_km	Yes	-0.0299	-1.1629	Approximately symmetric, Normal tails

2026-09-19 04:37:35 | INFO | All groups approximately follow a normal distribution.

2026-09-19 04:37:35 | INFO | Check homogeneity of variances across groups using Levene's test (median-centered)

2026-09-19 04:37:35 | DEBUG | Number of groups detected: 2

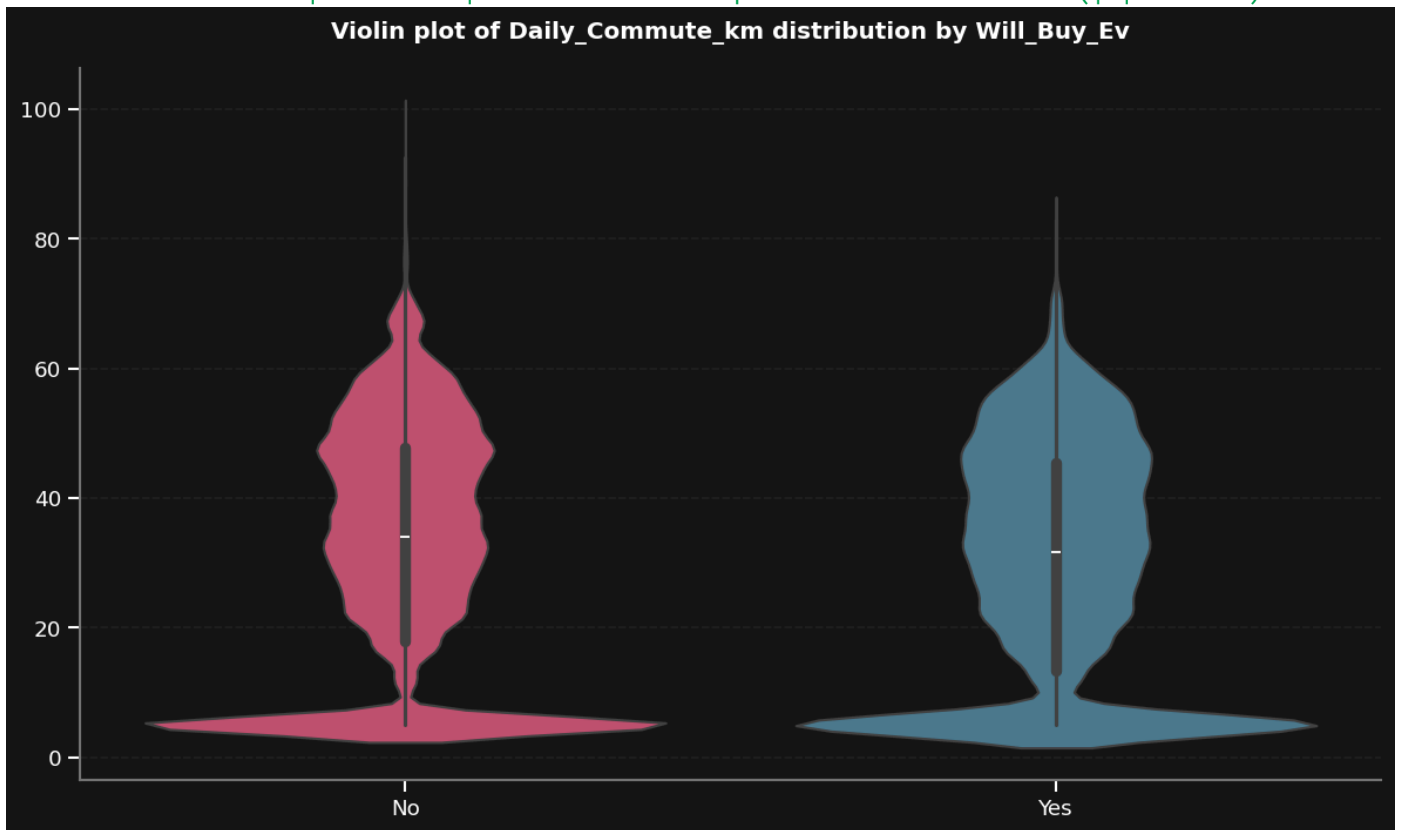
Homogeneity of Variance — Daily_Commute_km by Will_Buy_EV

	Metric	Value
0	Levene's Statistic	82.6108
1	p-value	0.000000
2	Max/Min Variance Ratio	1.07

```

2026-09-19 04:37:35 | INFO      | Interpretation:
2026-09-19 04:37:35 | INFO      | - Status: Statistically different variances, but practical
difference is small.
2026-09-19 04:37:35 | INFO      | - Recommendation: ANOVA or T-Test are acceptable, but Welch's
ANOVA or Welch's T-Test is recommended.
2026-09-19 04:37:35 | INFO      | Running Independent Two-Sample T-Test: 'Daily_Commute_km' ~
'Will_Buy_EV'
2026-09-19 04:37:35 | DEBUG     | Number of groups detected: 2
2026-09-19 04:37:35 | DEBUG     | Test Type: Welch's T-Test (unequal variances)
2026-09-19 04:37:35 | INFO      | Comparing groups: 'No' vs 'Yes'
2026-09-19 04:37:35 | INFO      | p-value: 0.000000
2026-09-19 04:37:35 | INFO      | Cohen's d: 0.122
2026-09-19 04:37:35 | DEBUG     | t-statistic: 38.398
2026-09-19 04:37:35 | DEBUG     | Group means → No: 32.553, Yes: 30.291
2026-09-19 04:37:35 | INFO      | Result: p-value ≤ 0.05 → Reject H0
2026-09-19 04:37:35 | INFO      | Conclusion: Significant difference between the two groups.
2026-09-19 04:37:35 | INFO      | Effect size interpretation: small effect (|d| = 0.122)

```



Distribution of Charging_Stations_Near_Home by EV Purchase Decision

- **Overall Mean:** 4.96
- **Overall Median:** 4.0
- **Overall Std:** 3.93

	Will_Buy_EV	Count	Mean	Median	Std
0	No	551886	4.989947	4.000000	3.919259
1	Yes	116779	4.820807	4.000000	3.959525

2026-09-19 04:37:38 | INFO | Checking normality of feature 'Charging_Stations_Near_Home' by groups of 'Will_Buy_EV'...

2026-09-19 04:37:38 | INFO | Evaluating distribution characteristics (skewness & kurtosis)...

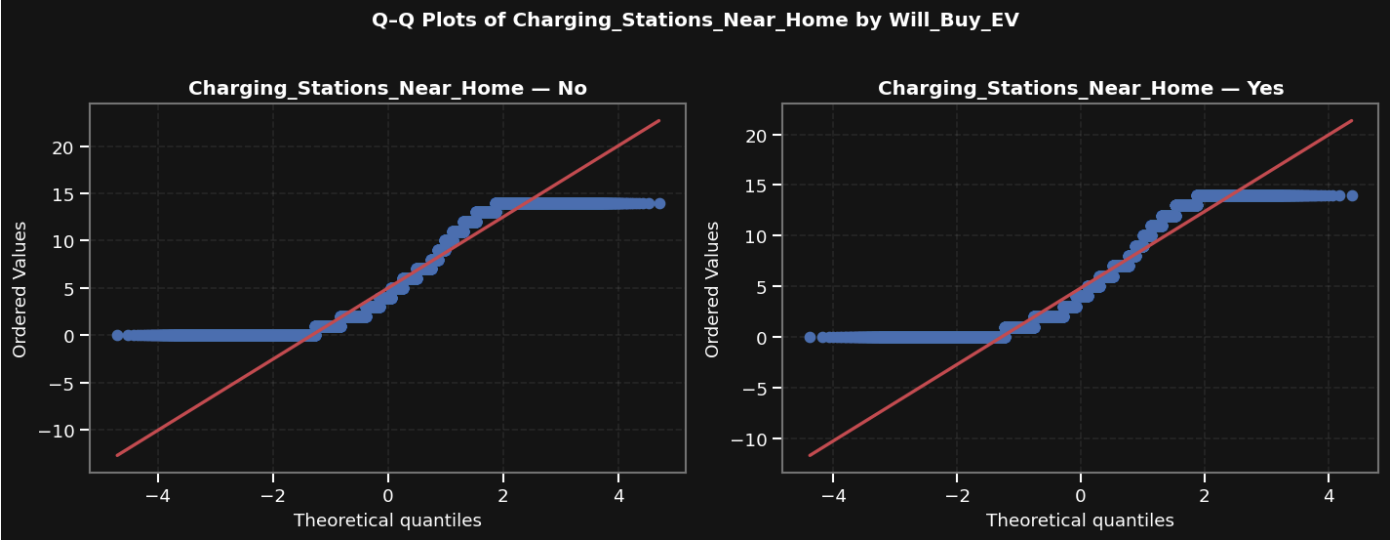
2026-09-19 04:37:38 | DEBUG | Number of groups detected: 2

2026-09-19 04:37:38 | DEBUG | Thresholds → |skew|≤0.5, |kurtosis|≤1.5

2026-09-19 04:37:38 | DEBUG | [Will_Buy_EV=No] skew=0.6897, kurtosis=-0.4864 → Moderately skewed, Normal tails

2026-09-19 04:37:38 | DEBUG | [Will_Buy_EV=Yes] skew=0.7438, kurtosis=-0.4564 → Moderately skewed, Normal tails

2026-09-19 04:37:38 | INFO | Generating Q-Q plots for visual assessment...



Skewness & Kurtosis of Charging_Stations_Near_Home by Will_Buy_EV

	Feature	Group	Skewness	Kurtosis	Remark
0	Charging_Stations_Near_Home	No	0.6897	-0.4864	Moderately skewed, Normal tails
1	Charging_Stations_Near_Home	Yes	0.7438	-0.4564	Moderately skewed, Normal tails

2026-09-19 04:37:40 | WARNING | Normality assumption violated in at least one group.

2026-09-19 04:37:40 | INFO | Recommendation: Use non-parametric tests (Kruskal-Wallis / Mann-Whitney U).

2026-09-19 04:37:40 | INFO | Running Mann-Whitney U Test: 'Charging_Stations_Near_Home' by 'Will_Buy_EV'...

2026-09-19 04:37:40 | DEBUG | Number of groups detected: 2

2026-09-19 04:37:40 | INFO | U statistic: 33248197432.0

2026-09-19 04:37:40 | INFO | p-value : 0.000000

2026-09-19 04:37:42 | INFO | Result: p-value ≤ 0.05 → Reject H0

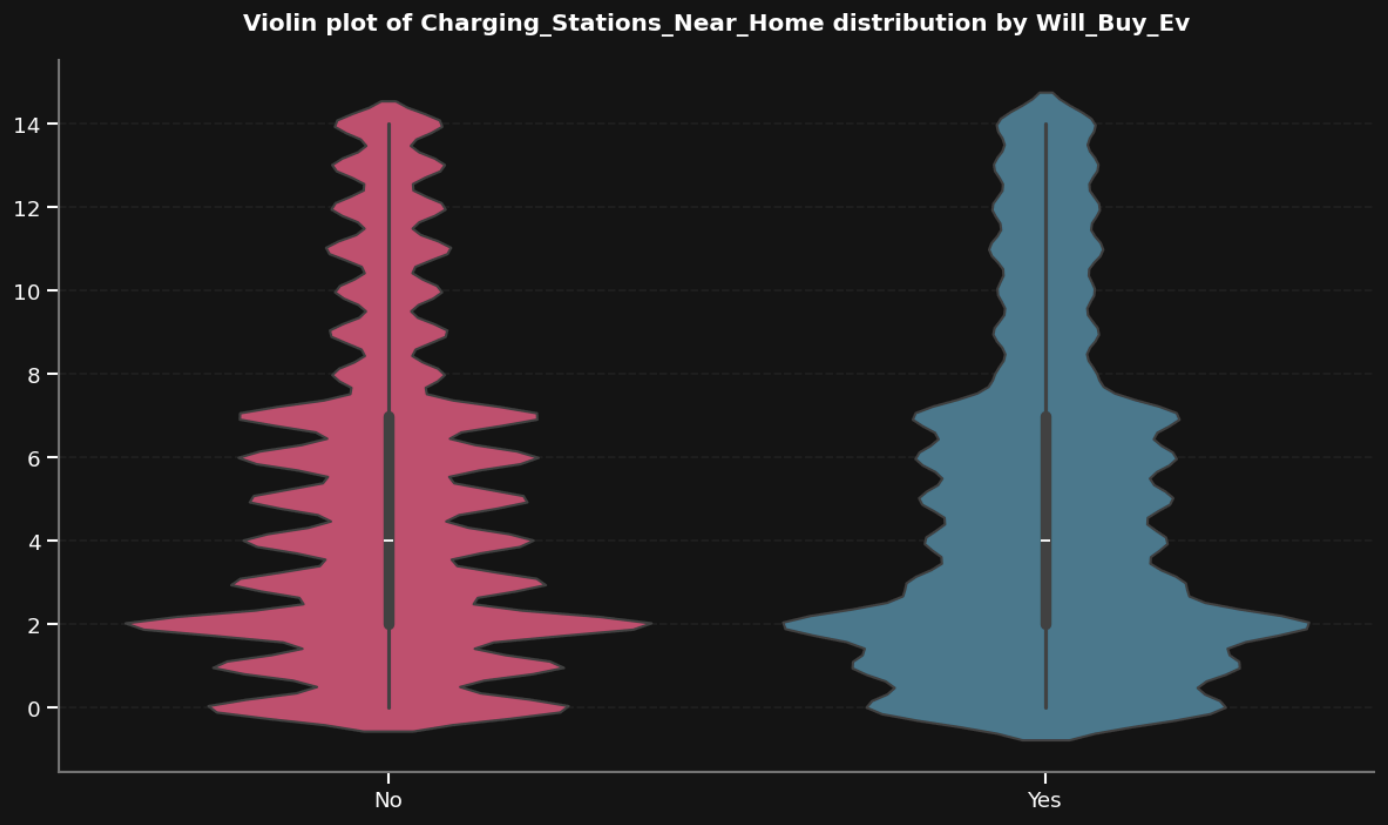
2026-09-19 04:37:42 | INFO | Conclusion: Significant difference between the two groups.

2026-09-19 04:37:42 | INFO | Effect size (r): 0.0209

2026-09-19 04:37:42 | INFO | Effect size interpretation: Negligible

2026-09-19 04:37:42 | INFO | Cliff's Delta: -0.0318

2026-09-19 04:37:42 | INFO | Effect size (Cliff's Delta) interpretation: Negligible

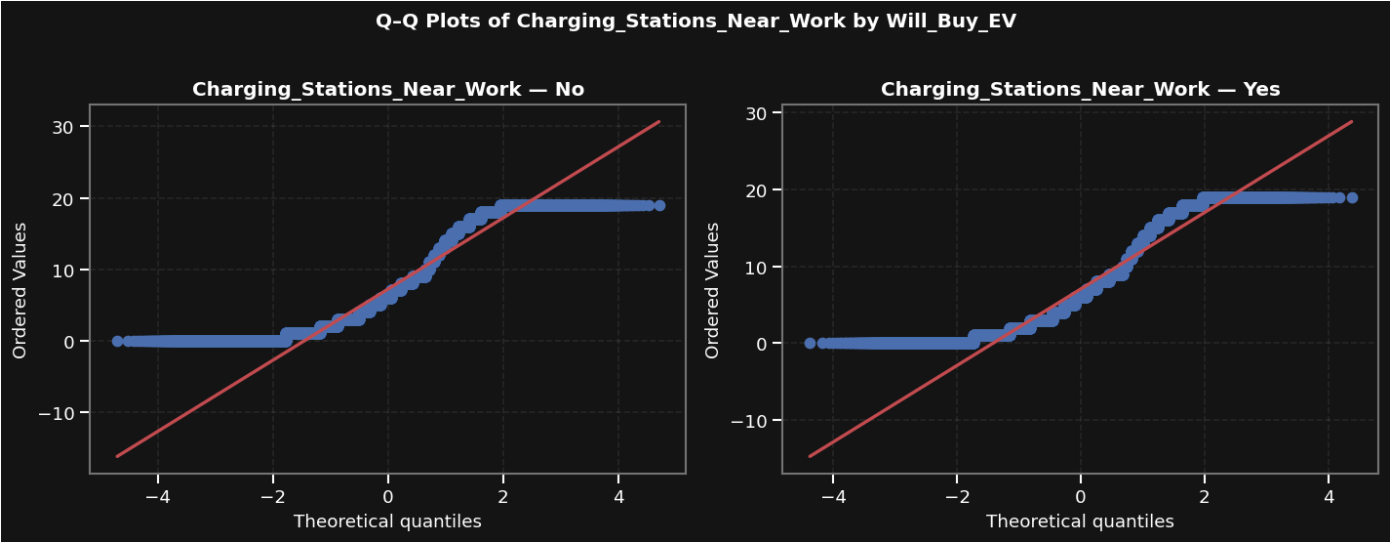


Distribution of Charging_Stations_Near_Work by EV Purchase Decision

- Overall Mean: 7.18
- Overall Median: 6.0
- Overall Std: 5.19

Will_Buy_EV		Count	Mean	Median	Std
0	No	551886	7.205490	6.000000	5.185977
1	Yes	116779	7.038432	6.000000	5.187498

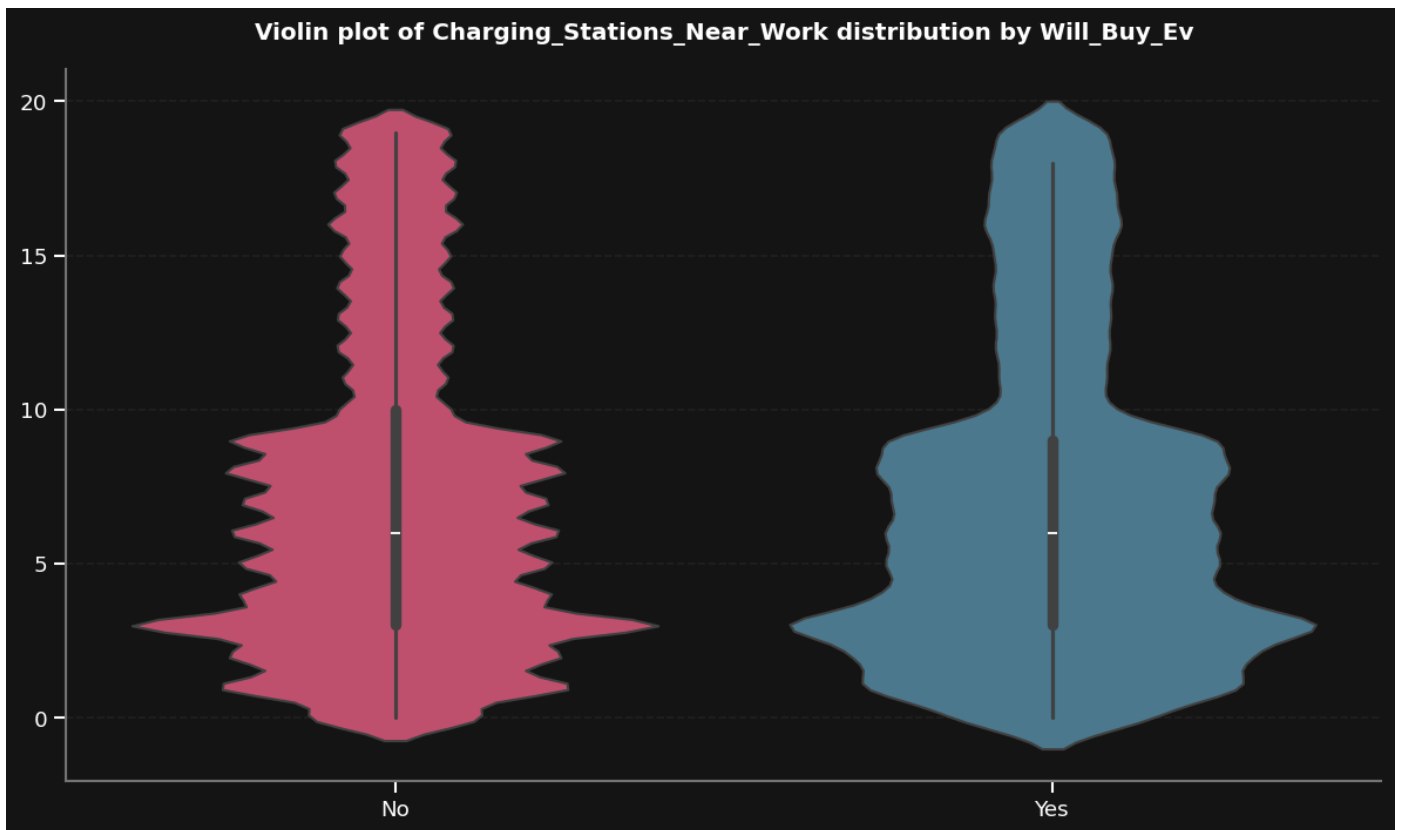
```
2026-09-19 04:37:45 | INFO | Checking normality of feature 'Charging_Stations_Near_Work' by
groups of 'Will_Buy_EV'...
2026-09-19 04:37:45 | INFO | Evaluating distribution characteristics (skewness &
kurtosis)...
2026-09-19 04:37:45 | DEBUG | Number of groups detected: 2
2026-09-19 04:37:45 | DEBUG | Thresholds → |skew|≤0.5, |kurtosis|≤1.5
2026-09-19 04:37:45 | DEBUG | [Will_Buy_EV=No] skew=0.6944, kurtosis=-0.4975 → Moderately
skewed, Normal tails
2026-09-19 04:37:45 | DEBUG | [Will_Buy_EV=Yes] skew=0.7283, kurtosis=-0.4476 → Moderately
skewed, Normal tails
2026-09-19 04:37:45 | INFO | Generating Q-Q plots for visual assessment...
```



Skewness & Kurtosis of Charging_Stations_Near_Work by Will_Buy_EV

	Feature	Group	Skewness	Kurtosis	Remark
0	Charging_Stations_Near_Work	No	0.6944	-0.4975	Moderately skewed, Normal tails
1	Charging_Stations_Near_Work	Yes	0.7283	-0.4476	Moderately skewed, Normal tails

```
2026-09-19 04:37:47 | WARNING | Normality assumption violated in at least one group.
2026-09-19 04:37:47 | INFO | Recommendation: Use non-parametric tests (Kruskal-Wallis /
Mann-Whitney U).
2026-09-19 04:37:47 | INFO | Running Mann-Whitney U Test: 'Charging_Stations_Near_Work' by
'Will_Buy_EV'...
2026-09-19 04:37:47 | DEBUG | Number of groups detected: 2
2026-09-19 04:37:47 | INFO | U statistic: 32908779219.5
2026-09-19 04:37:47 | INFO | p-value : 0.000000
2026-09-19 04:37:49 | INFO | Result: p-value ≤ 0.05 → Reject H0
2026-09-19 04:37:49 | INFO | Conclusion: Significant difference between the two groups.
2026-09-19 04:37:49 | INFO | Effect size (r): 0.0140
2026-09-19 04:37:49 | INFO | Effect size interpretation: Negligible
2026-09-19 04:37:49 | INFO | Cliff's Delta: -0.0212
2026-09-19 04:37:49 | INFO | Effect size (Cliff's Delta) interpretation: Negligible
```



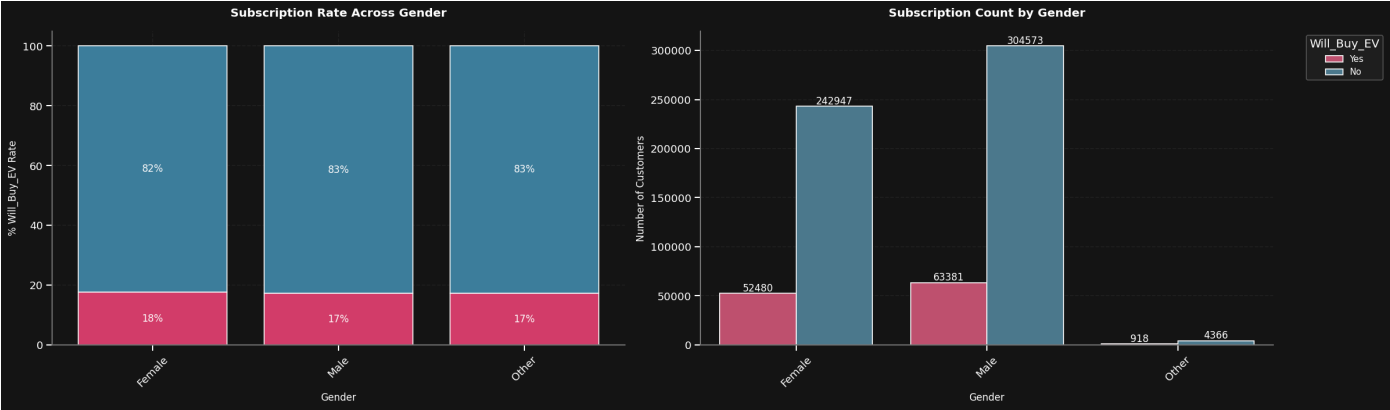
Numerical Features by EV Purchase Decision

- **Age:** Age differs statistically between EV buyers and non-buyers ($p < 0.001$), but the mean difference is only **~0.25 years** with a negligible effect size (Cohen's $d = 0.020$), suggesting that **age has limited practical impact on EV purchase decisions**.
- **Annual_Income_USD:** Income differs significantly between the two groups ($p < 0.001$), with EV buyers having substantially higher average income (**~98,827 vs. 81,795**) and a large effect size (Cohen's $d = 0.624$), suggesting that **income is an important factor in EV purchase decisions**.
- **Daily_Commute_km:** Daily commuting distance differs significantly ($p < 0.001$), but the effect size is small (Cohen's $d = 0.122$); non-EV buyers have a slightly longer average commute (**~32.55 km vs. ~30.29 km**), suggesting that **Daily_Commute_km has limited practical impact on EV purchase decisions**.
- **Charging_Stations_Near_Home:** The number of charging stations near home differs significantly between the two groups ($p < 0.001$), but the effect size is negligible ($r = 0.021$; Cliff's $\delta = -0.032$), suggesting **very limited practical impact on EV purchase decisions**.
- **Charging_Stations_Near_Work:** The number of charging stations near work differs significantly between the two groups ($p < 0.001$), but the effect size is negligible ($r = 0.014$; Cliff's $\delta = -0.021$), suggesting **very limited practical impact on EV purchase decisions**.

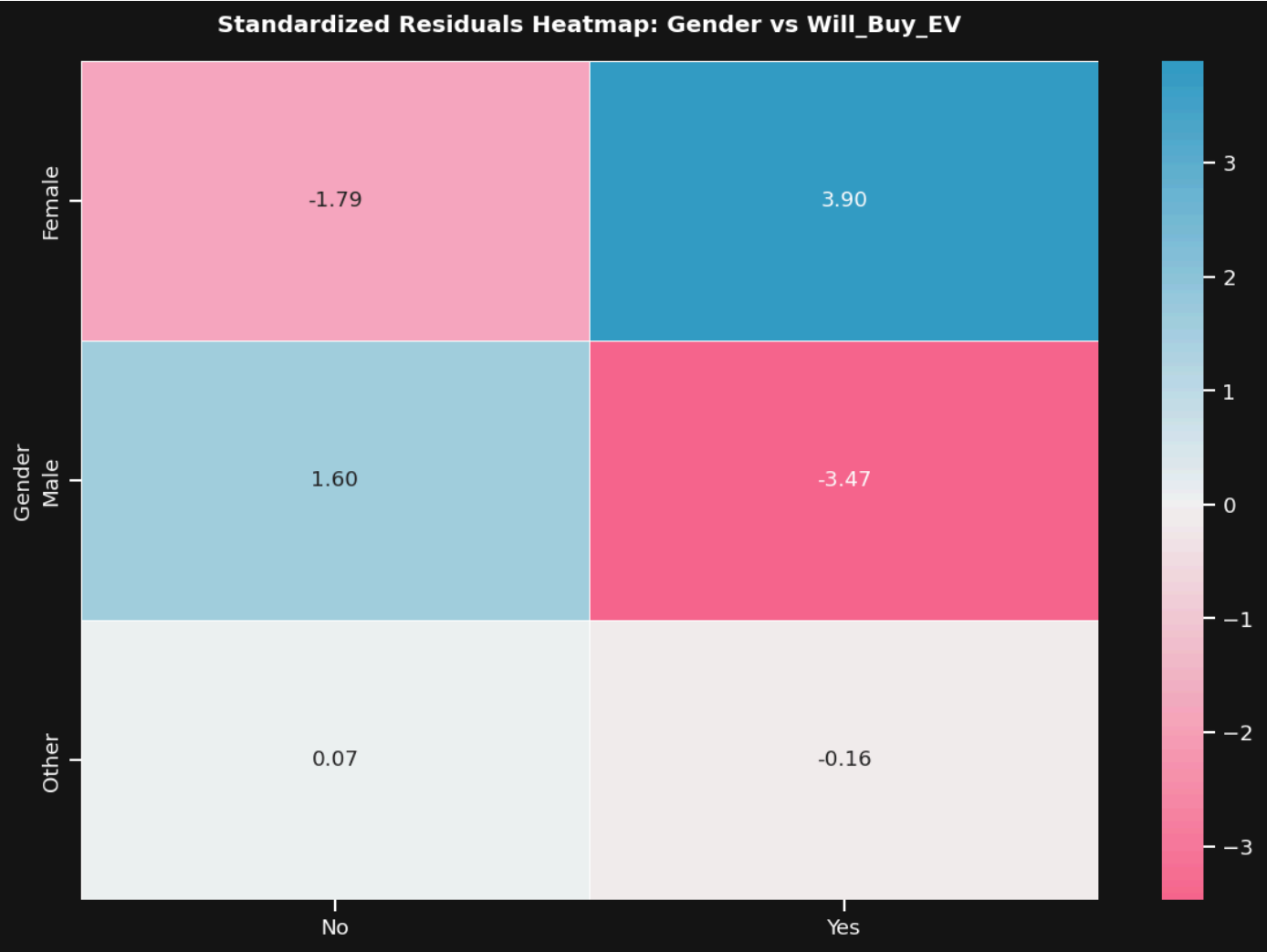
```
# Categorical Feature Analysis by Target
for feature in cat_features:
    # Display a title for the current feature.
    display(HTML(
        f"<h2 style='text-align:center; font-size:22px; color:#fde68a;'>"
        f"<b>Distribution of {feature.title()} by EV Purchase Decision</b>"
        f"</h2>"
    ))

    # Analyze the categorical feature across target classes.
    StatUtils.analyze_categorical_feature_vs_target(
        cat=feature,
        target_feature=config["target"]["feature"],
        logger=logger,
        df=df_plot
    )
```

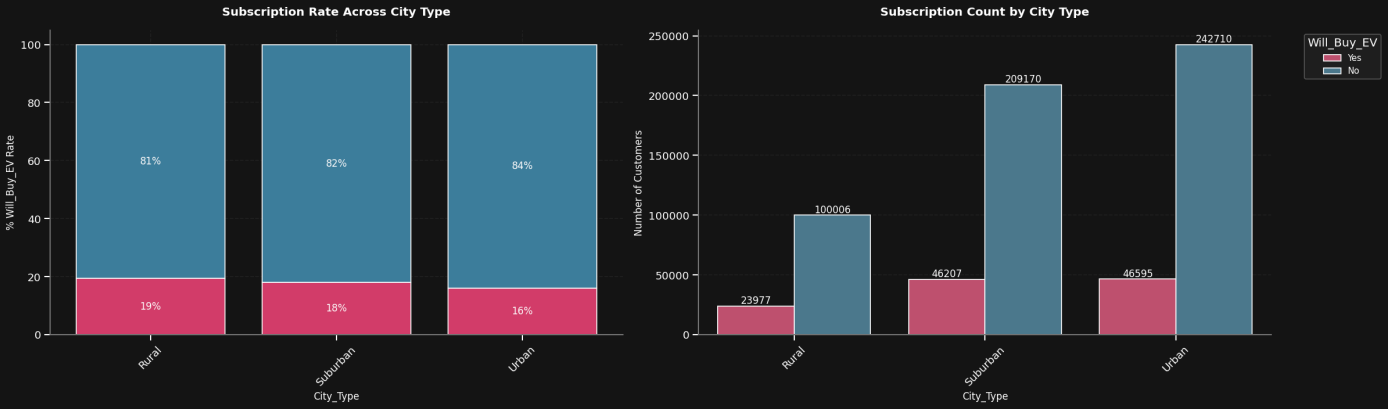
Distribution of Gender by EV Purchase Decision



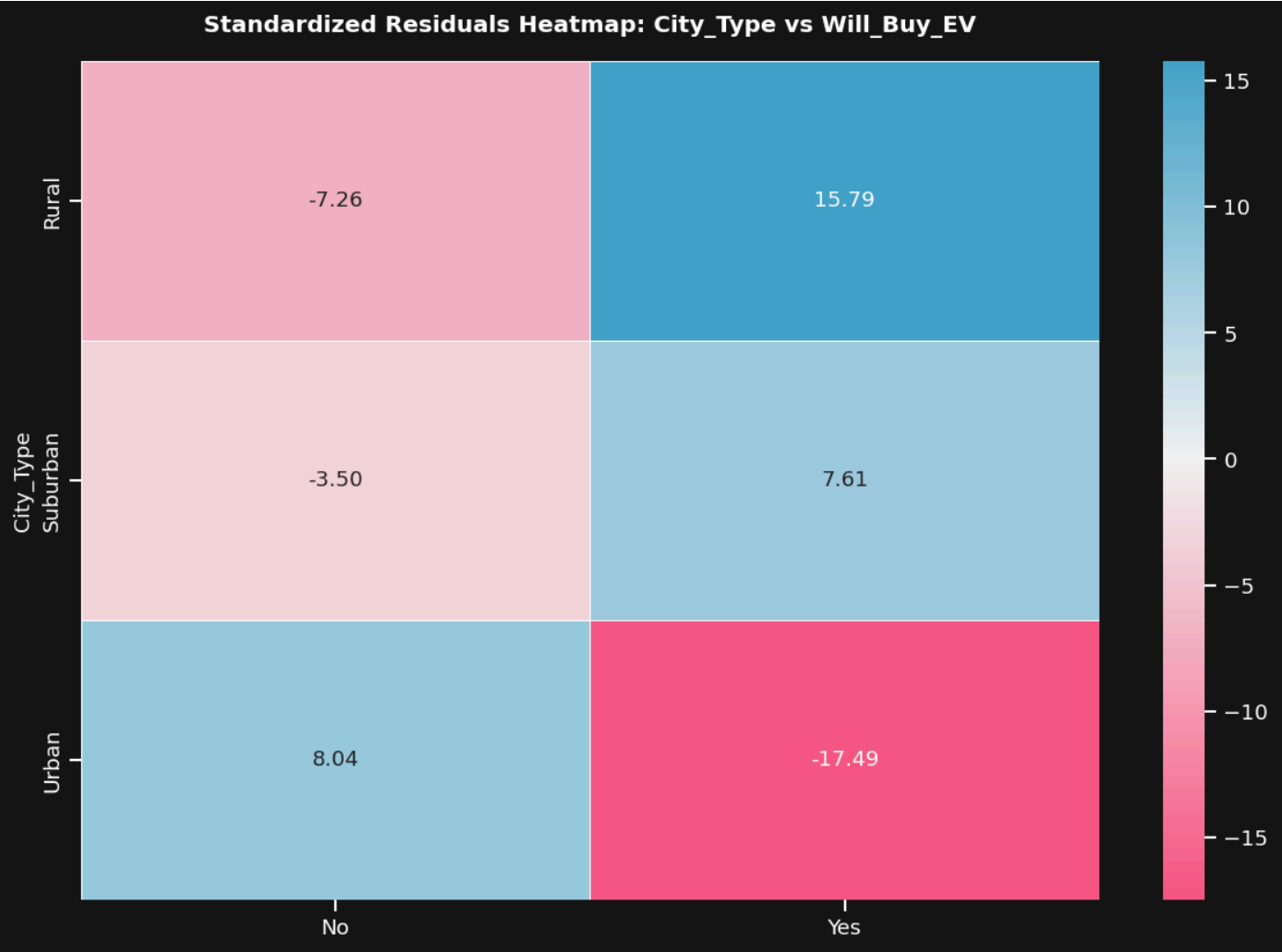
```
2026-09-19 04:37:53 | INFO | Chi-Square Test of Independence: 'Gender' vs. 'Will_Buy_EV'...
2026-09-19 04:37:53 | INFO | Chi-squared statistic: 33.041
2026-09-19 04:37:53 | INFO | Degrees of freedom: 2
2026-09-19 04:37:53 | INFO | p-value: 0.000000
2026-09-19 04:37:53 | INFO | Result: p-value < 0.05 → Reject H0
2026-09-19 04:37:53 | INFO | Variables 'Gender' and 'Will_Buy_EV' are statistically
associated.
2026-09-19 04:37:53 | DEBUG | Displaying standardized residuals heatmap.
```



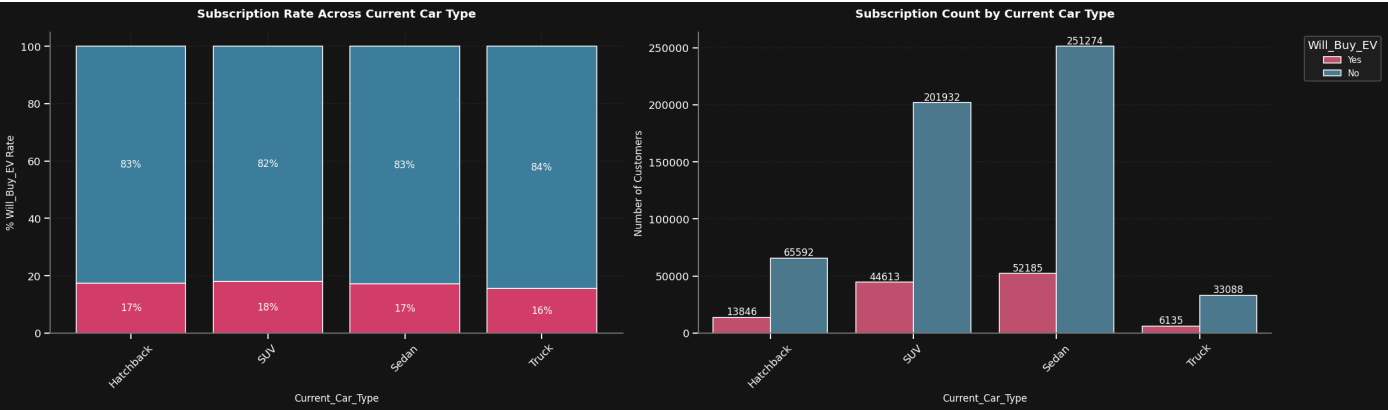
Distribution of City_Type by EV Purchase Decision



```
2026-09-19 04:37:55 | INFO | Chi-Square Test of Independence: 'City_Type' vs. 'Will_Buy_EV'...
2026-09-19 04:37:55 | INFO | Chi-squared statistic: 742.830
2026-09-19 04:37:55 | INFO | Degrees of freedom: 2
2026-09-19 04:37:55 | INFO | p-value: 0.000000
2026-09-19 04:37:55 | INFO | Result: p-value < 0.05 → Reject H0
2026-09-19 04:37:55 | INFO | Variables 'City_Type' and 'Will_Buy_EV' are statistically associated.
2026-09-19 04:37:55 | DEBUG | Displaying standardized residuals heatmap.
```



Distribution of Current_Car_Type by EV Purchase Decision



2026-09-19 04:37:56 | INFO | Chi-Square Test of Independence: 'Current_Car_Type' vs. 'Will_Buy_EV'...

2026-09-19 04:37:56 | INFO | Chi-squared statistic: 173.663

2026-09-19 04:37:56 | INFO | Degrees of freedom: 3

2026-09-19 04:37:56 | INFO | p-value: 0.000000

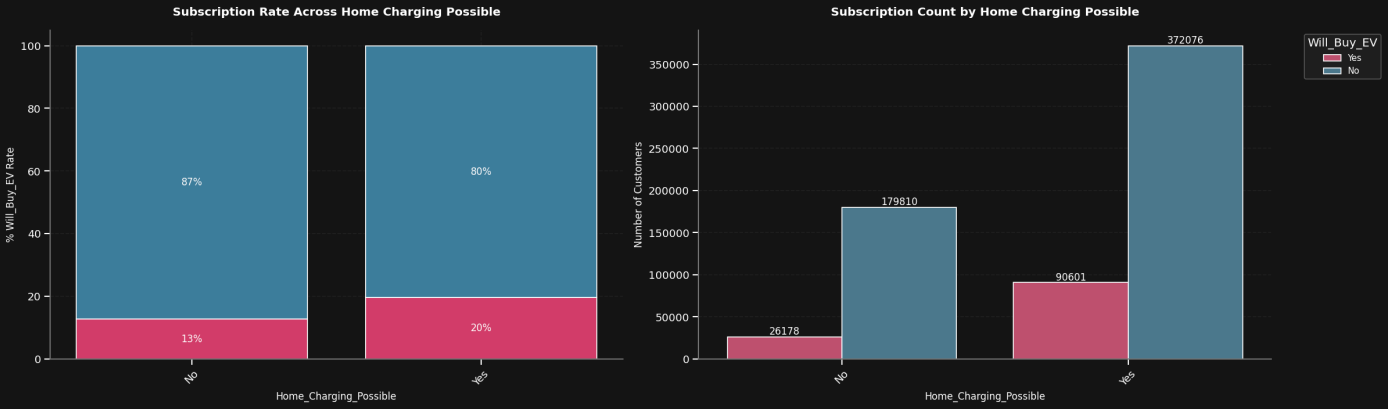
2026-09-19 04:37:56 | INFO | Result: p-value < 0.05 → Reject H₀

2026-09-19 04:37:56 | INFO | Variables 'Current_Car_Type' and 'Will_Buy_EV' are statistically associated.

2026-09-19 04:37:56 | DEBUG | Displaying standardized residuals heatmap.



Distribution of Home_Charging_Possible by EV Purchase Decision



2026-09-19 04:37:58 | INFO | Chi-Square Test of Independence: 'Home_Charging_Possible' vs. 'Will_Buy_EV'...

2026-09-19 04:37:58 | INFO | Chi-squared statistic: 4671.041

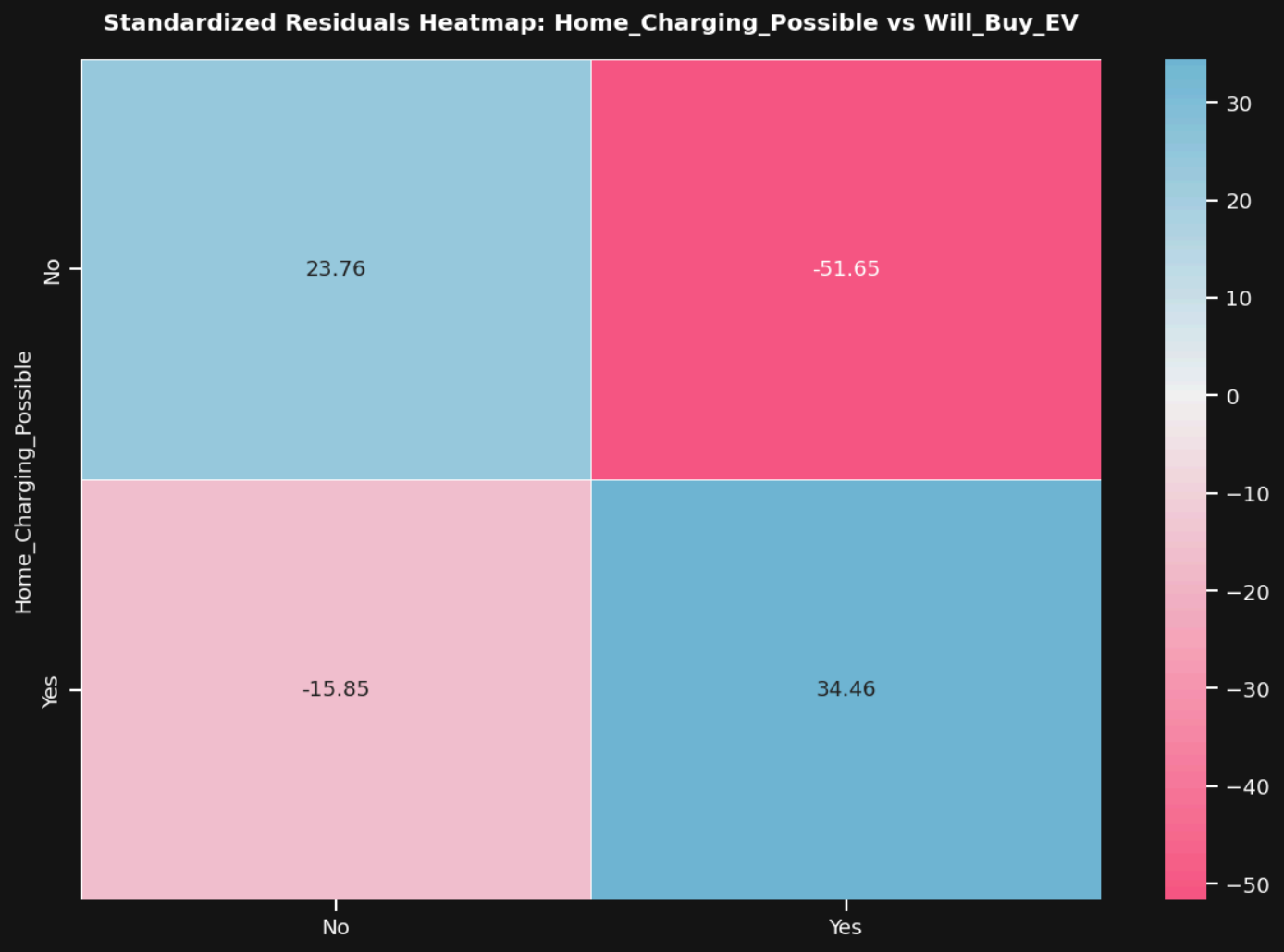
2026-09-19 04:37:58 | INFO | Degrees of freedom: 1

2026-09-19 04:37:58 | INFO | p-value: 0.000000

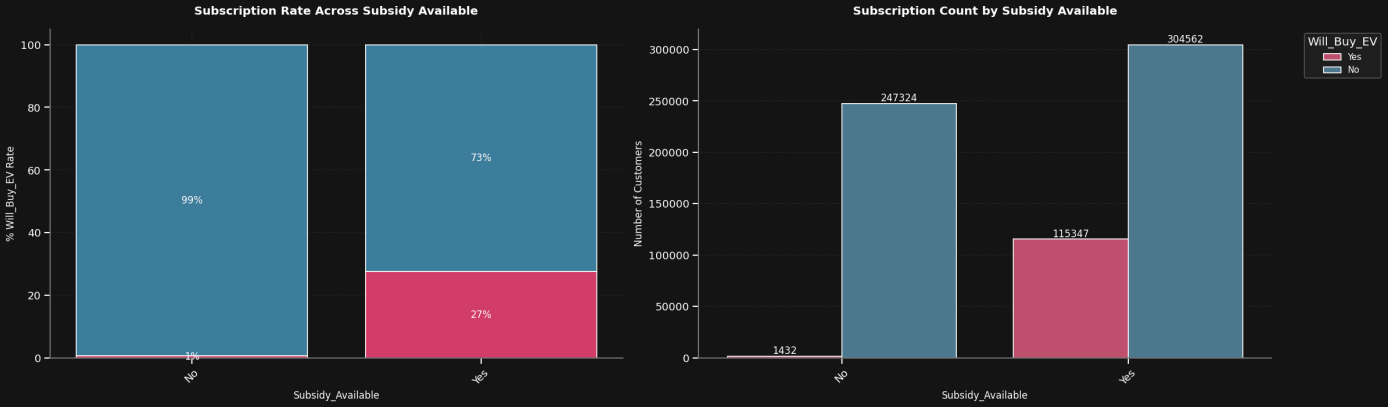
2026-09-19 04:37:58 | INFO | Result: p-value < 0.05 → Reject H₀

2026-09-19 04:37:58 | INFO | Variables 'Home_Charging_Possible' and 'Will_Buy_EV' are statistically associated.

2026-09-19 04:37:58 | DEBUG | Displaying standardized residuals heatmap.



Distribution of Subsidy_Available by EV Purchase Decision



2026-09-19 04:37:59 | INFO | Chi-Square Test of Independence: 'Subsidy_Available' vs. 'Will_Buy_EV'...

2026-09-19 04:37:59 | INFO | Chi-squared statistic: 78382.500

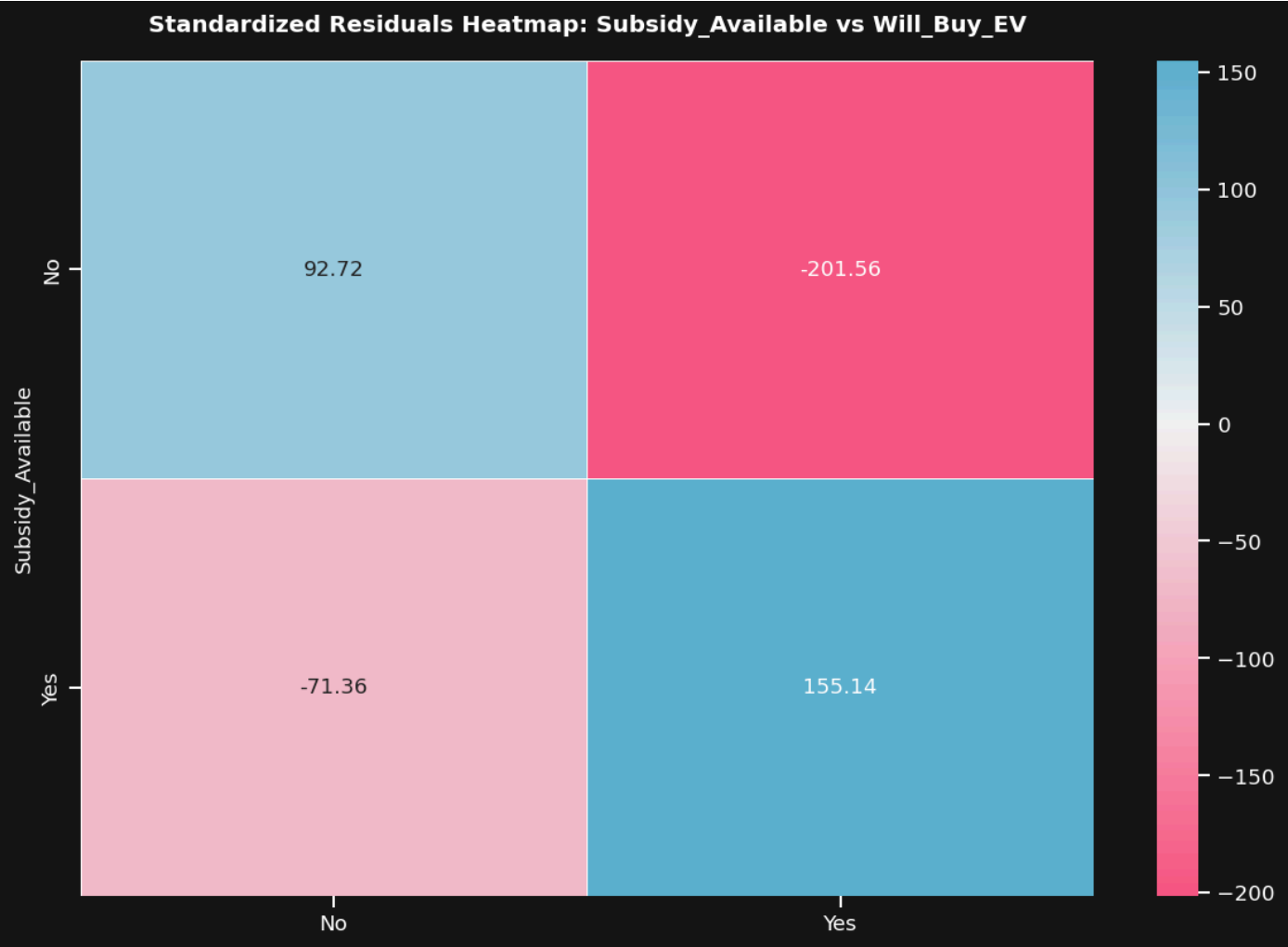
2026-09-19 04:37:59 | INFO | Degrees of freedom: 1

2026-09-19 04:37:59 | INFO | p-value: 0.000000

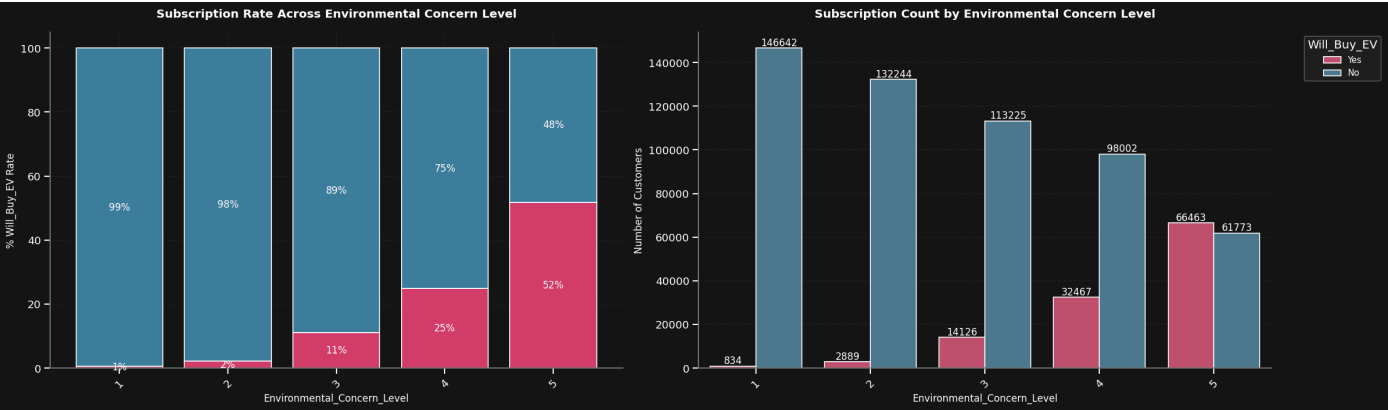
2026-09-19 04:37:59 | INFO | Result: p-value < 0.05 → Reject H₀

2026-09-19 04:37:59 | INFO | Variables 'Subsidy_Available' and 'Will_Buy_EV' are statistically associated.

2026-09-19 04:37:59 | DEBUG | Displaying standardized residuals heatmap.



Distribution of Environmental_Concern_Level by EV Purchase Decision



2026-09-19 04:38:01 | INFO | Chi-Square Test of Independence: 'Environmental_Concern_Level' vs. 'Will_Buy_EV'...

2026-09-19 04:38:01 | INFO | Chi-squared statistic: 164867.872

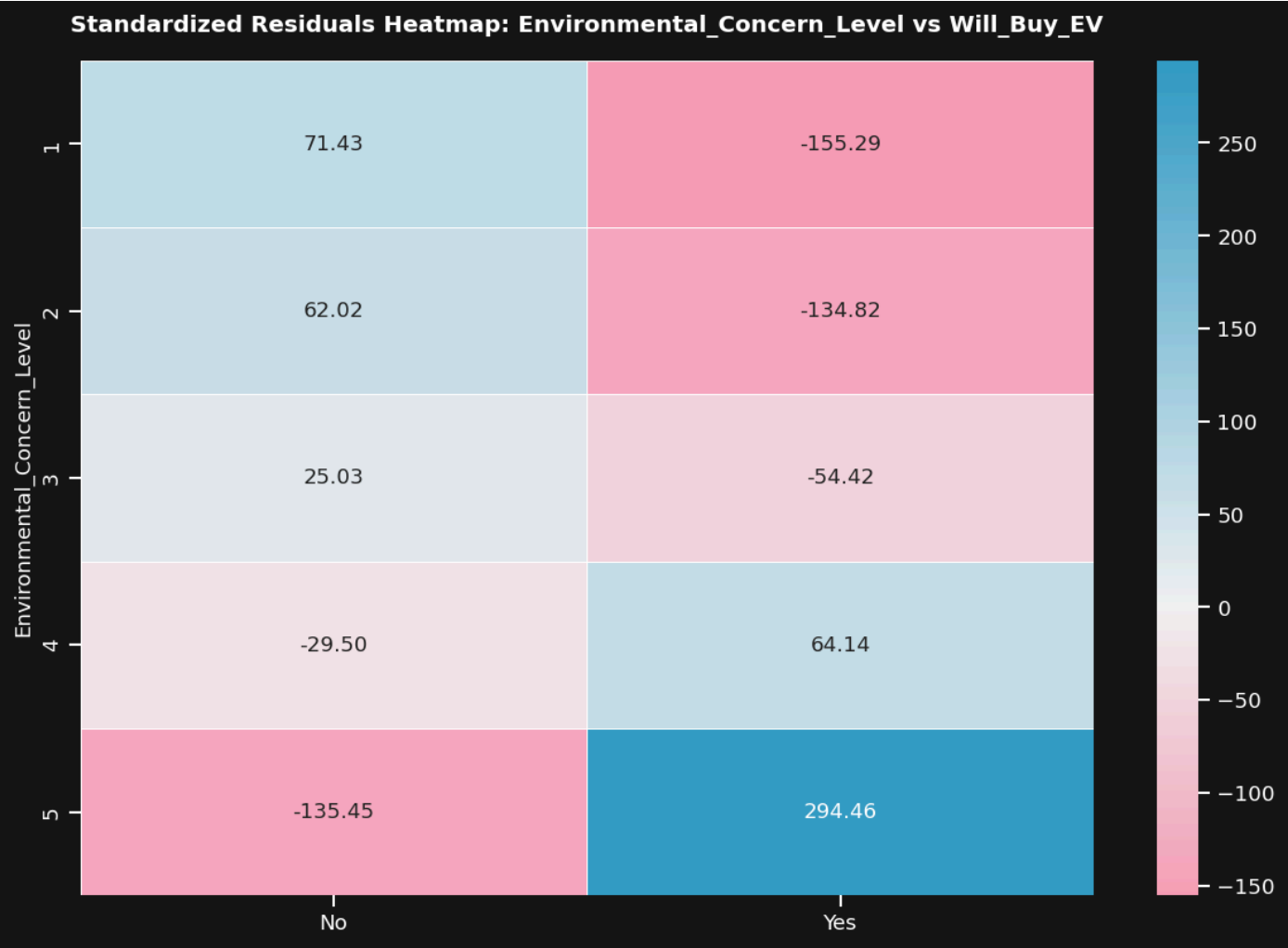
2026-09-19 04:38:01 | INFO | Degrees of freedom: 4

2026-09-19 04:38:01 | INFO | p-value: 0.000000

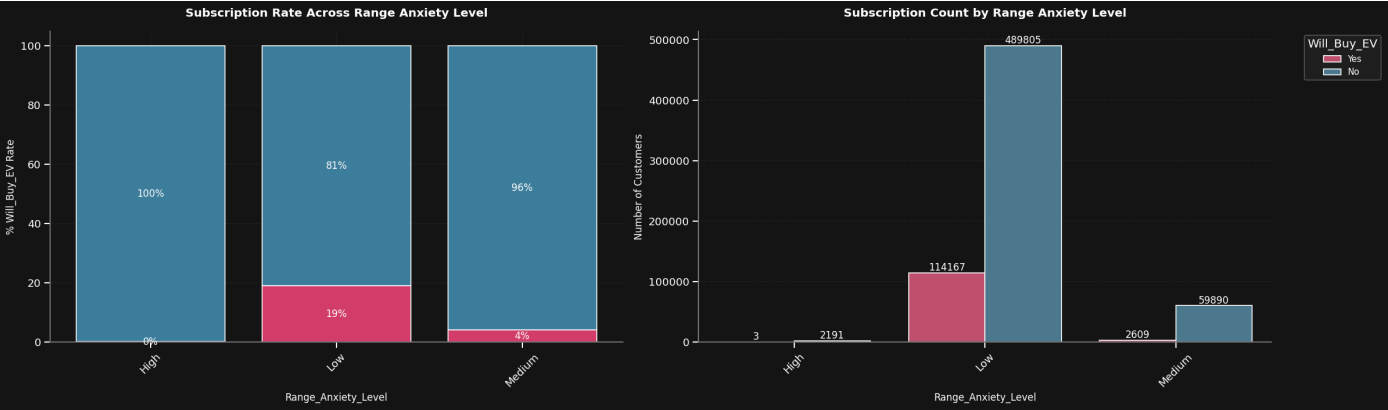
2026-09-19 04:38:01 | INFO | Result: p-value < 0.05 → Reject H₀

2026-09-19 04:38:01 | INFO | Variables 'Environmental_Concern_Level' and 'Will_Buy_EV' are statistically associated.

2026-09-19 04:38:01 | DEBUG | Displaying standardized residuals heatmap.



Distribution of Range_Anxiety_Level by EV Purchase Decision



2026-09-19 04:38:02 | INFO | Chi-Square Test of Independence: 'Range_Anxiety_Level' vs. 'Will_Buy_EV'...

2026-09-19 04:38:02 | INFO | Chi-squared statistic: 8981.913

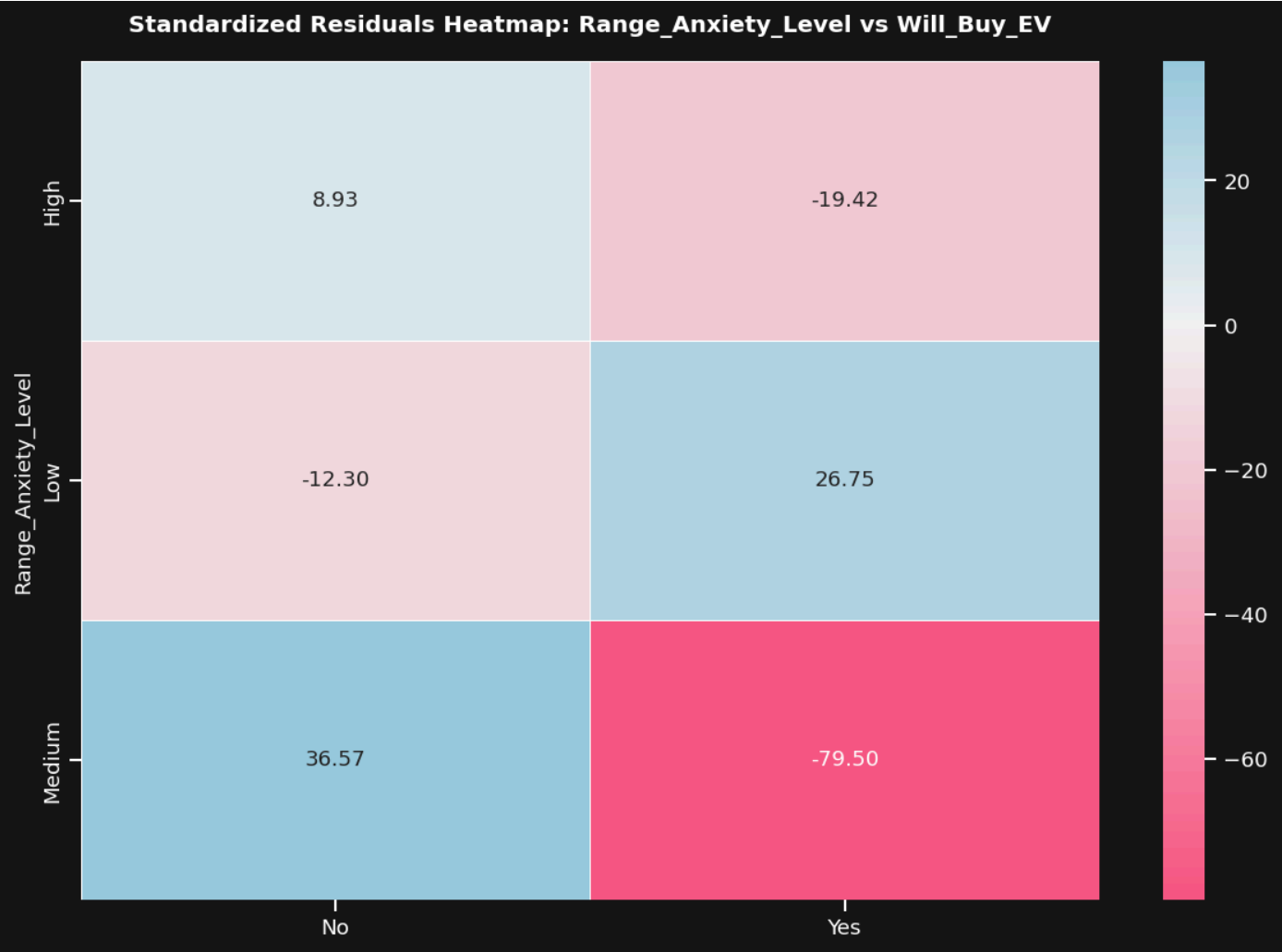
2026-09-19 04:38:02 | INFO | Degrees of freedom: 2

2026-09-19 04:38:02 | INFO | p-value: 0.000000

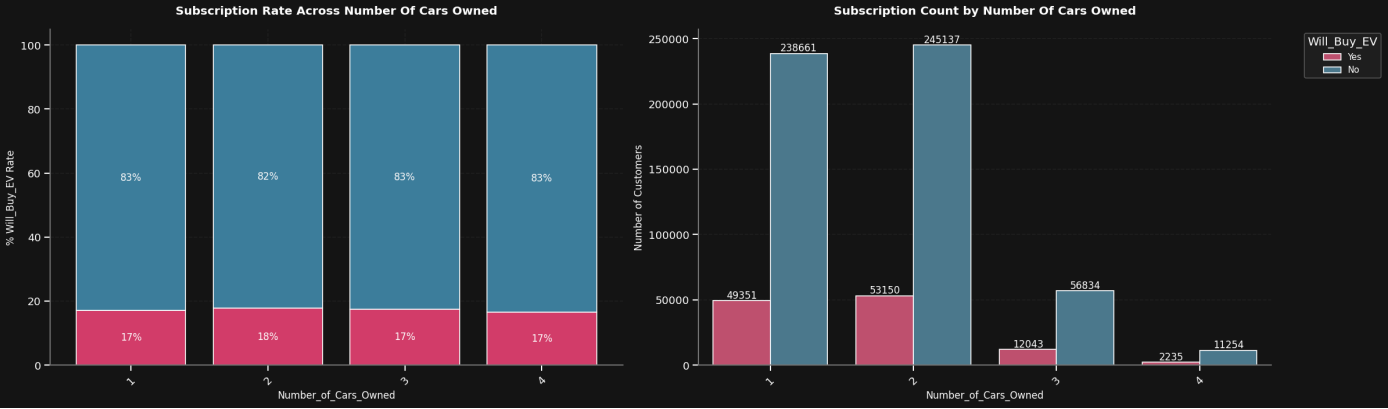
2026-09-19 04:38:02 | INFO | Result: p-value < 0.05 → Reject H₀

2026-09-19 04:38:02 | INFO | Variables 'Range_Anxiety_Level' and 'Will_Buy_EV' are statistically associated.

2026-09-19 04:38:02 | DEBUG | Displaying standardized residuals heatmap.



Distribution of Number_Of_Cars_Owned by EV Purchase Decision



2026-09-19 04:38:04 | INFO | Chi-Square Test of Independence: 'Number_of_Cars_Owned' vs. 'Will_Buy_EV'...

2026-09-19 04:38:04 | INFO | Chi-squared statistic: 55.129

2026-09-19 04:38:04 | INFO | Degrees of freedom: 3

2026-09-19 04:38:04 | INFO | p-value: 0.000000

2026-09-19 04:38:04 | INFO | Result: p-value < 0.05 → Reject H₀

2026-09-19 04:38:04 | INFO | Variables 'Number_of_Cars_Owned' and 'Will_Buy_EV' are statistically associated.

2026-09-19 04:38:04 | DEBUG | Displaying standardized residuals heatmap.

Standardized Residuals Heatmap: Number_of_Cars_Owned vs Will_Buy_EV





Categorical Features by EV Purchase Decision

- **Gender:** Significantly associated with EV purchase decisions ($\chi^2 = 33.04$, $p < 0.001$); **Female** shows higher-than-expected purchases, while **Male** is lower than expected.
- **City_Type:** Significantly associated with EV purchase decisions ($\chi^2 = 742.83$, $p < 0.001$); **Rural and Suburban** show higher-than-expected purchases, while **Urban** is substantially lower than expected.
- **Current_Car_Type:** Significantly associated with EV purchase decisions ($\chi^2 = 173.66$, $p < 0.001$); **SUV** shows higher-than-expected purchases, while **Truck** is substantially lower than expected.
- **Home_Charging_Possible:** Strongly differentiated across purchase decisions ($\chi^2 = 4671.04$, $p < 0.001$); EV purchase rate increases from approximately **13% to 20%** when home charging is available.
- **Subsidy_Available:** Shows a pronounced relationship with EV purchase decisions ($\chi^2 = 78,382.5$, $p < 0.001$); purchase rate rises sharply from approximately **1% without a subsidy to 27% with a subsidy**.
- **Environmental_Concern_Level:** Shows a pronounced relationship with EV purchase decisions ($\chi^2 = 164,867.87$, $p < 0.001$); purchase rate increases sharply from approximately **1% at level 1 to 52% at level 5**, indicating a strong positive adoption pattern.
- **Range_Anxiety_Level:** Significantly associated with EV purchase decisions ($\chi^2 = 8,981.91$, $p < 0.001$); purchase rate is highest for **Low** anxiety (~19%) and falls sharply for **Medium (~4%)** and **High (~0%)**.
- **Number_of_Cars_Owned:** Significantly associated with EV purchase decisions ($\chi^2 = 55.13$, $p < 0.001$); consumers owning **2 cars** show higher-than-expected purchases, while **1 and 4 cars** are lower than expected.
- **Overall:** Among the categorical variables, **Subsidy_Available**, **Environmental_Concern_Level**, **Home_Charging_Possible**, and **Range_Anxiety_Level** show the most pronounced differences in EV purchase rates across groups.

Overall Picture



Overall Picture

Feature	Key Insight
Age	Statistically different between buyers and non-buyers, but the effect is negligible; age has limited practical impact on EV purchase decisions.
Annual Income	EV buyers have substantially higher income, with a large effect size ; income is an important factor in EV purchase decisions.
Daily Commute	Non-buyers commute slightly farther on average, but the effect is small, indicating limited practical impact .
Charging Stations Near Home	Statistically different across groups, but the effect is negligible, suggesting limited practical impact on EV adoption.
Charging Stations Near Work	Also statistically significant but with a negligible effect, indicating limited practical impact .
Gender	Gender is significantly associated with EV purchase decisions, with Female purchases higher than expected and Male purchases lower than expected.
City Type	EV purchase patterns differ by residential area, with Rural and Suburban above expected levels and Urban below expected levels.
Current Car Type	SUV owners show higher-than-expected EV purchases, while Truck owners are substantially lower than expected.

Home Charging Possible	EV purchase rates increase from approximately 13% to 20% when home charging is possible, highlighting an important adoption factor .
Subsidy Available	Purchase rates rise sharply from approximately 1% to 27% when subsidies are available, indicating a major adoption driver .
Environmental Concern	EV purchase rates increase from approximately 1% at level 1 to 52% at level 5 , showing a strong positive adoption pattern .
Range Anxiety	Purchase rates are highest for Low anxiety (~19%) and fall sharply for Medium (~4%) and High (~0%) , highlighting range anxiety as a major adoption barrier .
Number of Cars Owned	Ownership is significantly associated with EV adoption, with the 2-car group above expected levels, while the overall differences are relatively modest.

- Strong practical relationship
- Moderate / noticeable relationship
- Limited practical impact

```

logger.info("=" * 80)
logger.info("DATA PREPROCESSING")
logger.info("=" * 80)

# Prepare Base Features
logger.info("Preparing base features...")
BASE = [
    c for c in df_train.columns
    if c not in [config["columns"]["train_test"], config["target"]["feature"],
config["columns"]["original"]]
]

logger.info(
    f"Data loaded | "
    f"Train: {df_train.shape} | "
    f"Test: {df_test.shape} | "
    f"Original: {df_original.shape}"
)

logger.info(f"Base features: {len(BASE)}")
logger.debug(f"Base feature list: {BASE}")

# Combine Train + Original Dataset
logger.info("Combining train and original datasets...")

```

```

train_before = len(df_train)

df_train = pd.concat([df_train, df_original], ignore_index=True)
logger.info(
    f"Train dataset combined | "
    f"Rows: {train_before:,} -> {len(df_train):,}"
)

# Feature Engineering
logger.info("Starting feature engineering...")

for df_name, df in [("Train", df_train), ("Test", df_test)]:
    logger.info(f"[{df_name}] Creating engineered features...")

    # 1. Total Charging Infrastructure
    # Combine the number of charging stations available
    # near home and near work.
    df["Total_Charging_Stations"] = (
        df["Charging_Stations_Near_Home"]
        + df["Charging_Stations_Near_Work"]
    )

    # 2. Charging Station Difference
    # Measure the difference between charging stations
    # near work and near home.
    #
    # Positive value -> more stations near work
    # Negative value -> more stations near home
    # Zero -> same number near home and work
    df["Charging_Station_Difference"] = (
        df["Charging_Stations_Near_Work"]
        - df["Charging_Stations_Near_Home"]
    )

    # 3. Commute per Charging Station
    # Relate daily commuting distance to the available
    # charging infrastructure.
    #
    # +1 prevents division by zero when there are
    # no charging stations.
    #
    # Higher value:
    # Longer commute with fewer charging stations.
    #
    # Lower value:
    # Shorter commute with more charging stations.
    df["Commute_per_Charging_Station"] = (
        df["Daily_Commute_km"]
        / (df["Total_Charging_Stations"] + 1)
    )

    # 4. Range Anxiety Score
    # Convert the ordinal categorical feature
    # Range_Anxiety_Level into an ordered numerical score.
    #
    # Low -> 1
    # Medium -> 2
    # High -> 3
    df["Range_Anxiety_Score"] = (

```

```

        df["Range_Anxiety_Level"]
        .map({"Low": 1, "Medium": 2, "High": 3}).astype("int8")
    )

    # 5. Range Anxiety × Daily Commute
    # Interaction feature between range anxiety and
    # daily commuting distance.
    df["Range_Anxiety_Commute"] = (
        df["Range_Anxiety_Score"].astype("float32")
        * df["Daily_Commute_km"]
    )

    # 6. Home Charging Score
    # Combine:
    # - Number of charging stations near home
    # - Whether home charging is possible
    # If home charging is not possible, the score becomes 0.
    df["Home_Charging_Score"] = (
        df["Charging_Stations_Near_Home"]
        * df["Home_Charging_Possible"].eq("Yes").astype("int8")
    )

# Final Preprocessing Summary
logger.info("=" * 80)
logger.info("DATA PREPROCESSING COMPLETED")
logger.info("=" * 80)

logger.info(
    f"Final dataset shapes | "
    f"Train: {df_train.shape} | "
    f"Test: {df_test.shape}"
)

logger.info(f"Categorical features processed: {len(cat_features)}")

```

```

2026-09-19 04:38:04 | INFO      |
=====
2026-09-19 04:38:04 | INFO      | DATA PREPROCESSING
2026-09-19 04:38:04 | INFO      |
=====
2026-09-19 04:38:04 | INFO      | Preparing base features...
2026-09-19 04:38:04 | INFO      | Data loaded | Train: (668665, 14) | Test: (286571, 13) |
Original: (10000, 14)
2026-09-19 04:38:04 | INFO      | Base features: 13
2026-09-19 04:38:04 | DEBUG     | Base feature list: ['Age', 'Annual_Income_USD',
'Daily_Commute_km', 'Number_of_Cars_Owned', 'Charging_Stations_Near_Home',
'Charging_Stations_Near_Work', 'Environmental_Concern_Level', 'Gender', 'City_Type',
'Current_Car_Type', 'Home_Charging_Possible', 'Subsidy_Available', 'Range_Anxiety_Level']
2026-09-19 04:38:04 | INFO      | Combining train and original datasets...
2026-09-19 04:38:04 | INFO      | Train dataset combined | Rows: 668,665 -> 678,665
2026-09-19 04:38:04 | INFO      | Starting feature engineering...
2026-09-19 04:38:04 | INFO      | [Train] Creating engineered features...
2026-09-19 04:38:04 | INFO      | [Test] Creating engineered features...
2026-09-19 04:38:04 | INFO      |
=====
2026-09-19 04:38:04 | INFO      | DATA PREPROCESSING COMPLETED
2026-09-19 04:38:04 | INFO      |
=====
2026-09-19 04:38:04 | INFO      | Final dataset shapes | Train: (678665, 20) | Test: (286571, 19)
2026-09-19 04:38:04 | INFO      | Categorical features processed: 8

```

```

BASE = num_features + cat_features

new_features = ["Total_Charging_Stations", "Charging_Station_Difference",
"Commute_per_Charging_Station",
                "Range_Anxiety_Commute", "Home_Charging_Score"]

FEATURES = BASE + new_features

logger.info(
    f"Feature composition | "
    f"Base: {len(BASE)} | "
    f"Engineered: {len(new_features)} | "
    f"Total: {len(FEATURES)}"
)

logger.debug(f"Final feature list: {FEATURES}")

# Reduce Memory Usage
logger.info("Reducing memory usage...")

df_train = Utils.reduce_mem_usage(cols=FEATURES, df=df_train)
df_test = Utils.reduce_mem_usage(cols=FEATURES, df=df_test)

logger.info(
    f"Memory optimization completed | "
    f"Train: {df_train.shape} | "
    f"Test: {df_test.shape}"
)

# Final Categorical Conversion
# Only original categorical features are converted.
# Missing indicators remain numerical binary features (0/1).
logger.info("Applying categorical conversion to original categorical features...")

# Numerical features
new_num_features = [
    "Age",
    "Annual_Income_USD",
    "Daily_Commute_km",
    "Charging_Stations_Near_Home",
    "Charging_Stations_Near_Work",
    "Total_Charging_Stations",
    "Charging_Station_Difference",
    "Commute_per_Charging_Station",
    "Range_Anxiety_Commute",
    "Home_Charging_Score"
]

# Categorical features
new_cat_features = [
    "Gender",
    "City_Type",
    "Current_Car_Type",
    "Home_Charging_Possible",
    "Subsidy_Available",
    "Environmental_Concern_Level",
    "Range_Anxiety_Level",
    "Number_of_Cars_Owned"
]

```

```

# Handle Missing Values

# Environmental_Concern_Level is an ordinal categorical feature.
# Missing values are kept as a separate category instead of
# imputing them with the most frequent value.
#
# This allows the model to distinguish:
# 1, 2, 3, 4, 5 -> known concern levels
# Missing      -> unavailable information

for df_name, df in [("Train", df_train), ("Test", df_test)]:
    for feature in new_cat_features:
        if feature in df.columns:
            missing_count = df[feature].isna().sum()

            if missing_count > 0:
                df[feature] = df[feature].astype("string").fillna("Missing")
                logger.info(
                    f"[{df_name}] {feature}: "
                    f"filled {missing_count:,} missing values "
                    f"with 'Missing'"
                )
            else:
                df[feature] = df[feature].astype("string")

TOTAL_FEATURES = new_num_features + new_cat_features

df_train = Utils.convert_cat(features=new_cat_features, df=df_train)
df_test = Utils.convert_cat(features=new_cat_features, df=df_test)

logger.info("Categorical conversion completed.")

# Final Preprocessing Summary
logger.info("*" * 80)
logger.info("DATA PREPROCESSING COMPLETED")
logger.info("=" * 80)
logger.info(f"Final Train Shape: {df_train.shape}")
logger.info(f"Final Test Shape: {df_test.shape}")
logger.info(f"Total Features: {len(FEATURES)}")
logger.info(f"Categorical Features: {len(cat_features)}")
logger.info("*" * 80)

```

```

2026-09-19 04:38:05 | INFO      | Feature composition | Base: 13 | Engineered: 5 | Total: 18
2026-09-19 04:38:05 | DEBUG    | Final feature list: ['Age', 'Annual_Income_USD',
'Daily_Commute_km', 'Charging_Stations_Near_Home', 'Charging_Stations_Near_Work', 'Gender',
'City_Type', 'Current_Car_Type', 'Home_Charging_Possible', 'Subsidy_Available',
'Environmental_Concern_Level', 'Range_Anxiety_Level', 'Number_of_Cars_Owned',
'Total_Charging_Stations', 'Charging_Station_Difference', 'Commute_per_Charging_Station',
'Range_Anxiety_Commute', 'Home_Charging_Score']
2026-09-19 04:38:05 | INFO      | Reducing memory usage...
2026-09-19 04:38:05 | INFO      | Memory optimization completed | Train: (678665, 20) | Test:
(286571, 19)
2026-09-19 04:38:05 | INFO      | Applying categorical conversion to original categorical
features...
2026-09-19 04:38:05 | INFO      | [Train] Environmental_Concern_Level: filled 184 missing values
with 'Missing'
2026-09-19 04:38:06 | INFO      | Categorical conversion completed.
2026-09-19 04:38:06 | INFO      |
*****
2026-09-19 04:38:06 | INFO      | DATA PREPROCESSING COMPLETED
2026-09-19 04:38:06 | INFO      |
=====
2026-09-19 04:38:06 | INFO      | Final Train Shape: (678665, 20)
2026-09-19 04:38:06 | INFO      | Final Test Shape: (286571, 19)
2026-09-19 04:38:06 | INFO      | Total Features: 18
2026-09-19 04:38:06 | INFO      | Categorical Features: 8
2026-09-19 04:38:06 | INFO      |
*****

```

Modeling

CatBoostClassifier

```

# Training Data
X = df_train[TOTAL_FEATURES]
y = df_train[config["target"]["feature"]]
logger.info(f"X Shape: {X.shape}")
logger.info(f"y Shape: {y.shape}")

logger.info("-" * 80)
logger.info("Starting CatBoost hyperparameter tuning")
study_cb = optuna.create_study(direction="maximize", study_name="catboost_churn_tuning")
study_cb.optimize(
    lambda trial: objective_catboost(
        trial=trial,
        X=X,
        y=y,
        cat_features=new_cat_features,
        logger=logger
    ),
    n_trials=50
)

```

```

logger.info("CatBoost hyperparameter tuning completed")
logger.info(f"Best CV AUC: {study_cb.best_value:.5f}")
logger.info(f"Best trial: {study_cb.best_trial.number}")
logger.info(f"Best parameters: {study_cb.best_params}")

logger.info("-" * 80)
logger.info("Initializing CatBoost model with best hyperparameters")

model_cb = CatBoostClassifier(**study_cb.best_params, loss_function="Logloss",
auto_class_weights="Balanced", random_seed=config["random_state"],
                             verbose = 0, thread_count=-1, eval_metric="Logloss",
allow_writing_files = False, task_type="GPU", devices="0")
logger.info("-" * 80)
logger.info("Starting CatBoost OOF evaluation")
logger.info(
    f"Training samples: {len(X):,} | "
    f"Features: {X.shape[1]} | "
    f"Categorical features: {len(new_cat_features)} | "
    f"CV Folds: {config["cv"]["n_splits"]}"
)

results_cb = run_oof_v2(model=model_cb, X=X, y=y, X_test=df_test[TOTAL_FEATURES],
                        cat_features=new_cat_features, logger=logger)
logger.info("CatBoost OOF evaluation completed")
logger.info("-" * 80)

```

```
2026-09-19 04:38:06 | INFO | X Shape: (678665, 18)
2026-09-19 04:38:06 | INFO | y Shape: (678665,)
2026-09-19 04:38:06 | INFO | -----
-----
2026-09-19 04:38:06 | INFO | Starting CatBoost hyperparameter tuning
2026-09-19 04:39:32 | DEBUG | Trial 0 | Fold 1/5 | AUC: 0.94062 | Best iteration: 1617
2026-09-19 04:40:57 | DEBUG | Trial 0 | Fold 2/5 | AUC: 0.93970 | Best iteration: 1657
2026-09-19 04:42:22 | DEBUG | Trial 0 | Fold 3/5 | AUC: 0.93985 | Best iteration: 1657
2026-09-19 04:43:47 | DEBUG | Trial 0 | Fold 4/5 | AUC: 0.93999 | Best iteration: 1657
2026-09-19 04:45:12 | DEBUG | Trial 0 | Fold 5/5 | AUC: 0.93838 | Best iteration: 1656
2026-09-19 04:45:12 | INFO | Trial 0 finished | Mean AUC: 0.93971 | Std: 0.00073
2026-09-19 04:47:17 | DEBUG | Trial 1 | Fold 1/5 | AUC: 0.94062 | Best iteration: 1468
2026-09-19 04:49:22 | DEBUG | Trial 1 | Fold 2/5 | AUC: 0.93978 | Best iteration: 1518
2026-09-19 04:51:26 | DEBUG | Trial 1 | Fold 3/5 | AUC: 0.93997 | Best iteration: 1349
2026-09-19 04:53:31 | DEBUG | Trial 1 | Fold 4/5 | AUC: 0.94007 | Best iteration: 1449
2026-09-19 04:55:37 | DEBUG | Trial 1 | Fold 5/5 | AUC: 0.93849 | Best iteration: 1580
2026-09-19 04:55:37 | INFO | Trial 1 finished | Mean AUC: 0.93979 | Std: 0.00071
2026-09-19 04:56:11 | DEBUG | Trial 2 | Fold 1/5 | AUC: 0.94196 | Best iteration: 1006
2026-09-19 04:56:46 | DEBUG | Trial 2 | Fold 2/5 | AUC: 0.94095 | Best iteration: 969
2026-09-19 04:57:20 | DEBUG | Trial 2 | Fold 3/5 | AUC: 0.94108 | Best iteration: 1004
2026-09-19 04:57:54 | DEBUG | Trial 2 | Fold 4/5 | AUC: 0.94132 | Best iteration: 977
2026-09-19 04:58:29 | DEBUG | Trial 2 | Fold 5/5 | AUC: 0.93964 | Best iteration: 938
2026-09-19 04:58:29 | INFO | Trial 2 finished | Mean AUC: 0.94099 | Std: 0.00076
2026-09-19 04:59:45 | DEBUG | Trial 3 | Fold 1/5 | AUC: 0.93921 | Best iteration: 1479
2026-09-19 05:01:01 | DEBUG | Trial 3 | Fold 2/5 | AUC: 0.93809 | Best iteration: 1478
2026-09-19 05:02:18 | DEBUG | Trial 3 | Fold 3/5 | AUC: 0.93829 | Best iteration: 1479
2026-09-19 05:03:34 | DEBUG | Trial 3 | Fold 4/5 | AUC: 0.93838 | Best iteration: 1475
2026-09-19 05:04:50 | DEBUG | Trial 3 | Fold 5/5 | AUC: 0.93670 | Best iteration: 1463
2026-09-19 05:04:51 | INFO | Trial 3 finished | Mean AUC: 0.93813 | Std: 0.00081
2026-09-19 05:06:38 | DEBUG | Trial 4 | Fold 1/5 | AUC: 0.94129 | Best iteration: 1118
2026-09-19 05:08:07 | DEBUG | Trial 4 | Fold 2/5 | AUC: 0.94024 | Best iteration: 879
2026-09-19 05:10:07 | DEBUG | Trial 4 | Fold 3/5 | AUC: 0.94060 | Best iteration: 1281
2026-09-19 05:12:01 | DEBUG | Trial 4 | Fold 4/5 | AUC: 0.94086 | Best iteration: 1220
2026-09-19 05:13:35 | DEBUG | Trial 4 | Fold 5/5 | AUC: 0.93902 | Best iteration: 937
2026-09-19 05:13:35 | INFO | Trial 4 finished | Mean AUC: 0.94040 | Std: 0.00077
2026-09-19 05:14:13 | DEBUG | Trial 5 | Fold 1/5 | AUC: 0.93914 | Best iteration: 884
2026-09-19 05:14:53 | DEBUG | Trial 5 | Fold 2/5 | AUC: 0.93817 | Best iteration: 906
2026-09-19 05:15:31 | DEBUG | Trial 5 | Fold 3/5 | AUC: 0.93868 | Best iteration: 906
2026-09-19 05:16:10 | DEBUG | Trial 5 | Fold 4/5 | AUC: 0.93859 | Best iteration: 906
2026-09-19 05:16:49 | DEBUG | Trial 5 | Fold 5/5 | AUC: 0.93661 | Best iteration: 906
2026-09-19 05:16:49 | INFO | Trial 5 finished | Mean AUC: 0.93824 | Std: 0.00087
2026-09-19 05:18:28 | DEBUG | Trial 6 | Fold 1/5 | AUC: 0.94036 | Best iteration: 1265
2026-09-19 05:20:07 | DEBUG | Trial 6 | Fold 2/5 | AUC: 0.93954 | Best iteration: 1302
2026-09-19 05:21:46 | DEBUG | Trial 6 | Fold 3/5 | AUC: 0.93961 | Best iteration: 1291
2026-09-19 05:23:26 | DEBUG | Trial 6 | Fold 4/5 | AUC: 0.93991 | Best iteration: 1234
2026-09-19 05:25:06 | DEBUG | Trial 6 | Fold 5/5 | AUC: 0.93802 | Best iteration: 1183
2026-09-19 05:25:06 | INFO | Trial 6 finished | Mean AUC: 0.93949 | Std: 0.00079
2026-09-19 05:25:50 | DEBUG | Trial 7 | Fold 1/5 | AUC: 0.93756 | Best iteration: 1286
2026-09-19 05:26:33 | DEBUG | Trial 7 | Fold 2/5 | AUC: 0.93644 | Best iteration: 1286
2026-09-19 05:27:17 | DEBUG | Trial 7 | Fold 3/5 | AUC: 0.93701 | Best iteration: 1286
2026-09-19 05:28:01 | DEBUG | Trial 7 | Fold 4/5 | AUC: 0.93700 | Best iteration: 1286
2026-09-19 05:28:46 | DEBUG | Trial 7 | Fold 5/5 | AUC: 0.93509 | Best iteration: 1286
2026-09-19 05:28:46 | INFO | Trial 7 finished | Mean AUC: 0.93662 | Std: 0.00084
2026-09-19 05:29:41 | DEBUG | Trial 8 | Fold 1/5 | AUC: 0.94173 | Best iteration: 574
2026-09-19 05:30:40 | DEBUG | Trial 8 | Fold 2/5 | AUC: 0.94084 | Best iteration: 648
2026-09-19 05:31:35 | DEBUG | Trial 8 | Fold 3/5 | AUC: 0.94100 | Best iteration: 563
2026-09-19 05:32:33 | DEBUG | Trial 8 | Fold 4/5 | AUC: 0.94127 | Best iteration: 626
2026-09-19 05:33:28 | DEBUG | Trial 8 | Fold 5/5 | AUC: 0.93953 | Best iteration: 578
2026-09-19 05:33:28 | INFO | Trial 8 finished | Mean AUC: 0.94087 | Std: 0.00074
2026-09-19 05:34:10 | DEBUG | Trial 9 | Fold 1/5 | AUC: 0.94099 | Best iteration: 363
```

2026-09-19 05:34:52	DEBUG	Trial 9	Fold 2/5	AUC: 0.93997	Best iteration: 276
2026-09-19 05:35:34	DEBUG	Trial 9	Fold 3/5	AUC: 0.94038	Best iteration: 398
2026-09-19 05:36:17	DEBUG	Trial 9	Fold 4/5	AUC: 0.94056	Best iteration: 474
2026-09-19 05:36:59	DEBUG	Trial 9	Fold 5/5	AUC: 0.93873	Best iteration: 353
2026-09-19 05:36:59	INFO	Trial 9 finished Mean AUC: 0.94013 Std: 0.00077			
2026-09-19 05:37:33	DEBUG	Trial 10	Fold 1/5	AUC: 0.94189	Best iteration: 895
2026-09-19 05:38:08	DEBUG	Trial 10	Fold 2/5	AUC: 0.94088	Best iteration: 946
2026-09-19 05:38:42	DEBUG	Trial 10	Fold 3/5	AUC: 0.94107	Best iteration: 983
2026-09-19 05:39:16	DEBUG	Trial 10	Fold 4/5	AUC: 0.94132	Best iteration: 997
2026-09-19 05:39:50	DEBUG	Trial 10	Fold 5/5	AUC: 0.93964	Best iteration: 998
2026-09-19 05:39:50	INFO	Trial 10 finished Mean AUC: 0.94096 Std: 0.00074			
2026-09-19 05:40:23	DEBUG	Trial 11	Fold 1/5	AUC: 0.94192	Best iteration: 934
2026-09-19 05:40:55	DEBUG	Trial 11	Fold 2/5	AUC: 0.94086	Best iteration: 944
2026-09-19 05:41:28	DEBUG	Trial 11	Fold 3/5	AUC: 0.94109	Best iteration: 942
2026-09-19 05:42:00	DEBUG	Trial 11	Fold 4/5	AUC: 0.94132	Best iteration: 945
2026-09-19 05:42:33	DEBUG	Trial 11	Fold 5/5	AUC: 0.93962	Best iteration: 923
2026-09-19 05:42:33	INFO	Trial 11 finished Mean AUC: 0.94096 Std: 0.00076			
2026-09-19 05:43:03	DEBUG	Trial 12	Fold 1/5	AUC: 0.94182	Best iteration: 884
2026-09-19 05:43:33	DEBUG	Trial 12	Fold 2/5	AUC: 0.94081	Best iteration: 878
2026-09-19 05:44:04	DEBUG	Trial 12	Fold 3/5	AUC: 0.94098	Best iteration: 884
2026-09-19 05:44:34	DEBUG	Trial 12	Fold 4/5	AUC: 0.94119	Best iteration: 884
2026-09-19 05:45:04	DEBUG	Trial 12	Fold 5/5	AUC: 0.93948	Best iteration: 880
2026-09-19 05:45:04	INFO	Trial 12 finished Mean AUC: 0.94086 Std: 0.00077			
2026-09-19 05:45:31	DEBUG	Trial 13	Fold 1/5	AUC: 0.94177	Best iteration: 625
2026-09-19 05:45:58	DEBUG	Trial 13	Fold 2/5	AUC: 0.94076	Best iteration: 625
2026-09-19 05:46:25	DEBUG	Trial 13	Fold 3/5	AUC: 0.94093	Best iteration: 623
2026-09-19 05:46:52	DEBUG	Trial 13	Fold 4/5	AUC: 0.94116	Best iteration: 624
2026-09-19 05:47:19	DEBUG	Trial 13	Fold 5/5	AUC: 0.93944	Best iteration: 625
2026-09-19 05:47:19	INFO	Trial 13 finished Mean AUC: 0.94081 Std: 0.00077			
2026-09-19 05:48:00	DEBUG	Trial 14	Fold 1/5	AUC: 0.94174	Best iteration: 663
2026-09-19 05:48:45	DEBUG	Trial 14	Fold 2/5	AUC: 0.94069	Best iteration: 918
2026-09-19 05:49:31	DEBUG	Trial 14	Fold 3/5	AUC: 0.94093	Best iteration: 856
2026-09-19 05:50:17	DEBUG	Trial 14	Fold 4/5	AUC: 0.94119	Best iteration: 893
2026-09-19 05:51:03	DEBUG	Trial 14	Fold 5/5	AUC: 0.93957	Best iteration: 997
2026-09-19 05:51:03	INFO	Trial 14 finished Mean AUC: 0.94082 Std: 0.00072			
2026-09-19 05:51:27	DEBUG	Trial 15	Fold 1/5	AUC: 0.94158	Best iteration: 707
2026-09-19 05:51:52	DEBUG	Trial 15	Fold 2/5	AUC: 0.94060	Best iteration: 707
2026-09-19 05:52:16	DEBUG	Trial 15	Fold 3/5	AUC: 0.94071	Best iteration: 707
2026-09-19 05:52:41	DEBUG	Trial 15	Fold 4/5	AUC: 0.94099	Best iteration: 707
2026-09-19 05:53:05	DEBUG	Trial 15	Fold 5/5	AUC: 0.93930	Best iteration: 706
2026-09-19 05:53:05	INFO	Trial 15 finished Mean AUC: 0.94064 Std: 0.00075			
2026-09-19 05:53:56	DEBUG	Trial 16	Fold 1/5	AUC: 0.94189	Best iteration: 1161
2026-09-19 05:54:47	DEBUG	Trial 16	Fold 2/5	AUC: 0.94077	Best iteration: 1167
2026-09-19 05:55:39	DEBUG	Trial 16	Fold 3/5	AUC: 0.94101	Best iteration: 1213
2026-09-19 05:56:30	DEBUG	Trial 16	Fold 4/5	AUC: 0.94124	Best iteration: 1214
2026-09-19 05:57:21	DEBUG	Trial 16	Fold 5/5	AUC: 0.93963	Best iteration: 1194
2026-09-19 05:57:21	INFO	Trial 16 finished Mean AUC: 0.94091 Std: 0.00074			
2026-09-19 05:57:51	DEBUG	Trial 17	Fold 1/5	AUC: 0.94155	Best iteration: 855
2026-09-19 05:58:20	DEBUG	Trial 17	Fold 2/5	AUC: 0.94059	Best iteration: 856
2026-09-19 05:58:49	DEBUG	Trial 17	Fold 3/5	AUC: 0.94072	Best iteration: 856
2026-09-19 05:59:18	DEBUG	Trial 17	Fold 4/5	AUC: 0.94094	Best iteration: 856
2026-09-19 05:59:48	DEBUG	Trial 17	Fold 5/5	AUC: 0.93930	Best iteration: 856
2026-09-19 05:59:48	INFO	Trial 17 finished Mean AUC: 0.94062 Std: 0.00074			
2026-09-19 06:00:45	DEBUG	Trial 18	Fold 1/5	AUC: 0.94171	Best iteration: 1021
2026-09-19 06:01:37	DEBUG	Trial 18	Fold 2/5	AUC: 0.94055	Best iteration: 720
2026-09-19 06:02:35	DEBUG	Trial 18	Fold 3/5	AUC: 0.94088	Best iteration: 1067
2026-09-19 06:03:32	DEBUG	Trial 18	Fold 4/5	AUC: 0.94116	Best iteration: 889
2026-09-19 06:04:29	DEBUG	Trial 18	Fold 5/5	AUC: 0.93943	Best iteration: 908
2026-09-19 06:04:29	INFO	Trial 18 finished Mean AUC: 0.94075 Std: 0.00076			
2026-09-19 06:05:52	DEBUG	Trial 19	Fold 1/5	AUC: 0.94169	Best iteration: 1954

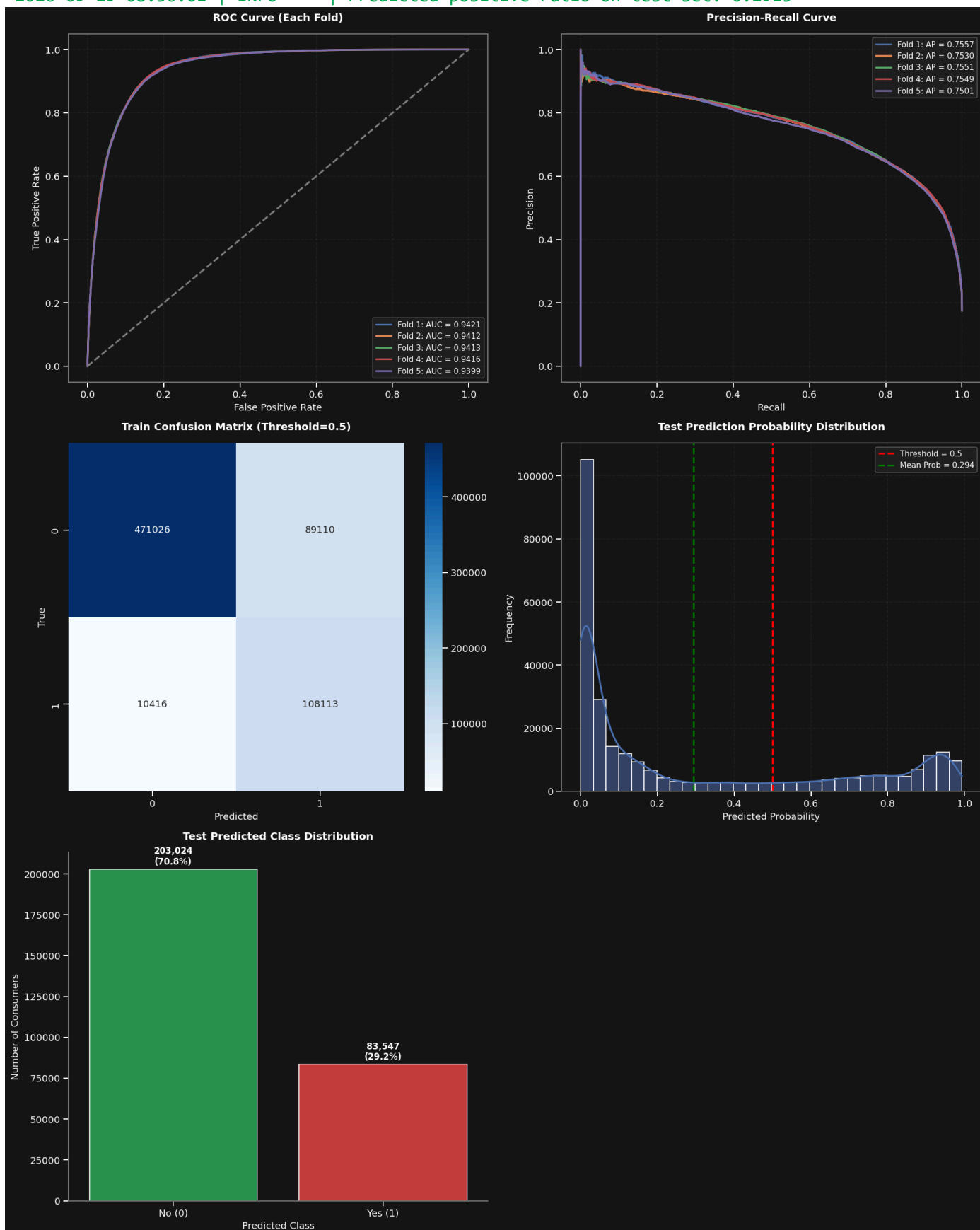
2026-09-19 06:07:14	DEBUG	Trial 19	Fold 2/5	AUC: 0.94063	Best iteration: 1932
2026-09-19 06:08:37	DEBUG	Trial 19	Fold 3/5	AUC: 0.94088	Best iteration: 1991
2026-09-19 06:10:00	DEBUG	Trial 19	Fold 4/5	AUC: 0.94113	Best iteration: 1987
2026-09-19 06:11:23	DEBUG	Trial 19	Fold 5/5	AUC: 0.93945	Best iteration: 1992
2026-09-19 06:11:23	INFO	Trial 19 finished Mean AUC: 0.94076 Std: 0.00074			
2026-09-19 06:12:11	DEBUG	Trial 20	Fold 1/5	AUC: 0.94209	Best iteration: 1310
2026-09-19 06:12:59	DEBUG	Trial 20	Fold 2/5	AUC: 0.94118	Best iteration: 1421
2026-09-19 06:13:46	DEBUG	Trial 20	Fold 3/5	AUC: 0.94128	Best iteration: 1415
2026-09-19 06:14:34	DEBUG	Trial 20	Fold 4/5	AUC: 0.94152	Best iteration: 1420
2026-09-19 06:15:22	DEBUG	Trial 20	Fold 5/5	AUC: 0.93990	Best iteration: 1421
2026-09-19 06:15:22	INFO	Trial 20 finished Mean AUC: 0.94119 Std: 0.00072			
2026-09-19 06:16:10	DEBUG	Trial 21	Fold 1/5	AUC: 0.94214	Best iteration: 1311
2026-09-19 06:16:59	DEBUG	Trial 21	Fold 2/5	AUC: 0.94121	Best iteration: 1440
2026-09-19 06:17:47	DEBUG	Trial 21	Fold 3/5	AUC: 0.94131	Best iteration: 1439
2026-09-19 06:18:36	DEBUG	Trial 21	Fold 4/5	AUC: 0.94156	Best iteration: 1434
2026-09-19 06:19:24	DEBUG	Trial 21	Fold 5/5	AUC: 0.93989	Best iteration: 1441
2026-09-19 06:19:24	INFO	Trial 21 finished Mean AUC: 0.94122 Std: 0.00074			
2026-09-19 06:20:14	DEBUG	Trial 22	Fold 1/5	AUC: 0.94212	Best iteration: 1457
2026-09-19 06:21:04	DEBUG	Trial 22	Fold 2/5	AUC: 0.94118	Best iteration: 1467
2026-09-19 06:21:55	DEBUG	Trial 22	Fold 3/5	AUC: 0.94133	Best iteration: 1493
2026-09-19 06:22:45	DEBUG	Trial 22	Fold 4/5	AUC: 0.94156	Best iteration: 1486
2026-09-19 06:23:35	DEBUG	Trial 22	Fold 5/5	AUC: 0.93989	Best iteration: 1479
2026-09-19 06:23:35	INFO	Trial 22 finished Mean AUC: 0.94122 Std: 0.00073			
2026-09-19 06:24:35	DEBUG	Trial 23	Fold 1/5	AUC: 0.94205	Best iteration: 1278
2026-09-19 06:25:27	DEBUG	Trial 23	Fold 2/5	AUC: 0.94105	Best iteration: 943
2026-09-19 06:26:27	DEBUG	Trial 23	Fold 3/5	AUC: 0.94132	Best iteration: 1370
2026-09-19 06:27:27	DEBUG	Trial 23	Fold 4/5	AUC: 0.94144	Best iteration: 1151
2026-09-19 06:28:27	DEBUG	Trial 23	Fold 5/5	AUC: 0.93980	Best iteration: 1118
2026-09-19 06:28:27	INFO	Trial 23 finished Mean AUC: 0.94113 Std: 0.00074			
2026-09-19 06:29:14	DEBUG	Trial 24	Fold 1/5	AUC: 0.94209	Best iteration: 1310
2026-09-19 06:30:02	DEBUG	Trial 24	Fold 2/5	AUC: 0.94107	Best iteration: 1187
2026-09-19 06:30:50	DEBUG	Trial 24	Fold 3/5	AUC: 0.94123	Best iteration: 1416
2026-09-19 06:31:38	DEBUG	Trial 24	Fold 4/5	AUC: 0.94148	Best iteration: 1401
2026-09-19 06:32:26	DEBUG	Trial 24	Fold 5/5	AUC: 0.93986	Best iteration: 1419
2026-09-19 06:32:26	INFO	Trial 24 finished Mean AUC: 0.94115 Std: 0.00073			
2026-09-19 06:33:18	DEBUG	Trial 25	Fold 1/5	AUC: 0.94208	Best iteration: 1544
2026-09-19 06:34:10	DEBUG	Trial 25	Fold 2/5	AUC: 0.94106	Best iteration: 1545
2026-09-19 06:35:02	DEBUG	Trial 25	Fold 3/5	AUC: 0.94114	Best iteration: 1541
2026-09-19 06:35:54	DEBUG	Trial 25	Fold 4/5	AUC: 0.94142	Best iteration: 1532
2026-09-19 06:36:46	DEBUG	Trial 25	Fold 5/5	AUC: 0.93979	Best iteration: 1541
2026-09-19 06:36:46	INFO	Trial 25 finished Mean AUC: 0.94110 Std: 0.00075			
2026-09-19 06:37:40	DEBUG	Trial 26	Fold 1/5	AUC: 0.94208	Best iteration: 972
2026-09-19 06:38:30	DEBUG	Trial 26	Fold 2/5	AUC: 0.94112	Best iteration: 872
2026-09-19 06:39:36	DEBUG	Trial 26	Fold 3/5	AUC: 0.94132	Best iteration: 1236
2026-09-19 06:40:25	DEBUG	Trial 26	Fold 4/5	AUC: 0.94151	Best iteration: 866
2026-09-19 06:41:31	DEBUG	Trial 26	Fold 5/5	AUC: 0.93996	Best iteration: 1240
2026-09-19 06:41:31	INFO	Trial 26 finished Mean AUC: 0.94120 Std: 0.00070			
2026-09-19 06:42:34	DEBUG	Trial 27	Fold 1/5	AUC: 0.94198	Best iteration: 1199
2026-09-19 06:43:29	DEBUG	Trial 27	Fold 2/5	AUC: 0.94091	Best iteration: 994
2026-09-19 06:44:43	DEBUG	Trial 27	Fold 3/5	AUC: 0.94116	Best iteration: 1435
2026-09-19 06:45:37	DEBUG	Trial 27	Fold 4/5	AUC: 0.94136	Best iteration: 926
2026-09-19 06:46:44	DEBUG	Trial 27	Fold 5/5	AUC: 0.93978	Best iteration: 1255
2026-09-19 06:46:44	INFO	Trial 27 finished Mean AUC: 0.94104 Std: 0.00072			
2026-09-19 06:48:18	DEBUG	Trial 28	Fold 1/5	AUC: 0.94011	Best iteration: 1849
2026-09-19 06:49:53	DEBUG	Trial 28	Fold 2/5	AUC: 0.93905	Best iteration: 1851
2026-09-19 06:51:28	DEBUG	Trial 28	Fold 3/5	AUC: 0.93928	Best iteration: 1841
2026-09-19 06:53:03	DEBUG	Trial 28	Fold 4/5	AUC: 0.93923	Best iteration: 1844
2026-09-19 06:54:38	DEBUG	Trial 28	Fold 5/5	AUC: 0.93773	Best iteration: 1851
2026-09-19 06:54:38	INFO	Trial 28 finished Mean AUC: 0.93908 Std: 0.00077			
2026-09-19 06:56:05	DEBUG	Trial 29	Fold 1/5	AUC: 0.93971	Best iteration: 1717

2026-09-19 06:57:33	DEBUG	Trial 29	Fold 2/5	AUC: 0.93882	Best iteration: 1717
2026-09-19 06:59:01	DEBUG	Trial 29	Fold 3/5	AUC: 0.93897	Best iteration: 1717
2026-09-19 07:00:30	DEBUG	Trial 29	Fold 4/5	AUC: 0.93885	Best iteration: 1717
2026-09-19 07:01:57	DEBUG	Trial 29	Fold 5/5	AUC: 0.93741	Best iteration: 1716
2026-09-19 07:01:58	INFO	Trial 29 finished Mean AUC: 0.93875 Std: 0.00074			
2026-09-19 07:03:03	DEBUG	Trial 30	Fold 1/5	AUC: 0.94181	Best iteration: 1558
2026-09-19 07:04:09	DEBUG	Trial 30	Fold 2/5	AUC: 0.94083	Best iteration: 1567
2026-09-19 07:05:15	DEBUG	Trial 30	Fold 3/5	AUC: 0.94105	Best iteration: 1565
2026-09-19 07:06:21	DEBUG	Trial 30	Fold 4/5	AUC: 0.94120	Best iteration: 1566
2026-09-19 07:07:26	DEBUG	Trial 30	Fold 5/5	AUC: 0.93955	Best iteration: 1560
2026-09-19 07:07:26	INFO	Trial 30 finished Mean AUC: 0.94089 Std: 0.00074			
2026-09-19 07:08:12	DEBUG	Trial 31	Fold 1/5	AUC: 0.94052	Best iteration: 1351
2026-09-19 07:08:58	DEBUG	Trial 31	Fold 2/5	AUC: 0.93950	Best iteration: 1351
2026-09-19 07:09:44	DEBUG	Trial 31	Fold 3/5	AUC: 0.93946	Best iteration: 1346
2026-09-19 07:10:30	DEBUG	Trial 31	Fold 4/5	AUC: 0.93964	Best iteration: 1351
2026-09-19 07:11:17	DEBUG	Trial 31	Fold 5/5	AUC: 0.93817	Best iteration: 1351
2026-09-19 07:11:18	INFO	Trial 31 finished Mean AUC: 0.93946 Std: 0.00075			
2026-09-19 07:12:16	DEBUG	Trial 32	Fold 1/5	AUC: 0.93979	Best iteration: 1679
2026-09-19 07:13:13	DEBUG	Trial 32	Fold 2/5	AUC: 0.93880	Best iteration: 1677
2026-09-19 07:14:09	DEBUG	Trial 32	Fold 3/5	AUC: 0.93893	Best iteration: 1676
2026-09-19 07:15:06	DEBUG	Trial 32	Fold 4/5	AUC: 0.93894	Best iteration: 1669
2026-09-19 07:16:02	DEBUG	Trial 32	Fold 5/5	AUC: 0.93746	Best iteration: 1679
2026-09-19 07:16:02	INFO	Trial 32 finished Mean AUC: 0.93878 Std: 0.00075			
2026-09-19 07:16:52	DEBUG	Trial 33	Fold 1/5	AUC: 0.94060	Best iteration: 894
2026-09-19 07:17:41	DEBUG	Trial 33	Fold 2/5	AUC: 0.93974	Best iteration: 882
2026-09-19 07:18:30	DEBUG	Trial 33	Fold 3/5	AUC: 0.93996	Best iteration: 1041
2026-09-19 07:19:16	DEBUG	Trial 33	Fold 4/5	AUC: 0.93995	Best iteration: 751
2026-09-19 07:20:07	DEBUG	Trial 33	Fold 5/5	AUC: 0.93839	Best iteration: 939
2026-09-19 07:20:07	INFO	Trial 33 finished Mean AUC: 0.93973 Std: 0.00073			
2026-09-19 07:20:59	DEBUG	Trial 34	Fold 1/5	AUC: 0.94210	Best iteration: 1482
2026-09-19 07:21:51	DEBUG	Trial 34	Fold 2/5	AUC: 0.94116	Best iteration: 1449
2026-09-19 07:22:42	DEBUG	Trial 34	Fold 3/5	AUC: 0.94134	Best iteration: 1482
2026-09-19 07:23:32	DEBUG	Trial 34	Fold 4/5	AUC: 0.94153	Best iteration: 1486
2026-09-19 07:24:23	DEBUG	Trial 34	Fold 5/5	AUC: 0.93987	Best iteration: 1479
2026-09-19 07:24:23	INFO	Trial 34 finished Mean AUC: 0.94120 Std: 0.00074			
2026-09-19 07:25:29	DEBUG	Trial 35	Fold 1/5	AUC: 0.94203	Best iteration: 1926
2026-09-19 07:26:32	DEBUG	Trial 35	Fold 2/5	AUC: 0.94101	Best iteration: 1553
2026-09-19 07:27:39	DEBUG	Trial 35	Fold 3/5	AUC: 0.94121	Best iteration: 1638
2026-09-19 07:28:46	DEBUG	Trial 35	Fold 4/5	AUC: 0.94146	Best iteration: 1905
2026-09-19 07:29:54	DEBUG	Trial 35	Fold 5/5	AUC: 0.93988	Best iteration: 1960
2026-09-19 07:29:54	INFO	Trial 35 finished Mean AUC: 0.94112 Std: 0.00071			
2026-09-19 07:30:40	DEBUG	Trial 36	Fold 1/5	AUC: 0.94018	Best iteration: 427
2026-09-19 07:31:45	DEBUG	Trial 36	Fold 2/5	AUC: 0.93945	Best iteration: 737
2026-09-19 07:32:50	DEBUG	Trial 36	Fold 3/5	AUC: 0.93980	Best iteration: 722
2026-09-19 07:33:51	DEBUG	Trial 36	Fold 4/5	AUC: 0.93985	Best iteration: 667
2026-09-19 07:34:52	DEBUG	Trial 36	Fold 5/5	AUC: 0.93810	Best iteration: 666
2026-09-19 07:34:52	INFO	Trial 36 finished Mean AUC: 0.93947 Std: 0.00073			
2026-09-19 07:36:01	DEBUG	Trial 37	Fold 1/5	AUC: 0.94192	Best iteration: 1431
2026-09-19 07:37:11	DEBUG	Trial 37	Fold 2/5	AUC: 0.94088	Best iteration: 1618
2026-09-19 07:38:20	DEBUG	Trial 37	Fold 3/5	AUC: 0.94106	Best iteration: 1639
2026-09-19 07:39:29	DEBUG	Trial 37	Fold 4/5	AUC: 0.94132	Best iteration: 1637
2026-09-19 07:40:39	DEBUG	Trial 37	Fold 5/5	AUC: 0.93967	Best iteration: 1655
2026-09-19 07:40:39	INFO	Trial 37 finished Mean AUC: 0.94097 Std: 0.00074			
2026-09-19 07:41:33	DEBUG	Trial 38	Fold 1/5	AUC: 0.94171	Best iteration: 1601
2026-09-19 07:42:28	DEBUG	Trial 38	Fold 2/5	AUC: 0.94070	Best iteration: 1585
2026-09-19 07:43:22	DEBUG	Trial 38	Fold 3/5	AUC: 0.94086	Best iteration: 1588
2026-09-19 07:44:17	DEBUG	Trial 38	Fold 4/5	AUC: 0.94107	Best iteration: 1582
2026-09-19 07:45:11	DEBUG	Trial 38	Fold 5/5	AUC: 0.93935	Best iteration: 1595
2026-09-19 07:45:11	INFO	Trial 38 finished Mean AUC: 0.94074 Std: 0.00077			
2026-09-19 07:46:09	DEBUG	Trial 39	Fold 1/5	AUC: 0.94206	Best iteration: 1554

2026-09-19 07:47:07	DEBUG	Trial 39	Fold 2/5	AUC: 0.94108	Best iteration: 1597
2026-09-19 07:48:06	DEBUG	Trial 39	Fold 3/5	AUC: 0.94129	Best iteration: 1620
2026-09-19 07:49:04	DEBUG	Trial 39	Fold 4/5	AUC: 0.94152	Best iteration: 1660
2026-09-19 07:50:03	DEBUG	Trial 39	Fold 5/5	AUC: 0.93986	Best iteration: 1428
2026-09-19 07:50:03	INFO	Trial 39 finished Mean AUC: 0.94116 Std: 0.00073			
2026-09-19 07:51:40	DEBUG	Trial 40	Fold 1/5	AUC: 0.93964	Best iteration: 1873
2026-09-19 07:53:17	DEBUG	Trial 40	Fold 2/5	AUC: 0.93868	Best iteration: 1874
2026-09-19 07:54:53	DEBUG	Trial 40	Fold 3/5	AUC: 0.93918	Best iteration: 1840
2026-09-19 07:56:30	DEBUG	Trial 40	Fold 4/5	AUC: 0.93894	Best iteration: 1807
2026-09-19 07:58:07	DEBUG	Trial 40	Fold 5/5	AUC: 0.93721	Best iteration: 1858
2026-09-19 07:58:07	INFO	Trial 40 finished Mean AUC: 0.93873 Std: 0.00082			
2026-09-19 07:58:52	DEBUG	Trial 41	Fold 1/5	AUC: 0.94212	Best iteration: 1311
2026-09-19 07:59:37	DEBUG	Trial 41	Fold 2/5	AUC: 0.94119	Best iteration: 1301
2026-09-19 08:00:22	DEBUG	Trial 41	Fold 3/5	AUC: 0.94132	Best iteration: 1332
2026-09-19 08:01:07	DEBUG	Trial 41	Fold 4/5	AUC: 0.94153	Best iteration: 1321
2026-09-19 08:01:53	DEBUG	Trial 41	Fold 5/5	AUC: 0.93986	Best iteration: 1320
2026-09-19 08:01:53	INFO	Trial 41 finished Mean AUC: 0.94121 Std: 0.00074			
2026-09-19 08:02:39	DEBUG	Trial 42	Fold 1/5	AUC: 0.94210	Best iteration: 1324
2026-09-19 08:03:25	DEBUG	Trial 42	Fold 2/5	AUC: 0.94116	Best iteration: 1334
2026-09-19 08:04:11	DEBUG	Trial 42	Fold 3/5	AUC: 0.94131	Best iteration: 1347
2026-09-19 08:04:57	DEBUG	Trial 42	Fold 4/5	AUC: 0.94152	Best iteration: 1362
2026-09-19 08:05:43	DEBUG	Trial 42	Fold 5/5	AUC: 0.93985	Best iteration: 1361
2026-09-19 08:05:43	INFO	Trial 42 finished Mean AUC: 0.94119 Std: 0.00074			
2026-09-19 08:06:27	DEBUG	Trial 43	Fold 1/5	AUC: 0.94210	Best iteration: 1100
2026-09-19 08:07:10	DEBUG	Trial 43	Fold 2/5	AUC: 0.94118	Best iteration: 1148
2026-09-19 08:07:53	DEBUG	Trial 43	Fold 3/5	AUC: 0.94131	Best iteration: 1258
2026-09-19 08:08:37	DEBUG	Trial 43	Fold 4/5	AUC: 0.94154	Best iteration: 1279
2026-09-19 08:09:20	DEBUG	Trial 43	Fold 5/5	AUC: 0.93990	Best iteration: 1280
2026-09-19 08:09:21	INFO	Trial 43 finished Mean AUC: 0.94121 Std: 0.00073			
2026-09-19 08:10:01	DEBUG	Trial 44	Fold 1/5	AUC: 0.94204	Best iteration: 1111
2026-09-19 08:10:41	DEBUG	Trial 44	Fold 2/5	AUC: 0.94110	Best iteration: 1106
2026-09-19 08:11:22	DEBUG	Trial 44	Fold 3/5	AUC: 0.94121	Best iteration: 1161
2026-09-19 08:12:02	DEBUG	Trial 44	Fold 4/5	AUC: 0.94146	Best iteration: 1175
2026-09-19 08:12:42	DEBUG	Trial 44	Fold 5/5	AUC: 0.93985	Best iteration: 1188
2026-09-19 08:12:42	INFO	Trial 44 finished Mean AUC: 0.94113 Std: 0.00072			
2026-09-19 08:13:24	DEBUG	Trial 45	Fold 1/5	AUC: 0.94093	Best iteration: 1233
2026-09-19 08:14:05	DEBUG	Trial 45	Fold 2/5	AUC: 0.93994	Best iteration: 1233
2026-09-19 08:14:47	DEBUG	Trial 45	Fold 3/5	AUC: 0.94017	Best iteration: 1233
2026-09-19 08:15:28	DEBUG	Trial 45	Fold 4/5	AUC: 0.94030	Best iteration: 1233
2026-09-19 08:16:09	DEBUG	Trial 45	Fold 5/5	AUC: 0.93866	Best iteration: 1233
2026-09-19 08:16:10	INFO	Trial 45 finished Mean AUC: 0.94000 Std: 0.00075			
2026-09-19 08:16:53	DEBUG	Trial 46	Fold 1/5	AUC: 0.94212	Best iteration: 1294
2026-09-19 08:17:37	DEBUG	Trial 46	Fold 2/5	AUC: 0.94112	Best iteration: 1145
2026-09-19 08:18:22	DEBUG	Trial 46	Fold 3/5	AUC: 0.94123	Best iteration: 1258
2026-09-19 08:19:06	DEBUG	Trial 46	Fold 4/5	AUC: 0.94147	Best iteration: 1297
2026-09-19 08:19:50	DEBUG	Trial 46	Fold 5/5	AUC: 0.93988	Best iteration: 1301
2026-09-19 08:19:50	INFO	Trial 46 finished Mean AUC: 0.94116 Std: 0.00073			
2026-09-19 08:20:39	DEBUG	Trial 47	Fold 1/5	AUC: 0.94146	Best iteration: 1444
2026-09-19 08:21:28	DEBUG	Trial 47	Fold 2/5	AUC: 0.94053	Best iteration: 1446
2026-09-19 08:22:17	DEBUG	Trial 47	Fold 3/5	AUC: 0.94065	Best iteration: 1446
2026-09-19 08:23:06	DEBUG	Trial 47	Fold 4/5	AUC: 0.94088	Best iteration: 1446
2026-09-19 08:23:54	DEBUG	Trial 47	Fold 5/5	AUC: 0.93913	Best iteration: 1444
2026-09-19 08:23:54	INFO	Trial 47 finished Mean AUC: 0.94053 Std: 0.00077			
2026-09-19 08:24:45	DEBUG	Trial 48	Fold 1/5	AUC: 0.94201	Best iteration: 1488
2026-09-19 08:25:35	DEBUG	Trial 48	Fold 2/5	AUC: 0.94103	Best iteration: 1464
2026-09-19 08:26:25	DEBUG	Trial 48	Fold 3/5	AUC: 0.94122	Best iteration: 1482
2026-09-19 08:27:15	DEBUG	Trial 48	Fold 4/5	AUC: 0.94146	Best iteration: 1486
2026-09-19 08:28:05	DEBUG	Trial 48	Fold 5/5	AUC: 0.93982	Best iteration: 1481
2026-09-19 08:28:05	INFO	Trial 48 finished Mean AUC: 0.94111 Std: 0.00072			
2026-09-19 08:28:51	DEBUG	Trial 49	Fold 1/5	AUC: 0.94204	Best iteration: 1353

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2026-09-19 08:29:36 | DEBUG | Trial 49 | Fold 2/5 | AUC: 0.94111 | Best iteration: 1359
2026-09-19 08:30:22 | DEBUG | Trial 49 | Fold 3/5 | AUC: 0.94122 | Best iteration: 1358
2026-09-19 08:31:08 | DEBUG | Trial 49 | Fold 4/5 | AUC: 0.94149 | Best iteration: 1361
2026-09-19 08:31:54 | DEBUG | Trial 49 | Fold 5/5 | AUC: 0.93976 | Best iteration: 1351
2026-09-19 08:31:54 | INFO | Trial 49 finished | Mean AUC: 0.94112 | Std: 0.00076
2026-09-19 08:31:54 | INFO | CatBoost hyperparameter tuning completed
2026-09-19 08:31:54 | INFO | Best CV AUC: 0.94122
2026-09-19 08:31:54 | INFO | Best trial: 21
2026-09-19 08:31:54 | INFO | Best parameters: {'bootstrap_type': 'Bernoulli', 'iterations':
1442, 'learning_rate': 0.06760076256960292, 'depth': 4, 'l2_leaf_reg': 0.09303745990444143,
'random_strength': 0.27712244457090135, 'border_count': 254, 'subsample': 0.7031975640363632}
2026-09-19 08:31:54 | INFO | -----
-----
2026-09-19 08:31:54 | INFO | Initializing CatBoost model with best hyperparameters
2026-09-19 08:31:54 | INFO | -----
-----
2026-09-19 08:31:54 | INFO | Starting CatBoost OOF evaluation
2026-09-19 08:31:54 | INFO | Training samples: 678,665 | Features: 18 | Categorical
features: 8 | CV Folds: 5
2026-09-19 08:31:54 | INFO | Starting OOF training with CatBoostClassifier
2026-09-19 08:31:54 | INFO | -----
-----
2026-09-19 08:31:54 | INFO | Starting Fold 1/5
2026-09-19 08:31:54 | DEBUG | Fold 1 | Train samples: 542,932 | Validation samples: 135,733
2026-09-19 08:32:42 | INFO | Fold 1 training completed | Best iteration: 1330
2026-09-19 08:32:43 | INFO | Fold 1 AUC: 0.94212
2026-09-19 08:32:43 | DEBUG | Fold 1 feature importance collected successfully
2026-09-19 08:32:43 | DEBUG | Fold 1 temporary memory released
2026-09-19 08:32:43 | INFO | -----
-----
2026-09-19 08:32:43 | INFO | Starting Fold 2/5
2026-09-19 08:32:43 | DEBUG | Fold 2 | Train samples: 542,932 | Validation samples: 135,733
2026-09-19 08:33:31 | INFO | Fold 2 training completed | Best iteration: 1439
2026-09-19 08:33:32 | INFO | Fold 2 AUC: 0.94118
2026-09-19 08:33:32 | DEBUG | Fold 2 feature importance collected successfully
2026-09-19 08:33:32 | DEBUG | Fold 2 temporary memory released
2026-09-19 08:33:32 | INFO | -----
-----
2026-09-19 08:33:32 | INFO | Starting Fold 3/5
2026-09-19 08:33:32 | DEBUG | Fold 3 | Train samples: 542,932 | Validation samples: 135,733
2026-09-19 08:34:20 | INFO | Fold 3 training completed | Best iteration: 1440
2026-09-19 08:34:21 | INFO | Fold 3 AUC: 0.94132
2026-09-19 08:34:21 | DEBUG | Fold 3 feature importance collected successfully
2026-09-19 08:34:21 | DEBUG | Fold 3 temporary memory released
2026-09-19 08:34:21 | INFO | -----
-----
2026-09-19 08:34:21 | INFO | Starting Fold 4/5
2026-09-19 08:34:21 | DEBUG | Fold 4 | Train samples: 542,932 | Validation samples: 135,733
2026-09-19 08:35:09 | INFO | Fold 4 training completed | Best iteration: 1433
2026-09-19 08:35:10 | INFO | Fold 4 AUC: 0.94158
2026-09-19 08:35:10 | DEBUG | Fold 4 feature importance collected successfully
2026-09-19 08:35:10 | DEBUG | Fold 4 temporary memory released
2026-09-19 08:35:10 | INFO | -----
-----
2026-09-19 08:35:10 | INFO | Starting Fold 5/5
2026-09-19 08:35:10 | DEBUG | Fold 5 | Train samples: 542,932 | Validation samples: 135,733
2026-09-19 08:35:58 | INFO | Fold 5 training completed | Best iteration: 1441
2026-09-19 08:35:59 | INFO | Fold 5 AUC: 0.93991
2026-09-19 08:35:59 | DEBUG | Fold 5 feature importance collected successfully
2026-09-19 08:35:59 | DEBUG | Fold 5 temporary memory released
```

```
2026-09-19 08:36:00 | INFO      | OOF training completed
2026-09-19 08:36:00 | INFO      | OOF AUC: 0.94121
2026-09-19 08:36:00 | INFO      | Fold AUCs: [np.float64(0.9421225544172911),
np.float64(0.9411824198733484), np.float64(0.9413179409658456), np.float64(0.9415816139407963),
np.float64(0.9399096218737587)]
2026-09-19 08:36:00 | INFO      | Mean Fold AUC: 0.94122 +/- 0.00073
2026-09-19 08:36:01 | INFO      | Predicted positive ratio on test set: 0.2915
```



2026-09-19 08:36:03 | INFO | CatBoost OOF evaluation completed

2026-09-19 08:36:03 | INFO | -----

XGBClassifier

```
logger.info("-" * 80)
logger.info("Starting XGBoost hyperparameter tuning")
study_xgb = optuna.create_study(direction="maximize", study_name="xgboost_churn_tuning")
study_xgb.optimize(
    lambda trial: objective_xgb(
        trial=trial,
        X=X,
        y=y,
        cat_features=new_cat_features,
        logger=logger
    ),
    n_trials=50
)

logger.info("XGBoost hyperparameter tuning completed")
logger.info(f"Best CV AUC: {study_xgb.best_value:.5f}")
logger.info(f"Best trial: {study_xgb.best_trial.number}")
logger.info(f"Best parameters: {study_xgb.best_params}")

logger.info("-" * 80)
logger.info("Initializing XGBoost model with best hyperparameters")

# Calculate class imbalance
n_negative = (y == 0).sum()
n_positive = (y == 1).sum()

if n_positive == 0:
    raise ValueError("Target contains no positive samples.")

scale_pos_weight = n_negative / n_positive

model_xgb = XGBClassifier(**study_xgb.best_params, objective="binary:logistic", metric="auc",
                           verbosity=0, n_jobs=-1, boosting_type="gbdt",
                           class_weight="balanced", random_state=config["random_state"],
                           tree_method="hist", device="cuda", enable_categorical=True,
                           scale_pos_weight = scale_pos_weight)

logger.info("-" * 80)
logger.info("Starting XGBoost OOF evaluation")
logger.info(
    f"Training samples: {len(X):,} | "
    f"Features: {X.shape[1]} | "
    f"Categorical features: {len(new_cat_features)} | "
    f"CV Folds: {config['cv']['n_splits']}"
)

results_xgb = run_oof_v2(model=model_xgb, X=X, y=y, X_test=df_test[TOTAL_FEATURES],
                          cat_features=new_cat_features, logger=logger)

logger.info("XGBoost OOF evaluation completed")
logger.info("-" * 80)
```

```
2026-09-19 08:36:03 | INFO | -----
-----
2026-09-19 08:36:03 | INFO | Starting XGBoost hyperparameter tuning
2026-09-19 08:36:03 | DEBUG | Trial 0 | scale_pos_weight: 4.7257
2026-09-19 08:36:03 | DEBUG | Trial 0 | Starting Fold 1/5
2026-09-19 08:36:11 | DEBUG | Trial 0 | Fold 1/5 | AUC: 0.94171 | Best iteration: 420
2026-09-19 08:36:11 | DEBUG | Trial 0 | Starting Fold 2/5
2026-09-19 08:36:18 | DEBUG | Trial 0 | Fold 2/5 | AUC: 0.94080 | Best iteration: 369
2026-09-19 08:36:18 | DEBUG | Trial 0 | Starting Fold 3/5
2026-09-19 08:36:25 | DEBUG | Trial 0 | Fold 3/5 | AUC: 0.94107 | Best iteration: 437
2026-09-19 08:36:26 | DEBUG | Trial 0 | Starting Fold 4/5
2026-09-19 08:36:33 | DEBUG | Trial 0 | Fold 4/5 | AUC: 0.94129 | Best iteration: 451
2026-09-19 08:36:33 | DEBUG | Trial 0 | Starting Fold 5/5
2026-09-19 08:36:41 | DEBUG | Trial 0 | Fold 5/5 | AUC: 0.93962 | Best iteration: 454
2026-09-19 08:36:41 | INFO | Trial 0 finished | Mean AUC: 0.94090 | Std: 0.00071
2026-09-19 08:36:41 | DEBUG | Trial 1 | scale_pos_weight: 4.7257
2026-09-19 08:36:41 | DEBUG | Trial 1 | Starting Fold 1/5
2026-09-19 08:36:47 | DEBUG | Trial 1 | Fold 1/5 | AUC: 0.94209 | Best iteration: 1090
2026-09-19 08:36:47 | DEBUG | Trial 1 | Starting Fold 2/5
2026-09-19 08:36:53 | DEBUG | Trial 1 | Fold 2/5 | AUC: 0.94134 | Best iteration: 956
2026-09-19 08:36:53 | DEBUG | Trial 1 | Starting Fold 3/5
2026-09-19 08:36:59 | DEBUG | Trial 1 | Fold 3/5 | AUC: 0.94148 | Best iteration: 870
2026-09-19 08:37:00 | DEBUG | Trial 1 | Starting Fold 4/5
2026-09-19 08:37:06 | DEBUG | Trial 1 | Fold 4/5 | AUC: 0.94158 | Best iteration: 1139
2026-09-19 08:37:06 | DEBUG | Trial 1 | Starting Fold 5/5
2026-09-19 08:37:12 | DEBUG | Trial 1 | Fold 5/5 | AUC: 0.94012 | Best iteration: 1066
2026-09-19 08:37:12 | INFO | Trial 1 finished | Mean AUC: 0.94132 | Std: 0.00065
2026-09-19 08:37:12 | DEBUG | Trial 2 | scale_pos_weight: 4.7257
2026-09-19 08:37:12 | DEBUG | Trial 2 | Starting Fold 1/5
2026-09-19 08:37:16 | DEBUG | Trial 2 | Fold 1/5 | AUC: 0.94199 | Best iteration: 517
2026-09-19 08:37:16 | DEBUG | Trial 2 | Starting Fold 2/5
2026-09-19 08:37:20 | DEBUG | Trial 2 | Fold 2/5 | AUC: 0.94123 | Best iteration: 517
2026-09-19 08:37:20 | DEBUG | Trial 2 | Starting Fold 3/5
2026-09-19 08:37:24 | DEBUG | Trial 2 | Fold 3/5 | AUC: 0.94135 | Best iteration: 514
2026-09-19 08:37:24 | DEBUG | Trial 2 | Starting Fold 4/5
2026-09-19 08:37:28 | DEBUG | Trial 2 | Fold 4/5 | AUC: 0.94148 | Best iteration: 516
2026-09-19 08:37:28 | DEBUG | Trial 2 | Starting Fold 5/5
2026-09-19 08:37:31 | DEBUG | Trial 2 | Fold 5/5 | AUC: 0.93988 | Best iteration: 515
2026-09-19 08:37:31 | INFO | Trial 2 finished | Mean AUC: 0.94119 | Std: 0.00070
2026-09-19 08:37:31 | DEBUG | Trial 3 | scale_pos_weight: 4.7257
2026-09-19 08:37:32 | DEBUG | Trial 3 | Starting Fold 1/5
2026-09-19 08:37:35 | DEBUG | Trial 3 | Fold 1/5 | AUC: 0.94047 | Best iteration: 496
2026-09-19 08:37:35 | DEBUG | Trial 3 | Starting Fold 2/5
2026-09-19 08:37:38 | DEBUG | Trial 3 | Fold 2/5 | AUC: 0.93964 | Best iteration: 503
2026-09-19 08:37:39 | DEBUG | Trial 3 | Starting Fold 3/5
2026-09-19 08:37:42 | DEBUG | Trial 3 | Fold 3/5 | AUC: 0.93990 | Best iteration: 496
2026-09-19 08:37:42 | DEBUG | Trial 3 | Starting Fold 4/5
2026-09-19 08:37:45 | DEBUG | Trial 3 | Fold 4/5 | AUC: 0.93993 | Best iteration: 503
2026-09-19 08:37:46 | DEBUG | Trial 3 | Starting Fold 5/5
2026-09-19 08:37:49 | DEBUG | Trial 3 | Fold 5/5 | AUC: 0.93818 | Best iteration: 501
2026-09-19 08:37:49 | INFO | Trial 3 finished | Mean AUC: 0.93962 | Std: 0.00077
2026-09-19 08:37:49 | DEBUG | Trial 4 | scale_pos_weight: 4.7257
2026-09-19 08:37:49 | DEBUG | Trial 4 | Starting Fold 1/5
2026-09-19 08:37:55 | DEBUG | Trial 4 | Fold 1/5 | AUC: 0.94182 | Best iteration: 341
2026-09-19 08:37:55 | DEBUG | Trial 4 | Starting Fold 2/5
2026-09-19 08:38:02 | DEBUG | Trial 4 | Fold 2/5 | AUC: 0.94103 | Best iteration: 367
2026-09-19 08:38:02 | DEBUG | Trial 4 | Starting Fold 3/5
2026-09-19 08:38:08 | DEBUG | Trial 4 | Fold 3/5 | AUC: 0.94115 | Best iteration: 352
2026-09-19 08:38:08 | DEBUG | Trial 4 | Starting Fold 4/5
2026-09-19 08:38:14 | DEBUG | Trial 4 | Fold 4/5 | AUC: 0.94130 | Best iteration: 363
```

2026-09-19 08:38:14	DEBUG	Trial 4	Starting Fold 5/5
2026-09-19 08:38:20	DEBUG	Trial 4	Fold 5/5 AUC: 0.93975 Best iteration: 352
2026-09-19 08:38:21	INFO	Trial 4	finished Mean AUC: 0.94101 Std: 0.00069
2026-09-19 08:38:21	DEBUG	Trial 5	scale_pos_weight: 4.7257
2026-09-19 08:38:21	DEBUG	Trial 5	Starting Fold 1/5
2026-09-19 08:38:29	DEBUG	Trial 5	Fold 1/5 AUC: 0.94207 Best iteration: 1329
2026-09-19 08:38:29	DEBUG	Trial 5	Starting Fold 2/5
2026-09-19 08:38:37	DEBUG	Trial 5	Fold 2/5 AUC: 0.94125 Best iteration: 1301
2026-09-19 08:38:37	DEBUG	Trial 5	Starting Fold 3/5
2026-09-19 08:38:45	DEBUG	Trial 5	Fold 3/5 AUC: 0.94146 Best iteration: 1302
2026-09-19 08:38:45	DEBUG	Trial 5	Starting Fold 4/5
2026-09-19 08:38:53	DEBUG	Trial 5	Fold 4/5 AUC: 0.94157 Best iteration: 1274
2026-09-19 08:38:53	DEBUG	Trial 5	Starting Fold 5/5
2026-09-19 08:39:01	DEBUG	Trial 5	Fold 5/5 AUC: 0.94004 Best iteration: 1322
2026-09-19 08:39:02	INFO	Trial 5	finished Mean AUC: 0.94128 Std: 0.00068
2026-09-19 08:39:02	DEBUG	Trial 6	scale_pos_weight: 4.7257
2026-09-19 08:39:02	DEBUG	Trial 6	Starting Fold 1/5
2026-09-19 08:39:11	DEBUG	Trial 6	Fold 1/5 AUC: 0.94145 Best iteration: 363
2026-09-19 08:39:11	DEBUG	Trial 6	Starting Fold 2/5
2026-09-19 08:39:21	DEBUG	Trial 6	Fold 2/5 AUC: 0.94057 Best iteration: 355
2026-09-19 08:39:21	DEBUG	Trial 6	Starting Fold 3/5
2026-09-19 08:39:30	DEBUG	Trial 6	Fold 3/5 AUC: 0.94070 Best iteration: 323
2026-09-19 08:39:30	DEBUG	Trial 6	Starting Fold 4/5
2026-09-19 08:39:39	DEBUG	Trial 6	Fold 4/5 AUC: 0.94096 Best iteration: 345
2026-09-19 08:39:40	DEBUG	Trial 6	Starting Fold 5/5
2026-09-19 08:39:51	DEBUG	Trial 6	Fold 5/5 AUC: 0.93930 Best iteration: 466
2026-09-19 08:39:51	INFO	Trial 6	finished Mean AUC: 0.94059 Std: 0.00071
2026-09-19 08:39:51	DEBUG	Trial 7	scale_pos_weight: 4.7257
2026-09-19 08:39:51	DEBUG	Trial 7	Starting Fold 1/5
2026-09-19 08:39:57	DEBUG	Trial 7	Fold 1/5 AUC: 0.94127 Best iteration: 972
2026-09-19 08:39:57	DEBUG	Trial 7	Starting Fold 2/5
2026-09-19 08:40:04	DEBUG	Trial 7	Fold 2/5 AUC: 0.94037 Best iteration: 972
2026-09-19 08:40:04	DEBUG	Trial 7	Starting Fold 3/5
2026-09-19 08:40:10	DEBUG	Trial 7	Fold 3/5 AUC: 0.94059 Best iteration: 972
2026-09-19 08:40:10	DEBUG	Trial 7	Starting Fold 4/5
2026-09-19 08:40:17	DEBUG	Trial 7	Fold 4/5 AUC: 0.94073 Best iteration: 972
2026-09-19 08:40:17	DEBUG	Trial 7	Starting Fold 5/5
2026-09-19 08:40:23	DEBUG	Trial 7	Fold 5/5 AUC: 0.93904 Best iteration: 972
2026-09-19 08:40:23	INFO	Trial 7	finished Mean AUC: 0.94040 Std: 0.00074
2026-09-19 08:40:23	DEBUG	Trial 8	scale_pos_weight: 4.7257
2026-09-19 08:40:24	DEBUG	Trial 8	Starting Fold 1/5
2026-09-19 08:40:41	DEBUG	Trial 8	Fold 1/5 AUC: 0.94134 Best iteration: 669
2026-09-19 08:40:41	DEBUG	Trial 8	Starting Fold 2/5
2026-09-19 08:41:00	DEBUG	Trial 8	Fold 2/5 AUC: 0.94035 Best iteration: 747
2026-09-19 08:41:00	DEBUG	Trial 8	Starting Fold 3/5
2026-09-19 08:41:18	DEBUG	Trial 8	Fold 3/5 AUC: 0.94064 Best iteration: 700
2026-09-19 08:41:18	DEBUG	Trial 8	Starting Fold 4/5
2026-09-19 08:41:37	DEBUG	Trial 8	Fold 4/5 AUC: 0.94084 Best iteration: 710
2026-09-19 08:41:37	DEBUG	Trial 8	Starting Fold 5/5
2026-09-19 08:41:56	DEBUG	Trial 8	Fold 5/5 AUC: 0.93913 Best iteration: 756
2026-09-19 08:41:56	INFO	Trial 8	finished Mean AUC: 0.94046 Std: 0.00074
2026-09-19 08:41:56	DEBUG	Trial 9	scale_pos_weight: 4.7257
2026-09-19 08:41:56	DEBUG	Trial 9	Starting Fold 1/5
2026-09-19 08:41:59	DEBUG	Trial 9	Fold 1/5 AUC: 0.94203 Best iteration: 503
2026-09-19 08:41:59	DEBUG	Trial 9	Starting Fold 2/5
2026-09-19 08:42:02	DEBUG	Trial 9	Fold 2/5 AUC: 0.94130 Best iteration: 500
2026-09-19 08:42:02	DEBUG	Trial 9	Starting Fold 3/5
2026-09-19 08:42:06	DEBUG	Trial 9	Fold 3/5 AUC: 0.94145 Best iteration: 494
2026-09-19 08:42:06	DEBUG	Trial 9	Starting Fold 4/5
2026-09-19 08:42:09	DEBUG	Trial 9	Fold 4/5 AUC: 0.94157 Best iteration: 496

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2026-09-19 08:42:09 | DEBUG | Trial 9 | Starting Fold 5/5
2026-09-19 08:42:12 | DEBUG | Trial 9 | Fold 5/5 | AUC: 0.94004 | Best iteration: 503
2026-09-19 08:42:12 | INFO | Trial 9 finished | Mean AUC: 0.94128 | Std: 0.00067
2026-09-19 08:42:12 | DEBUG | Trial 10 | scale_pos_weight: 4.7257
2026-09-19 08:42:12 | DEBUG | Trial 10 | Starting Fold 1/5
2026-09-19 08:42:18 | DEBUG | Trial 10 | Fold 1/5 | AUC: 0.94194 | Best iteration: 1685
2026-09-19 08:42:18 | DEBUG | Trial 10 | Starting Fold 2/5
2026-09-19 08:42:24 | DEBUG | Trial 10 | Fold 2/5 | AUC: 0.94112 | Best iteration: 1685
2026-09-19 08:42:24 | DEBUG | Trial 10 | Starting Fold 3/5
2026-09-19 08:42:29 | DEBUG | Trial 10 | Fold 3/5 | AUC: 0.94120 | Best iteration: 1689
2026-09-19 08:42:30 | DEBUG | Trial 10 | Starting Fold 4/5
2026-09-19 08:42:35 | DEBUG | Trial 10 | Fold 4/5 | AUC: 0.94135 | Best iteration: 1676
2026-09-19 08:42:35 | DEBUG | Trial 10 | Starting Fold 5/5
2026-09-19 08:42:41 | DEBUG | Trial 10 | Fold 5/5 | AUC: 0.93974 | Best iteration: 1682
2026-09-19 08:42:41 | INFO | Trial 10 finished | Mean AUC: 0.94107 | Std: 0.00072
2026-09-19 08:42:41 | DEBUG | Trial 11 | scale_pos_weight: 4.7257
2026-09-19 08:42:41 | DEBUG | Trial 11 | Starting Fold 1/5
2026-09-19 08:42:45 | DEBUG | Trial 11 | Fold 1/5 | AUC: 0.94210 | Best iteration: 743
2026-09-19 08:42:45 | DEBUG | Trial 11 | Starting Fold 2/5
2026-09-19 08:42:48 | DEBUG | Trial 11 | Fold 2/5 | AUC: 0.94129 | Best iteration: 741
2026-09-19 08:42:49 | DEBUG | Trial 11 | Starting Fold 3/5
2026-09-19 08:42:52 | DEBUG | Trial 11 | Fold 3/5 | AUC: 0.94144 | Best iteration: 743
2026-09-19 08:42:52 | DEBUG | Trial 11 | Starting Fold 4/5
2026-09-19 08:42:56 | DEBUG | Trial 11 | Fold 4/5 | AUC: 0.94152 | Best iteration: 743
2026-09-19 08:42:56 | DEBUG | Trial 11 | Starting Fold 5/5
2026-09-19 08:43:00 | DEBUG | Trial 11 | Fold 5/5 | AUC: 0.94005 | Best iteration: 743
2026-09-19 08:43:00 | INFO | Trial 11 finished | Mean AUC: 0.94128 | Std: 0.00067
2026-09-19 08:43:00 | DEBUG | Trial 12 | scale_pos_weight: 4.7257
2026-09-19 08:43:00 | DEBUG | Trial 12 | Starting Fold 1/5
2026-09-19 08:43:04 | DEBUG | Trial 12 | Fold 1/5 | AUC: 0.94200 | Best iteration: 830
2026-09-19 08:43:04 | DEBUG | Trial 12 | Starting Fold 2/5
2026-09-19 08:43:08 | DEBUG | Trial 12 | Fold 2/5 | AUC: 0.94121 | Best iteration: 829
2026-09-19 08:43:08 | DEBUG | Trial 12 | Starting Fold 3/5
2026-09-19 08:43:12 | DEBUG | Trial 12 | Fold 3/5 | AUC: 0.94133 | Best iteration: 825
2026-09-19 08:43:12 | DEBUG | Trial 12 | Starting Fold 4/5
2026-09-19 08:43:16 | DEBUG | Trial 12 | Fold 4/5 | AUC: 0.94143 | Best iteration: 830
2026-09-19 08:43:17 | DEBUG | Trial 12 | Starting Fold 5/5
2026-09-19 08:43:20 | DEBUG | Trial 12 | Fold 5/5 | AUC: 0.93988 | Best iteration: 830
2026-09-19 08:43:21 | INFO | Trial 12 finished | Mean AUC: 0.94117 | Std: 0.00070
2026-09-19 08:43:21 | DEBUG | Trial 13 | scale_pos_weight: 4.7257
2026-09-19 08:43:21 | DEBUG | Trial 13 | Starting Fold 1/5
2026-09-19 08:43:26 | DEBUG | Trial 13 | Fold 1/5 | AUC: 0.94185 | Best iteration: 364
2026-09-19 08:43:26 | DEBUG | Trial 13 | Starting Fold 2/5
2026-09-19 08:43:31 | DEBUG | Trial 13 | Fold 2/5 | AUC: 0.94103 | Best iteration: 373
2026-09-19 08:43:31 | DEBUG | Trial 13 | Starting Fold 3/5
2026-09-19 08:43:36 | DEBUG | Trial 13 | Fold 3/5 | AUC: 0.94123 | Best iteration: 313
2026-09-19 08:43:36 | DEBUG | Trial 13 | Starting Fold 4/5
2026-09-19 08:43:42 | DEBUG | Trial 13 | Fold 4/5 | AUC: 0.94141 | Best iteration: 379
2026-09-19 08:43:42 | DEBUG | Trial 13 | Starting Fold 5/5
2026-09-19 08:43:47 | DEBUG | Trial 13 | Fold 5/5 | AUC: 0.93983 | Best iteration: 402
2026-09-19 08:43:47 | INFO | Trial 13 finished | Mean AUC: 0.94107 | Std: 0.00068
2026-09-19 08:43:47 | DEBUG | Trial 14 | scale_pos_weight: 4.7257
2026-09-19 08:43:47 | DEBUG | Trial 14 | Starting Fold 1/5
2026-09-19 08:43:55 | DEBUG | Trial 14 | Fold 1/5 | AUC: 0.94212 | Best iteration: 1907
2026-09-19 08:43:55 | DEBUG | Trial 14 | Starting Fold 2/5
2026-09-19 08:44:02 | DEBUG | Trial 14 | Fold 2/5 | AUC: 0.94136 | Best iteration: 1887
2026-09-19 08:44:03 | DEBUG | Trial 14 | Starting Fold 3/5
2026-09-19 08:44:10 | DEBUG | Trial 14 | Fold 3/5 | AUC: 0.94145 | Best iteration: 1955
2026-09-19 08:44:10 | DEBUG | Trial 14 | Starting Fold 4/5
2026-09-19 08:44:18 | DEBUG | Trial 14 | Fold 4/5 | AUC: 0.94156 | Best iteration: 1935
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2026-09-19 08:44:18	DEBUG	Trial 14	Starting Fold 5/5
2026-09-19 08:44:25	DEBUG	Trial 14	Fold 5/5 AUC: 0.94006 Best iteration: 1953
2026-09-19 08:44:25	INFO	Trial 14	finished Mean AUC: 0.94131 Std: 0.00068
2026-09-19 08:44:25	DEBUG	Trial 15	scale_pos_weight: 4.7257
2026-09-19 08:44:25	DEBUG	Trial 15	Starting Fold 1/5
2026-09-19 08:44:33	DEBUG	Trial 15	Fold 1/5 AUC: 0.94197 Best iteration: 1957
2026-09-19 08:44:33	DEBUG	Trial 15	Starting Fold 2/5
2026-09-19 08:44:40	DEBUG	Trial 15	Fold 2/5 AUC: 0.94111 Best iteration: 1956
2026-09-19 08:44:40	DEBUG	Trial 15	Starting Fold 3/5
2026-09-19 08:44:47	DEBUG	Trial 15	Fold 3/5 AUC: 0.94122 Best iteration: 1963
2026-09-19 08:44:47	DEBUG	Trial 15	Starting Fold 4/5
2026-09-19 08:44:54	DEBUG	Trial 15	Fold 4/5 AUC: 0.94140 Best iteration: 1972
2026-09-19 08:44:54	DEBUG	Trial 15	Starting Fold 5/5
2026-09-19 08:45:02	DEBUG	Trial 15	Fold 5/5 AUC: 0.93981 Best iteration: 1969
2026-09-19 08:45:02	INFO	Trial 15	finished Mean AUC: 0.94110 Std: 0.00071
2026-09-19 08:45:02	DEBUG	Trial 16	scale_pos_weight: 4.7257
2026-09-19 08:45:02	DEBUG	Trial 16	Starting Fold 1/5
2026-09-19 08:45:13	DEBUG	Trial 16	Fold 1/5 AUC: 0.94202 Best iteration: 1536
2026-09-19 08:45:13	DEBUG	Trial 16	Starting Fold 2/5
2026-09-19 08:45:24	DEBUG	Trial 16	Fold 2/5 AUC: 0.94118 Best iteration: 1303
2026-09-19 08:45:24	DEBUG	Trial 16	Starting Fold 3/5
2026-09-19 08:45:34	DEBUG	Trial 16	Fold 3/5 AUC: 0.94140 Best iteration: 1328
2026-09-19 08:45:34	DEBUG	Trial 16	Starting Fold 4/5
2026-09-19 08:45:46	DEBUG	Trial 16	Fold 4/5 AUC: 0.94152 Best iteration: 1596
2026-09-19 08:45:46	DEBUG	Trial 16	Starting Fold 5/5
2026-09-19 08:45:57	DEBUG	Trial 16	Fold 5/5 AUC: 0.93998 Best iteration: 1486
2026-09-19 08:45:57	INFO	Trial 16	finished Mean AUC: 0.94122 Std: 0.00068
2026-09-19 08:45:57	DEBUG	Trial 17	scale_pos_weight: 4.7257
2026-09-19 08:45:57	DEBUG	Trial 17	Starting Fold 1/5
2026-09-19 08:46:05	DEBUG	Trial 17	Fold 1/5 AUC: 0.94212 Best iteration: 1406
2026-09-19 08:46:05	DEBUG	Trial 17	Starting Fold 2/5
2026-09-19 08:46:13	DEBUG	Trial 17	Fold 2/5 AUC: 0.94132 Best iteration: 1331
2026-09-19 08:46:13	DEBUG	Trial 17	Starting Fold 3/5
2026-09-19 08:46:21	DEBUG	Trial 17	Fold 3/5 AUC: 0.94149 Best iteration: 1502
2026-09-19 08:46:22	DEBUG	Trial 17	Starting Fold 4/5
2026-09-19 08:46:30	DEBUG	Trial 17	Fold 4/5 AUC: 0.94159 Best iteration: 1320
2026-09-19 08:46:30	DEBUG	Trial 17	Starting Fold 5/5
2026-09-19 08:46:38	DEBUG	Trial 17	Fold 5/5 AUC: 0.94014 Best iteration: 1555
2026-09-19 08:46:38	INFO	Trial 17	finished Mean AUC: 0.94133 Std: 0.00065
2026-09-19 08:46:38	DEBUG	Trial 18	scale_pos_weight: 4.7257
2026-09-19 08:46:38	DEBUG	Trial 18	Starting Fold 1/5
2026-09-19 08:46:44	DEBUG	Trial 18	Fold 1/5 AUC: 0.94193 Best iteration: 565
2026-09-19 08:46:45	DEBUG	Trial 18	Starting Fold 2/5
2026-09-19 08:46:51	DEBUG	Trial 18	Fold 2/5 AUC: 0.94107 Best iteration: 611
2026-09-19 08:46:51	DEBUG	Trial 18	Starting Fold 3/5
2026-09-19 08:46:58	DEBUG	Trial 18	Fold 3/5 AUC: 0.94131 Best iteration: 603
2026-09-19 08:46:58	DEBUG	Trial 18	Starting Fold 4/5
2026-09-19 08:47:04	DEBUG	Trial 18	Fold 4/5 AUC: 0.94140 Best iteration: 476
2026-09-19 08:47:04	DEBUG	Trial 18	Starting Fold 5/5
2026-09-19 08:47:11	DEBUG	Trial 18	Fold 5/5 AUC: 0.93985 Best iteration: 677
2026-09-19 08:47:12	INFO	Trial 18	finished Mean AUC: 0.94111 Std: 0.00069
2026-09-19 08:47:12	DEBUG	Trial 19	scale_pos_weight: 4.7257
2026-09-19 08:47:12	DEBUG	Trial 19	Starting Fold 1/5
2026-09-19 08:47:17	DEBUG	Trial 19	Fold 1/5 AUC: 0.94204 Best iteration: 693
2026-09-19 08:47:17	DEBUG	Trial 19	Starting Fold 2/5
2026-09-19 08:47:22	DEBUG	Trial 19	Fold 2/5 AUC: 0.94128 Best iteration: 638
2026-09-19 08:47:22	DEBUG	Trial 19	Starting Fold 3/5
2026-09-19 08:47:28	DEBUG	Trial 19	Fold 3/5 AUC: 0.94149 Best iteration: 1006
2026-09-19 08:47:29	DEBUG	Trial 19	Starting Fold 4/5
2026-09-19 08:47:34	DEBUG	Trial 19	Fold 4/5 AUC: 0.94157 Best iteration: 797

2026-09-19 08:47:34	DEBUG	Trial 19	Starting Fold 5/5
2026-09-19 08:47:40	DEBUG	Trial 19	Fold 5/5 AUC: 0.94008 Best iteration: 1073
2026-09-19 08:47:41	INFO	Trial 19	finished Mean AUC: 0.94129 Std: 0.00066
2026-09-19 08:47:41	DEBUG	Trial 20	scale_pos_weight: 4.7257
2026-09-19 08:47:41	DEBUG	Trial 20	Starting Fold 1/5
2026-09-19 08:47:51	DEBUG	Trial 20	Fold 1/5 AUC: 0.94176 Best iteration: 871
2026-09-19 08:47:51	DEBUG	Trial 20	Starting Fold 2/5
2026-09-19 08:48:01	DEBUG	Trial 20	Fold 2/5 AUC: 0.94095 Best iteration: 852
2026-09-19 08:48:01	DEBUG	Trial 20	Starting Fold 3/5
2026-09-19 08:48:12	DEBUG	Trial 20	Fold 3/5 AUC: 0.94117 Best iteration: 864
2026-09-19 08:48:12	DEBUG	Trial 20	Starting Fold 4/5
2026-09-19 08:48:21	DEBUG	Trial 20	Fold 4/5 AUC: 0.94132 Best iteration: 780
2026-09-19 08:48:21	DEBUG	Trial 20	Starting Fold 5/5
2026-09-19 08:48:32	DEBUG	Trial 20	Fold 5/5 AUC: 0.93969 Best iteration: 921
2026-09-19 08:48:32	INFO	Trial 20	finished Mean AUC: 0.94098 Std: 0.00070
2026-09-19 08:48:32	DEBUG	Trial 21	scale_pos_weight: 4.7257
2026-09-19 08:48:32	DEBUG	Trial 21	Starting Fold 1/5
2026-09-19 08:48:39	DEBUG	Trial 21	Fold 1/5 AUC: 0.94205 Best iteration: 1806
2026-09-19 08:48:39	DEBUG	Trial 21	Starting Fold 2/5
2026-09-19 08:48:46	DEBUG	Trial 21	Fold 2/5 AUC: 0.94127 Best iteration: 1847
2026-09-19 08:48:46	DEBUG	Trial 21	Starting Fold 3/5
2026-09-19 08:48:53	DEBUG	Trial 21	Fold 3/5 AUC: 0.94136 Best iteration: 1842
2026-09-19 08:48:54	DEBUG	Trial 21	Starting Fold 4/5
2026-09-19 08:49:01	DEBUG	Trial 21	Fold 4/5 AUC: 0.94145 Best iteration: 1848
2026-09-19 08:49:01	DEBUG	Trial 21	Starting Fold 5/5
2026-09-19 08:49:08	DEBUG	Trial 21	Fold 5/5 AUC: 0.93990 Best iteration: 1835
2026-09-19 08:49:08	INFO	Trial 21	finished Mean AUC: 0.94121 Std: 0.00071
2026-09-19 08:49:08	DEBUG	Trial 22	scale_pos_weight: 4.7257
2026-09-19 08:49:08	DEBUG	Trial 22	Starting Fold 1/5
2026-09-19 08:49:15	DEBUG	Trial 22	Fold 1/5 AUC: 0.94216 Best iteration: 1127
2026-09-19 08:49:15	DEBUG	Trial 22	Starting Fold 2/5
2026-09-19 08:49:22	DEBUG	Trial 22	Fold 2/5 AUC: 0.94130 Best iteration: 960
2026-09-19 08:49:22	DEBUG	Trial 22	Starting Fold 3/5
2026-09-19 08:49:29	DEBUG	Trial 22	Fold 3/5 AUC: 0.94152 Best iteration: 1122
2026-09-19 08:49:29	DEBUG	Trial 22	Starting Fold 4/5
2026-09-19 08:49:37	DEBUG	Trial 22	Fold 4/5 AUC: 0.94159 Best iteration: 1220
2026-09-19 08:49:37	DEBUG	Trial 22	Starting Fold 5/5
2026-09-19 08:49:45	DEBUG	Trial 22	Fold 5/5 AUC: 0.94014 Best iteration: 1294
2026-09-19 08:49:45	INFO	Trial 22	finished Mean AUC: 0.94134 Std: 0.00067
2026-09-19 08:49:45	DEBUG	Trial 23	scale_pos_weight: 4.7257
2026-09-19 08:49:45	DEBUG	Trial 23	Starting Fold 1/5
2026-09-19 08:49:51	DEBUG	Trial 23	Fold 1/5 AUC: 0.94202 Best iteration: 750
2026-09-19 08:49:51	DEBUG	Trial 23	Starting Fold 2/5
2026-09-19 08:49:57	DEBUG	Trial 23	Fold 2/5 AUC: 0.94130 Best iteration: 633
2026-09-19 08:49:57	DEBUG	Trial 23	Starting Fold 3/5
2026-09-19 08:50:03	DEBUG	Trial 23	Fold 3/5 AUC: 0.94149 Best iteration: 680
2026-09-19 08:50:03	DEBUG	Trial 23	Starting Fold 4/5
2026-09-19 08:50:09	DEBUG	Trial 23	Fold 4/5 AUC: 0.94154 Best iteration: 591
2026-09-19 08:50:09	DEBUG	Trial 23	Starting Fold 5/5
2026-09-19 08:50:15	DEBUG	Trial 23	Fold 5/5 AUC: 0.94004 Best iteration: 659
2026-09-19 08:50:15	INFO	Trial 23	finished Mean AUC: 0.94128 Std: 0.00067
2026-09-19 08:50:15	DEBUG	Trial 24	scale_pos_weight: 4.7257
2026-09-19 08:50:15	DEBUG	Trial 24	Starting Fold 1/5
2026-09-19 08:50:21	DEBUG	Trial 24	Fold 1/5 AUC: 0.94204 Best iteration: 681
2026-09-19 08:50:21	DEBUG	Trial 24	Starting Fold 2/5
2026-09-19 08:50:26	DEBUG	Trial 24	Fold 2/5 AUC: 0.94131 Best iteration: 635
2026-09-19 08:50:26	DEBUG	Trial 24	Starting Fold 3/5
2026-09-19 08:50:31	DEBUG	Trial 24	Fold 3/5 AUC: 0.94139 Best iteration: 634
2026-09-19 08:50:31	DEBUG	Trial 24	Starting Fold 4/5
2026-09-19 08:50:36	DEBUG	Trial 24	Fold 4/5 AUC: 0.94154 Best iteration: 604

2026-09-19 08:50:36	DEBUG	Trial 24 Starting Fold 5/5
2026-09-19 08:50:42	DEBUG	Trial 24 Fold 5/5 AUC: 0.94007 Best iteration: 851
2026-09-19 08:50:42	INFO	Trial 24 finished Mean AUC: 0.94127 Std: 0.00065
2026-09-19 08:50:42	DEBUG	Trial 25 scale_pos_weight: 4.7257
2026-09-19 08:50:42	DEBUG	Trial 25 Starting Fold 1/5
2026-09-19 08:50:49	DEBUG	Trial 25 Fold 1/5 AUC: 0.94202 Best iteration: 877
2026-09-19 08:50:49	DEBUG	Trial 25 Starting Fold 2/5
2026-09-19 08:50:55	DEBUG	Trial 25 Fold 2/5 AUC: 0.94119 Best iteration: 657
2026-09-19 08:50:55	DEBUG	Trial 25 Starting Fold 3/5
2026-09-19 08:51:02	DEBUG	Trial 25 Fold 3/5 AUC: 0.94141 Best iteration: 876
2026-09-19 08:51:03	DEBUG	Trial 25 Starting Fold 4/5
2026-09-19 08:51:09	DEBUG	Trial 25 Fold 4/5 AUC: 0.94151 Best iteration: 749
2026-09-19 08:51:09	DEBUG	Trial 25 Starting Fold 5/5
2026-09-19 08:51:16	DEBUG	Trial 25 Fold 5/5 AUC: 0.94005 Best iteration: 912
2026-09-19 08:51:17	INFO	Trial 25 finished Mean AUC: 0.94124 Std: 0.00065
2026-09-19 08:51:17	DEBUG	Trial 26 scale_pos_weight: 4.7257
2026-09-19 08:51:17	DEBUG	Trial 26 Starting Fold 1/5
2026-09-19 08:51:24	DEBUG	Trial 26 Fold 1/5 AUC: 0.94212 Best iteration: 1505
2026-09-19 08:51:24	DEBUG	Trial 26 Starting Fold 2/5
2026-09-19 08:51:31	DEBUG	Trial 26 Fold 2/5 AUC: 0.94131 Best iteration: 1448
2026-09-19 08:51:31	DEBUG	Trial 26 Starting Fold 3/5
2026-09-19 08:51:39	DEBUG	Trial 26 Fold 3/5 AUC: 0.94146 Best iteration: 1502
2026-09-19 08:51:39	DEBUG	Trial 26 Starting Fold 4/5
2026-09-19 08:51:46	DEBUG	Trial 26 Fold 4/5 AUC: 0.94158 Best iteration: 1452
2026-09-19 08:51:46	DEBUG	Trial 26 Starting Fold 5/5
2026-09-19 08:51:53	DEBUG	Trial 26 Fold 5/5 AUC: 0.94009 Best iteration: 1505
2026-09-19 08:51:54	INFO	Trial 26 finished Mean AUC: 0.94131 Std: 0.00067
2026-09-19 08:51:54	DEBUG	Trial 27 scale_pos_weight: 4.7257
2026-09-19 08:51:54	DEBUG	Trial 27 Starting Fold 1/5
2026-09-19 08:51:59	DEBUG	Trial 27 Fold 1/5 AUC: 0.94194 Best iteration: 357
2026-09-19 08:51:59	DEBUG	Trial 27 Starting Fold 2/5
2026-09-19 08:52:04	DEBUG	Trial 27 Fold 2/5 AUC: 0.94112 Best iteration: 330
2026-09-19 08:52:04	DEBUG	Trial 27 Starting Fold 3/5
2026-09-19 08:52:09	DEBUG	Trial 27 Fold 3/5 AUC: 0.94125 Best iteration: 347
2026-09-19 08:52:09	DEBUG	Trial 27 Starting Fold 4/5
2026-09-19 08:52:14	DEBUG	Trial 27 Fold 4/5 AUC: 0.94139 Best iteration: 370
2026-09-19 08:52:14	DEBUG	Trial 27 Starting Fold 5/5
2026-09-19 08:52:20	DEBUG	Trial 27 Fold 5/5 AUC: 0.93982 Best iteration: 404
2026-09-19 08:52:20	INFO	Trial 27 finished Mean AUC: 0.94110 Std: 0.00070
2026-09-19 08:52:20	DEBUG	Trial 28 scale_pos_weight: 4.7257
2026-09-19 08:52:20	DEBUG	Trial 28 Starting Fold 1/5
2026-09-19 08:52:24	DEBUG	Trial 28 Fold 1/5 AUC: 0.94189 Best iteration: 1074
2026-09-19 08:52:25	DEBUG	Trial 28 Starting Fold 2/5
2026-09-19 08:52:29	DEBUG	Trial 28 Fold 2/5 AUC: 0.94106 Best iteration: 1075
2026-09-19 08:52:29	DEBUG	Trial 28 Starting Fold 3/5
2026-09-19 08:52:34	DEBUG	Trial 28 Fold 3/5 AUC: 0.94121 Best iteration: 1056
2026-09-19 08:52:34	DEBUG	Trial 28 Starting Fold 4/5
2026-09-19 08:52:38	DEBUG	Trial 28 Fold 4/5 AUC: 0.94133 Best iteration: 1075
2026-09-19 08:52:39	DEBUG	Trial 28 Starting Fold 5/5
2026-09-19 08:52:43	DEBUG	Trial 28 Fold 5/5 AUC: 0.93970 Best iteration: 1053
2026-09-19 08:52:43	INFO	Trial 28 finished Mean AUC: 0.94104 Std: 0.00072
2026-09-19 08:52:43	DEBUG	Trial 29 scale_pos_weight: 4.7257
2026-09-19 08:52:43	DEBUG	Trial 29 Starting Fold 1/5
2026-09-19 08:52:51	DEBUG	Trial 29 Fold 1/5 AUC: 0.94173 Best iteration: 577
2026-09-19 08:52:51	DEBUG	Trial 29 Starting Fold 2/5
2026-09-19 08:52:59	DEBUG	Trial 29 Fold 2/5 AUC: 0.94090 Best iteration: 648
2026-09-19 08:52:59	DEBUG	Trial 29 Starting Fold 3/5
2026-09-19 08:53:07	DEBUG	Trial 29 Fold 3/5 AUC: 0.94117 Best iteration: 648
2026-09-19 08:53:07	DEBUG	Trial 29 Starting Fold 4/5
2026-09-19 08:53:14	DEBUG	Trial 29 Fold 4/5 AUC: 0.94124 Best iteration: 643

2026-09-19 08:53:15	DEBUG	Trial 29 Starting Fold 5/5
2026-09-19 08:53:22	DEBUG	Trial 29 Fold 5/5 AUC: 0.93967 Best iteration: 684
2026-09-19 08:53:23	INFO	Trial 29 finished Mean AUC: 0.94094 Std: 0.00069
2026-09-19 08:53:23	DEBUG	Trial 30 scale_pos_weight: 4.7257
2026-09-19 08:53:23	DEBUG	Trial 30 Starting Fold 1/5
2026-09-19 08:53:30	DEBUG	Trial 30 Fold 1/5 AUC: 0.94202 Best iteration: 1175
2026-09-19 08:53:30	DEBUG	Trial 30 Starting Fold 2/5
2026-09-19 08:53:38	DEBUG	Trial 30 Fold 2/5 AUC: 0.94120 Best iteration: 1121
2026-09-19 08:53:38	DEBUG	Trial 30 Starting Fold 3/5
2026-09-19 08:53:45	DEBUG	Trial 30 Fold 3/5 AUC: 0.94142 Best iteration: 1214
2026-09-19 08:53:45	DEBUG	Trial 30 Starting Fold 4/5
2026-09-19 08:53:53	DEBUG	Trial 30 Fold 4/5 AUC: 0.94147 Best iteration: 1081
2026-09-19 08:53:53	DEBUG	Trial 30 Starting Fold 5/5
2026-09-19 08:54:00	DEBUG	Trial 30 Fold 5/5 AUC: 0.93998 Best iteration: 1206
2026-09-19 08:54:00	INFO	Trial 30 finished Mean AUC: 0.94122 Std: 0.00067
2026-09-19 08:54:00	DEBUG	Trial 31 scale_pos_weight: 4.7257
2026-09-19 08:54:00	DEBUG	Trial 31 Starting Fold 1/5
2026-09-19 08:54:08	DEBUG	Trial 31 Fold 1/5 AUC: 0.94207 Best iteration: 1517
2026-09-19 08:54:08	DEBUG	Trial 31 Starting Fold 2/5
2026-09-19 08:54:15	DEBUG	Trial 31 Fold 2/5 AUC: 0.94128 Best iteration: 1574
2026-09-19 08:54:15	DEBUG	Trial 31 Starting Fold 3/5
2026-09-19 08:54:22	DEBUG	Trial 31 Fold 3/5 AUC: 0.94139 Best iteration: 1603
2026-09-19 08:54:23	DEBUG	Trial 31 Starting Fold 4/5
2026-09-19 08:54:30	DEBUG	Trial 31 Fold 4/5 AUC: 0.94156 Best iteration: 1590
2026-09-19 08:54:30	DEBUG	Trial 31 Starting Fold 5/5
2026-09-19 08:54:37	DEBUG	Trial 31 Fold 5/5 AUC: 0.93999 Best iteration: 1594
2026-09-19 08:54:37	INFO	Trial 31 finished Mean AUC: 0.94126 Std: 0.00069
2026-09-19 08:54:37	DEBUG	Trial 32 scale_pos_weight: 4.7257
2026-09-19 08:54:37	DEBUG	Trial 32 Starting Fold 1/5
2026-09-19 08:54:43	DEBUG	Trial 32 Fold 1/5 AUC: 0.94212 Best iteration: 874
2026-09-19 08:54:43	DEBUG	Trial 32 Starting Fold 2/5
2026-09-19 08:54:50	DEBUG	Trial 32 Fold 2/5 AUC: 0.94132 Best iteration: 916
2026-09-19 08:54:50	DEBUG	Trial 32 Starting Fold 3/5
2026-09-19 08:54:57	DEBUG	Trial 32 Fold 3/5 AUC: 0.94146 Best iteration: 1285
2026-09-19 08:54:57	DEBUG	Trial 32 Starting Fold 4/5
2026-09-19 08:55:03	DEBUG	Trial 32 Fold 4/5 AUC: 0.94161 Best iteration: 908
2026-09-19 08:55:04	DEBUG	Trial 32 Starting Fold 5/5
2026-09-19 08:55:10	DEBUG	Trial 32 Fold 5/5 AUC: 0.94013 Best iteration: 1054
2026-09-19 08:55:10	INFO	Trial 32 finished Mean AUC: 0.94133 Std: 0.00066
2026-09-19 08:55:10	DEBUG	Trial 33 scale_pos_weight: 4.7257
2026-09-19 08:55:10	DEBUG	Trial 33 Starting Fold 1/5
2026-09-19 08:55:16	DEBUG	Trial 33 Fold 1/5 AUC: 0.94217 Best iteration: 1371
2026-09-19 08:55:17	DEBUG	Trial 33 Starting Fold 2/5
2026-09-19 08:55:22	DEBUG	Trial 33 Fold 2/5 AUC: 0.94139 Best iteration: 1381
2026-09-19 08:55:23	DEBUG	Trial 33 Starting Fold 3/5
2026-09-19 08:55:29	DEBUG	Trial 33 Fold 3/5 AUC: 0.94151 Best iteration: 1377
2026-09-19 08:55:29	DEBUG	Trial 33 Starting Fold 4/5
2026-09-19 08:55:35	DEBUG	Trial 33 Fold 4/5 AUC: 0.94158 Best iteration: 1379
2026-09-19 08:55:35	DEBUG	Trial 33 Starting Fold 5/5
2026-09-19 08:55:41	DEBUG	Trial 33 Fold 5/5 AUC: 0.94020 Best iteration: 1390
2026-09-19 08:55:41	INFO	Trial 33 finished Mean AUC: 0.94137 Std: 0.00064
2026-09-19 08:55:41	DEBUG	Trial 34 scale_pos_weight: 4.7257
2026-09-19 08:55:41	DEBUG	Trial 34 Starting Fold 1/5
2026-09-19 08:55:47	DEBUG	Trial 34 Fold 1/5 AUC: 0.94215 Best iteration: 1409
2026-09-19 08:55:47	DEBUG	Trial 34 Starting Fold 2/5
2026-09-19 08:55:53	DEBUG	Trial 34 Fold 2/5 AUC: 0.94139 Best iteration: 1372
2026-09-19 08:55:53	DEBUG	Trial 34 Starting Fold 3/5
2026-09-19 08:55:59	DEBUG	Trial 34 Fold 3/5 AUC: 0.94151 Best iteration: 1404
2026-09-19 08:56:00	DEBUG	Trial 34 Starting Fold 4/5
2026-09-19 08:56:06	DEBUG	Trial 34 Fold 4/5 AUC: 0.94163 Best iteration: 1385

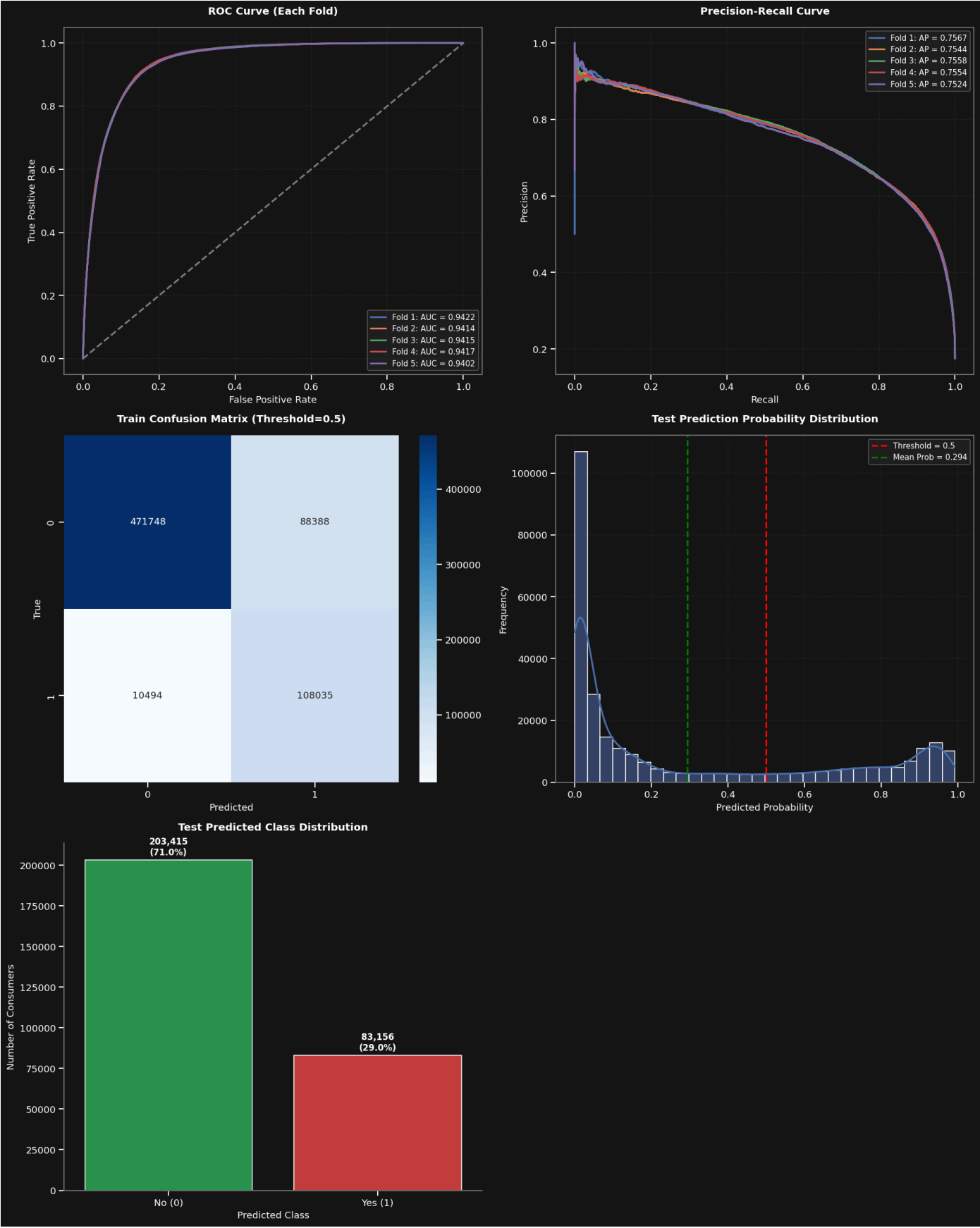
2026-09-19 08:56:06	DEBUG	Trial 34 Starting Fold 5/5
2026-09-19 08:56:12	DEBUG	Trial 34 Fold 5/5 AUC: 0.94016 Best iteration: 1403
2026-09-19 08:56:12	INFO	Trial 34 finished Mean AUC: 0.94137 Std: 0.00066
2026-09-19 08:56:12	DEBUG	Trial 35 scale_pos_weight: 4.7257
2026-09-19 08:56:12	DEBUG	Trial 35 Starting Fold 1/5
2026-09-19 08:56:18	DEBUG	Trial 35 Fold 1/5 AUC: 0.94211 Best iteration: 1405
2026-09-19 08:56:18	DEBUG	Trial 35 Starting Fold 2/5
2026-09-19 08:56:24	DEBUG	Trial 35 Fold 2/5 AUC: 0.94134 Best iteration: 1409
2026-09-19 08:56:24	DEBUG	Trial 35 Starting Fold 3/5
2026-09-19 08:56:30	DEBUG	Trial 35 Fold 3/5 AUC: 0.94148 Best iteration: 1410
2026-09-19 08:56:30	DEBUG	Trial 35 Starting Fold 4/5
2026-09-19 08:56:36	DEBUG	Trial 35 Fold 4/5 AUC: 0.94153 Best iteration: 1411
2026-09-19 08:56:37	DEBUG	Trial 35 Starting Fold 5/5
2026-09-19 08:56:43	DEBUG	Trial 35 Fold 5/5 AUC: 0.94009 Best iteration: 1409
2026-09-19 08:56:43	INFO	Trial 35 finished Mean AUC: 0.94131 Std: 0.00067
2026-09-19 08:56:43	DEBUG	Trial 36 scale_pos_weight: 4.7257
2026-09-19 08:56:43	DEBUG	Trial 36 Starting Fold 1/5
2026-09-19 08:56:48	DEBUG	Trial 36 Fold 1/5 AUC: 0.94206 Best iteration: 1373
2026-09-19 08:56:49	DEBUG	Trial 36 Starting Fold 2/5
2026-09-19 08:56:54	DEBUG	Trial 36 Fold 2/5 AUC: 0.94127 Best iteration: 1374
2026-09-19 08:56:54	DEBUG	Trial 36 Starting Fold 3/5
2026-09-19 08:57:00	DEBUG	Trial 36 Fold 3/5 AUC: 0.94137 Best iteration: 1373
2026-09-19 08:57:00	DEBUG	Trial 36 Starting Fold 4/5
2026-09-19 08:57:05	DEBUG	Trial 36 Fold 4/5 AUC: 0.94151 Best iteration: 1375
2026-09-19 08:57:05	DEBUG	Trial 36 Starting Fold 5/5
2026-09-19 08:57:11	DEBUG	Trial 36 Fold 5/5 AUC: 0.93997 Best iteration: 1373
2026-09-19 08:57:11	INFO	Trial 36 finished Mean AUC: 0.94123 Std: 0.00069
2026-09-19 08:57:11	DEBUG	Trial 37 scale_pos_weight: 4.7257
2026-09-19 08:57:11	DEBUG	Trial 37 Starting Fold 1/5
2026-09-19 08:57:16	DEBUG	Trial 37 Fold 1/5 AUC: 0.94221 Best iteration: 1279
2026-09-19 08:57:17	DEBUG	Trial 37 Starting Fold 2/5
2026-09-19 08:57:22	DEBUG	Trial 37 Fold 2/5 AUC: 0.94145 Best iteration: 1280
2026-09-19 08:57:22	DEBUG	Trial 37 Starting Fold 3/5
2026-09-19 08:57:28	DEBUG	Trial 37 Fold 3/5 AUC: 0.94152 Best iteration: 1133
2026-09-19 08:57:28	DEBUG	Trial 37 Starting Fold 4/5
2026-09-19 08:57:34	DEBUG	Trial 37 Fold 4/5 AUC: 0.94167 Best iteration: 1277
2026-09-19 08:57:34	DEBUG	Trial 37 Starting Fold 5/5
2026-09-19 08:57:39	DEBUG	Trial 37 Fold 5/5 AUC: 0.94022 Best iteration: 1280
2026-09-19 08:57:39	INFO	Trial 37 finished Mean AUC: 0.94141 Std: 0.00065
2026-09-19 08:57:39	DEBUG	Trial 38 scale_pos_weight: 4.7257
2026-09-19 08:57:39	DEBUG	Trial 38 Starting Fold 1/5
2026-09-19 08:57:44	DEBUG	Trial 38 Fold 1/5 AUC: 0.94218 Best iteration: 1060
2026-09-19 08:57:44	DEBUG	Trial 38 Starting Fold 2/5
2026-09-19 08:57:49	DEBUG	Trial 38 Fold 2/5 AUC: 0.94141 Best iteration: 1063
2026-09-19 08:57:49	DEBUG	Trial 38 Starting Fold 3/5
2026-09-19 08:57:54	DEBUG	Trial 38 Fold 3/5 AUC: 0.94150 Best iteration: 1063
2026-09-19 08:57:54	DEBUG	Trial 38 Starting Fold 4/5
2026-09-19 08:57:59	DEBUG	Trial 38 Fold 4/5 AUC: 0.94160 Best iteration: 1062
2026-09-19 08:57:59	DEBUG	Trial 38 Starting Fold 5/5
2026-09-19 08:58:04	DEBUG	Trial 38 Fold 5/5 AUC: 0.94018 Best iteration: 1063
2026-09-19 08:58:04	INFO	Trial 38 finished Mean AUC: 0.94138 Std: 0.00066
2026-09-19 08:58:04	DEBUG	Trial 39 scale_pos_weight: 4.7257
2026-09-19 08:58:04	DEBUG	Trial 39 Starting Fold 1/5
2026-09-19 08:58:09	DEBUG	Trial 39 Fold 1/5 AUC: 0.94214 Best iteration: 1052
2026-09-19 08:58:09	DEBUG	Trial 39 Starting Fold 2/5
2026-09-19 08:58:14	DEBUG	Trial 39 Fold 2/5 AUC: 0.94138 Best iteration: 1035
2026-09-19 08:58:14	DEBUG	Trial 39 Starting Fold 3/5
2026-09-19 08:58:19	DEBUG	Trial 39 Fold 3/5 AUC: 0.94154 Best iteration: 1037
2026-09-19 08:58:19	DEBUG	Trial 39 Starting Fold 4/5
2026-09-19 08:58:24	DEBUG	Trial 39 Fold 4/5 AUC: 0.94157 Best iteration: 1031

2026-09-19 08:58:24	DEBUG	Trial 39 Starting Fold 5/5
2026-09-19 08:58:29	DEBUG	Trial 39 Fold 5/5 AUC: 0.94011 Best iteration: 1052
2026-09-19 08:58:29	INFO	Trial 39 finished Mean AUC: 0.94135 Std: 0.00067
2026-09-19 08:58:29	DEBUG	Trial 40 scale_pos_weight: 4.7257
2026-09-19 08:58:29	DEBUG	Trial 40 Starting Fold 1/5
2026-09-19 08:58:35	DEBUG	Trial 40 Fold 1/5 AUC: 0.94218 Best iteration: 1274
2026-09-19 08:58:35	DEBUG	Trial 40 Starting Fold 2/5
2026-09-19 08:58:40	DEBUG	Trial 40 Fold 2/5 AUC: 0.94142 Best iteration: 1266
2026-09-19 08:58:40	DEBUG	Trial 40 Starting Fold 3/5
2026-09-19 08:58:46	DEBUG	Trial 40 Fold 3/5 AUC: 0.94154 Best iteration: 1283
2026-09-19 08:58:46	DEBUG	Trial 40 Starting Fold 4/5
2026-09-19 08:58:52	DEBUG	Trial 40 Fold 4/5 AUC: 0.94164 Best iteration: 1241
2026-09-19 08:58:52	DEBUG	Trial 40 Starting Fold 5/5
2026-09-19 08:58:58	DEBUG	Trial 40 Fold 5/5 AUC: 0.94017 Best iteration: 1283
2026-09-19 08:58:58	INFO	Trial 40 finished Mean AUC: 0.94139 Std: 0.00066
2026-09-19 08:58:58	DEBUG	Trial 41 scale_pos_weight: 4.7257
2026-09-19 08:58:58	DEBUG	Trial 41 Starting Fold 1/5
2026-09-19 08:59:03	DEBUG	Trial 41 Fold 1/5 AUC: 0.94221 Best iteration: 1213
2026-09-19 08:59:04	DEBUG	Trial 41 Starting Fold 2/5
2026-09-19 08:59:09	DEBUG	Trial 41 Fold 2/5 AUC: 0.94144 Best iteration: 1290
2026-09-19 08:59:09	DEBUG	Trial 41 Starting Fold 3/5
2026-09-19 08:59:15	DEBUG	Trial 41 Fold 3/5 AUC: 0.94156 Best iteration: 1184
2026-09-19 08:59:15	DEBUG	Trial 41 Starting Fold 4/5
2026-09-19 08:59:21	DEBUG	Trial 41 Fold 4/5 AUC: 0.94163 Best iteration: 1282
2026-09-19 08:59:21	DEBUG	Trial 41 Starting Fold 5/5
2026-09-19 08:59:26	DEBUG	Trial 41 Fold 5/5 AUC: 0.94019 Best iteration: 1289
2026-09-19 08:59:27	INFO	Trial 41 finished Mean AUC: 0.94141 Std: 0.00066
2026-09-19 08:59:27	DEBUG	Trial 42 scale_pos_weight: 4.7257
2026-09-19 08:59:27	DEBUG	Trial 42 Starting Fold 1/5
2026-09-19 08:59:32	DEBUG	Trial 42 Fold 1/5 AUC: 0.94219 Best iteration: 1120
2026-09-19 08:59:32	DEBUG	Trial 42 Starting Fold 2/5
2026-09-19 08:59:38	DEBUG	Trial 42 Fold 2/5 AUC: 0.94146 Best iteration: 1236
2026-09-19 08:59:38	DEBUG	Trial 42 Starting Fold 3/5
2026-09-19 08:59:44	DEBUG	Trial 42 Fold 3/5 AUC: 0.94156 Best iteration: 1256
2026-09-19 08:59:44	DEBUG	Trial 42 Starting Fold 4/5
2026-09-19 08:59:49	DEBUG	Trial 42 Fold 4/5 AUC: 0.94164 Best iteration: 1268
2026-09-19 08:59:49	DEBUG	Trial 42 Starting Fold 5/5
2026-09-19 08:59:55	DEBUG	Trial 42 Fold 5/5 AUC: 0.94020 Best iteration: 1238
2026-09-19 08:59:55	INFO	Trial 42 finished Mean AUC: 0.94141 Std: 0.00066
2026-09-19 08:59:55	DEBUG	Trial 43 scale_pos_weight: 4.7257
2026-09-19 08:59:55	DEBUG	Trial 43 Starting Fold 1/5
2026-09-19 09:00:01	DEBUG	Trial 43 Fold 1/5 AUC: 0.94220 Best iteration: 1263
2026-09-19 09:00:01	DEBUG	Trial 43 Starting Fold 2/5
2026-09-19 09:00:07	DEBUG	Trial 43 Fold 2/5 AUC: 0.94146 Best iteration: 1193
2026-09-19 09:00:07	DEBUG	Trial 43 Starting Fold 3/5
2026-09-19 09:00:12	DEBUG	Trial 43 Fold 3/5 AUC: 0.94156 Best iteration: 1260
2026-09-19 09:00:12	DEBUG	Trial 43 Starting Fold 4/5
2026-09-19 09:00:18	DEBUG	Trial 43 Fold 4/5 AUC: 0.94163 Best iteration: 1279
2026-09-19 09:00:18	DEBUG	Trial 43 Starting Fold 5/5
2026-09-19 09:00:24	DEBUG	Trial 43 Fold 5/5 AUC: 0.94017 Best iteration: 1278
2026-09-19 09:00:24	INFO	Trial 43 finished Mean AUC: 0.94141 Std: 0.00067
2026-09-19 09:00:24	DEBUG	Trial 44 scale_pos_weight: 4.7257
2026-09-19 09:00:24	DEBUG	Trial 44 Starting Fold 1/5
2026-09-19 09:00:30	DEBUG	Trial 44 Fold 1/5 AUC: 0.94218 Best iteration: 1213
2026-09-19 09:00:30	DEBUG	Trial 44 Starting Fold 2/5
2026-09-19 09:00:35	DEBUG	Trial 44 Fold 2/5 AUC: 0.94143 Best iteration: 1304
2026-09-19 09:00:36	DEBUG	Trial 44 Starting Fold 3/5
2026-09-19 09:00:41	DEBUG	Trial 44 Fold 3/5 AUC: 0.94154 Best iteration: 1133
2026-09-19 09:00:41	DEBUG	Trial 44 Starting Fold 4/5
2026-09-19 09:00:47	DEBUG	Trial 44 Fold 4/5 AUC: 0.94163 Best iteration: 1211

2026-09-19 09:00:47	DEBUG	Trial 44 Starting Fold 5/5
2026-09-19 09:00:53	DEBUG	Trial 44 Fold 5/5 AUC: 0.94019 Best iteration: 1283
2026-09-19 09:00:53	INFO	Trial 44 finished Mean AUC: 0.94139 Std: 0.00065
2026-09-19 09:00:53	DEBUG	Trial 45 scale_pos_weight: 4.7257
2026-09-19 09:00:53	DEBUG	Trial 45 Starting Fold 1/5
2026-09-19 09:00:58	DEBUG	Trial 45 Fold 1/5 AUC: 0.94219 Best iteration: 1145
2026-09-19 09:00:59	DEBUG	Trial 45 Starting Fold 2/5
2026-09-19 09:01:04	DEBUG	Trial 45 Fold 2/5 AUC: 0.94142 Best iteration: 1125
2026-09-19 09:01:04	DEBUG	Trial 45 Starting Fold 3/5
2026-09-19 09:01:09	DEBUG	Trial 45 Fold 3/5 AUC: 0.94157 Best iteration: 1145
2026-09-19 09:01:09	DEBUG	Trial 45 Starting Fold 4/5
2026-09-19 09:01:14	DEBUG	Trial 45 Fold 4/5 AUC: 0.94156 Best iteration: 1145
2026-09-19 09:01:14	DEBUG	Trial 45 Starting Fold 5/5
2026-09-19 09:01:20	DEBUG	Trial 45 Fold 5/5 AUC: 0.94013 Best iteration: 979
2026-09-19 09:01:20	INFO	Trial 45 finished Mean AUC: 0.94138 Std: 0.00068
2026-09-19 09:01:20	DEBUG	Trial 46 scale_pos_weight: 4.7257
2026-09-19 09:01:20	DEBUG	Trial 46 Starting Fold 1/5
2026-09-19 09:01:25	DEBUG	Trial 46 Fold 1/5 AUC: 0.94215 Best iteration: 1235
2026-09-19 09:01:25	DEBUG	Trial 46 Starting Fold 2/5
2026-09-19 09:01:31	DEBUG	Trial 46 Fold 2/5 AUC: 0.94142 Best iteration: 1237
2026-09-19 09:01:31	DEBUG	Trial 46 Starting Fold 3/5
2026-09-19 09:01:36	DEBUG	Trial 46 Fold 3/5 AUC: 0.94152 Best iteration: 1235
2026-09-19 09:01:37	DEBUG	Trial 46 Starting Fold 4/5
2026-09-19 09:01:42	DEBUG	Trial 46 Fold 4/5 AUC: 0.94161 Best iteration: 1237
2026-09-19 09:01:42	DEBUG	Trial 46 Starting Fold 5/5
2026-09-19 09:01:47	DEBUG	Trial 46 Fold 5/5 AUC: 0.94009 Best iteration: 1237
2026-09-19 09:01:48	INFO	Trial 46 finished Mean AUC: 0.94136 Std: 0.00068
2026-09-19 09:01:48	DEBUG	Trial 47 scale_pos_weight: 4.7257
2026-09-19 09:01:48	DEBUG	Trial 47 Starting Fold 1/5
2026-09-19 09:01:53	DEBUG	Trial 47 Fold 1/5 AUC: 0.94215 Best iteration: 1041
2026-09-19 09:01:54	DEBUG	Trial 47 Starting Fold 2/5
2026-09-19 09:01:59	DEBUG	Trial 47 Fold 2/5 AUC: 0.94135 Best iteration: 1075
2026-09-19 09:02:00	DEBUG	Trial 47 Starting Fold 3/5
2026-09-19 09:02:05	DEBUG	Trial 47 Fold 3/5 AUC: 0.94146 Best iteration: 1022
2026-09-19 09:02:05	DEBUG	Trial 47 Starting Fold 4/5
2026-09-19 09:02:11	DEBUG	Trial 47 Fold 4/5 AUC: 0.94158 Best iteration: 1232
2026-09-19 09:02:11	DEBUG	Trial 47 Starting Fold 5/5
2026-09-19 09:02:17	DEBUG	Trial 47 Fold 5/5 AUC: 0.94012 Best iteration: 1146
2026-09-19 09:02:17	INFO	Trial 47 finished Mean AUC: 0.94133 Std: 0.00067
2026-09-19 09:02:17	DEBUG	Trial 48 scale_pos_weight: 4.7257
2026-09-19 09:02:17	DEBUG	Trial 48 Starting Fold 1/5
2026-09-19 09:02:22	DEBUG	Trial 48 Fold 1/5 AUC: 0.94216 Best iteration: 1130
2026-09-19 09:02:23	DEBUG	Trial 48 Starting Fold 2/5
2026-09-19 09:02:28	DEBUG	Trial 48 Fold 2/5 AUC: 0.94142 Best iteration: 1141
2026-09-19 09:02:28	DEBUG	Trial 48 Starting Fold 3/5
2026-09-19 09:02:33	DEBUG	Trial 48 Fold 3/5 AUC: 0.94151 Best iteration: 1133
2026-09-19 09:02:33	DEBUG	Trial 48 Starting Fold 4/5
2026-09-19 09:02:38	DEBUG	Trial 48 Fold 4/5 AUC: 0.94157 Best iteration: 1142
2026-09-19 09:02:38	DEBUG	Trial 48 Starting Fold 5/5
2026-09-19 09:02:43	DEBUG	Trial 48 Fold 5/5 AUC: 0.94008 Best iteration: 1115
2026-09-19 09:02:43	INFO	Trial 48 finished Mean AUC: 0.94135 Std: 0.00069
2026-09-19 09:02:43	DEBUG	Trial 49 scale_pos_weight: 4.7257
2026-09-19 09:02:43	DEBUG	Trial 49 Starting Fold 1/5
2026-09-19 09:02:49	DEBUG	Trial 49 Fold 1/5 AUC: 0.94210 Best iteration: 662
2026-09-19 09:02:49	DEBUG	Trial 49 Starting Fold 2/5
2026-09-19 09:02:54	DEBUG	Trial 49 Fold 2/5 AUC: 0.94130 Best iteration: 624
2026-09-19 09:02:54	DEBUG	Trial 49 Starting Fold 3/5
2026-09-19 09:02:59	DEBUG	Trial 49 Fold 3/5 AUC: 0.94151 Best iteration: 754
2026-09-19 09:02:59	DEBUG	Trial 49 Starting Fold 4/5
2026-09-19 09:03:04	DEBUG	Trial 49 Fold 4/5 AUC: 0.94159 Best iteration: 697

```
2026-09-19 09:03:05 | DEBUG | Trial 49 | Starting Fold 5/5
2026-09-19 09:03:10 | DEBUG | Trial 49 | Fold 5/5 | AUC: 0.94010 | Best iteration: 815
2026-09-19 09:03:10 | INFO | Trial 49 finished | Mean AUC: 0.94132 | Std: 0.00067
2026-09-19 09:03:10 | INFO | XGBoost hyperparameter tuning completed
2026-09-19 09:03:10 | INFO | Best CV AUC: 0.94141
2026-09-19 09:03:10 | INFO | Best trial: 37
2026-09-19 09:03:10 | INFO | Best parameters: {'n_estimators': 1282, 'learning_rate':
0.06902561705824775, 'max_depth': 3, 'min_child_weight': 1.5479285111899654, 'subsample':
0.8665573343589501, 'colsample_bytree': 0.6586794044381585, 'gamma': 1.1223798819241955,
'reg_alpha': 0.15438375483848316, 'reg_lambda': 7.229801222090218}
2026-09-19 09:03:10 | INFO | -----
-----
2026-09-19 09:03:10 | INFO | Initializing XGBoost model with best hyperparameters
2026-09-19 09:03:10 | INFO | -----
-----
2026-09-19 09:03:10 | INFO | Starting XGBoost OOF evaluation
2026-09-19 09:03:10 | INFO | Training samples: 678,665 | Features: 18 | Categorical
features: 8 | CV Folds: 5
2026-09-19 09:03:10 | INFO | Starting OOF training with XGBClassifier
2026-09-19 09:03:10 | INFO | -----
-----
2026-09-19 09:03:10 | INFO | Starting Fold 1/5
2026-09-19 09:03:10 | DEBUG | Fold 1 | Train samples: 542,932 | Validation samples: 135,733
2026-09-19 09:03:15 | INFO | Fold 1 training completed | Best iteration: None
2026-09-19 09:03:15 | INFO | Fold 1 AUC: 0.94221
2026-09-19 09:03:16 | DEBUG | Fold 1 feature importance collected successfully
2026-09-19 09:03:16 | DEBUG | Fold 1 temporary memory released
2026-09-19 09:03:16 | INFO | -----
-----
2026-09-19 09:03:16 | INFO | Starting Fold 2/5
2026-09-19 09:03:16 | DEBUG | Fold 2 | Train samples: 542,932 | Validation samples: 135,733
2026-09-19 09:03:21 | INFO | Fold 2 training completed | Best iteration: None
2026-09-19 09:03:21 | INFO | Fold 2 AUC: 0.94144
2026-09-19 09:03:21 | DEBUG | Fold 2 feature importance collected successfully
2026-09-19 09:03:21 | DEBUG | Fold 2 temporary memory released
2026-09-19 09:03:21 | INFO | -----
-----
2026-09-19 09:03:21 | INFO | Starting Fold 3/5
2026-09-19 09:03:21 | DEBUG | Fold 3 | Train samples: 542,932 | Validation samples: 135,733
2026-09-19 09:03:27 | INFO | Fold 3 training completed | Best iteration: None
2026-09-19 09:03:27 | INFO | Fold 3 AUC: 0.94152
2026-09-19 09:03:27 | DEBUG | Fold 3 feature importance collected successfully
2026-09-19 09:03:27 | DEBUG | Fold 3 temporary memory released
2026-09-19 09:03:27 | INFO | -----
-----
2026-09-19 09:03:27 | INFO | Starting Fold 4/5
2026-09-19 09:03:27 | DEBUG | Fold 4 | Train samples: 542,932 | Validation samples: 135,733
2026-09-19 09:03:32 | INFO | Fold 4 training completed | Best iteration: None
2026-09-19 09:03:33 | INFO | Fold 4 AUC: 0.94166
2026-09-19 09:03:33 | DEBUG | Fold 4 feature importance collected successfully
2026-09-19 09:03:33 | DEBUG | Fold 4 temporary memory released
2026-09-19 09:03:33 | INFO | -----
-----
2026-09-19 09:03:33 | INFO | Starting Fold 5/5
2026-09-19 09:03:33 | DEBUG | Fold 5 | Train samples: 542,932 | Validation samples: 135,733
2026-09-19 09:03:38 | INFO | Fold 5 training completed | Best iteration: None
2026-09-19 09:03:38 | INFO | Fold 5 AUC: 0.94022
2026-09-19 09:03:38 | DEBUG | Fold 5 feature importance collected successfully
2026-09-19 09:03:38 | DEBUG | Fold 5 temporary memory released
2026-09-19 09:03:39 | INFO | OOF training completed
```

2026-09-19 09:03:39 | INFO | OOF AUC: 0.94141
2026-09-19 09:03:39 | INFO | Fold AUCs: [np.float64(0.9422055473491135),
np.float64(0.9414445420401152), np.float64(0.941523081428095), np.float64(0.9416648149787257),
np.float64(0.9402168835350194)]
2026-09-19 09:03:39 | INFO | Mean Fold AUC: 0.94141 +/- 0.00065
2026-09-19 09:03:40 | INFO | Predicted positive ratio on test set: 0.2902



2026-09-19 09:03:42 | INFO | XGBoost OOF evaluation completed

2026-09-19 09:03:42 | INFO | -----

LGBMClassifier

```
logger.info("-" * 80)
logger.info("Starting LightGBM hyperparameter tuning")
study_lgbm = optuna.create_study(direction="maximize", study_name="lightgbm_churn_tuning")
study_lgbm.optimize(
    lambda trial: objective_lgbm(
        trial=trial,
        X=X,
        y=y,
        cat_features=new_cat_features,
        logger=logger
    ),
    n_trials=50
)

logger.info("LightGBM hyperparameter tuning completed")
logger.info(f"Best CV AUC: {study_lgbm.best_value:.5f}")
logger.info(f"Best trial: {study_lgbm.best_trial.number}")
logger.info(f"Best parameters: {study_lgbm.best_params}")

logger.info("-" * 80)
logger.info("Initializing LightGBM model with best hyperparameters")

model_lgbm = lgb.LGBMClassifier(**study_lgbm.best_params, objective="binary", metric="auc",
                                verbosity=-1, n_jobs=-1, boosting_type="gbdt",
                                class_weight="balanced", random_state=config["random_state"])

logger.info("-" * 80)
logger.info("Starting LightGBM OOF evaluation")
logger.info(
    f"Training samples: {len(X):,} | "
    f"Features: {X.shape[1]} | "
    f"Categorical features: {len(new_cat_features)} | "
    f"CV Folds: {config['cv']['n_splits']}"
)

results_lgbm = run_oof_v2(model=model_lgbm, X=X, y=y, X_test=df_test[TOTAL_FEATURES],
                          cat_features=new_cat_features, logger=logger)
logger.info("LightGBM OOF evaluation completed")
logger.info("-" * 80)
```

```
2026-09-19 09:03:42 | INFO | -----
-----
2026-09-19 09:03:42 | INFO | Starting LightGBM hyperparameter tuning
2026-09-19 09:04:32 | DEBUG | Trial 0 | Fold 1/5 | AUC: 0.94171 | Best iteration: 864
2026-09-19 09:05:33 | DEBUG | Trial 0 | Fold 2/5 | AUC: 0.94067 | Best iteration: 1235
2026-09-19 09:06:29 | DEBUG | Trial 0 | Fold 3/5 | AUC: 0.94098 | Best iteration: 1007
2026-09-19 09:07:31 | DEBUG | Trial 0 | Fold 4/5 | AUC: 0.94106 | Best iteration: 1230
2026-09-19 09:08:26 | DEBUG | Trial 0 | Fold 5/5 | AUC: 0.93959 | Best iteration: 993
2026-09-19 09:08:26 | INFO | Trial 0 finished | Mean AUC: 0.94080 | Std: 0.00070
2026-09-19 09:09:04 | DEBUG | Trial 1 | Fold 1/5 | AUC: 0.94125 | Best iteration: 707
2026-09-19 09:09:43 | DEBUG | Trial 1 | Fold 2/5 | AUC: 0.94046 | Best iteration: 692
2026-09-19 09:10:20 | DEBUG | Trial 1 | Fold 3/5 | AUC: 0.94065 | Best iteration: 705
2026-09-19 09:10:58 | DEBUG | Trial 1 | Fold 4/5 | AUC: 0.94080 | Best iteration: 691
2026-09-19 09:11:35 | DEBUG | Trial 1 | Fold 5/5 | AUC: 0.93903 | Best iteration: 706
2026-09-19 09:11:35 | INFO | Trial 1 finished | Mean AUC: 0.94044 | Std: 0.00075
2026-09-19 09:12:04 | DEBUG | Trial 2 | Fold 1/5 | AUC: 0.94174 | Best iteration: 409
2026-09-19 09:12:36 | DEBUG | Trial 2 | Fold 2/5 | AUC: 0.94081 | Best iteration: 522
2026-09-19 09:13:10 | DEBUG | Trial 2 | Fold 3/5 | AUC: 0.94113 | Best iteration: 515
2026-09-19 09:13:44 | DEBUG | Trial 2 | Fold 4/5 | AUC: 0.94127 | Best iteration: 538
2026-09-19 09:14:18 | DEBUG | Trial 2 | Fold 5/5 | AUC: 0.93972 | Best iteration: 543
2026-09-19 09:14:18 | INFO | Trial 2 finished | Mean AUC: 0.94094 | Std: 0.00068
2026-09-19 09:15:08 | DEBUG | Trial 3 | Fold 1/5 | AUC: 0.94190 | Best iteration: 840
2026-09-19 09:15:58 | DEBUG | Trial 3 | Fold 2/5 | AUC: 0.94097 | Best iteration: 807
2026-09-19 09:16:47 | DEBUG | Trial 3 | Fold 3/5 | AUC: 0.94121 | Best iteration: 876
2026-09-19 09:17:36 | DEBUG | Trial 3 | Fold 4/5 | AUC: 0.94146 | Best iteration: 795
2026-09-19 09:18:26 | DEBUG | Trial 3 | Fold 5/5 | AUC: 0.93971 | Best iteration: 853
2026-09-19 09:18:26 | INFO | Trial 3 finished | Mean AUC: 0.94105 | Std: 0.00074
2026-09-19 09:19:01 | DEBUG | Trial 4 | Fold 1/5 | AUC: 0.94158 | Best iteration: 636
2026-09-19 09:19:36 | DEBUG | Trial 4 | Fold 2/5 | AUC: 0.94077 | Best iteration: 636
2026-09-19 09:20:11 | DEBUG | Trial 4 | Fold 3/5 | AUC: 0.94095 | Best iteration: 637
2026-09-19 09:20:47 | DEBUG | Trial 4 | Fold 4/5 | AUC: 0.94110 | Best iteration: 637
2026-09-19 09:21:23 | DEBUG | Trial 4 | Fold 5/5 | AUC: 0.93937 | Best iteration: 637
2026-09-19 09:21:23 | INFO | Trial 4 finished | Mean AUC: 0.94075 | Std: 0.00074
2026-09-19 09:21:56 | DEBUG | Trial 5 | Fold 1/5 | AUC: 0.94043 | Best iteration: 587
2026-09-19 09:22:29 | DEBUG | Trial 5 | Fold 2/5 | AUC: 0.93954 | Best iteration: 587
2026-09-19 09:23:01 | DEBUG | Trial 5 | Fold 3/5 | AUC: 0.93983 | Best iteration: 587
2026-09-19 09:23:34 | DEBUG | Trial 5 | Fold 4/5 | AUC: 0.93994 | Best iteration: 587
2026-09-19 09:24:07 | DEBUG | Trial 5 | Fold 5/5 | AUC: 0.93811 | Best iteration: 587
2026-09-19 09:24:07 | INFO | Trial 5 finished | Mean AUC: 0.93957 | Std: 0.00079
2026-09-19 09:24:43 | DEBUG | Trial 6 | Fold 1/5 | AUC: 0.94158 | Best iteration: 707
2026-09-19 09:25:21 | DEBUG | Trial 6 | Fold 2/5 | AUC: 0.94068 | Best iteration: 707
2026-09-19 09:25:57 | DEBUG | Trial 6 | Fold 3/5 | AUC: 0.94088 | Best iteration: 707
2026-09-19 09:26:33 | DEBUG | Trial 6 | Fold 4/5 | AUC: 0.94113 | Best iteration: 707
2026-09-19 09:27:08 | DEBUG | Trial 6 | Fold 5/5 | AUC: 0.93943 | Best iteration: 706
2026-09-19 09:27:08 | INFO | Trial 6 finished | Mean AUC: 0.94074 | Std: 0.00072
2026-09-19 09:28:18 | DEBUG | Trial 7 | Fold 1/5 | AUC: 0.94217 | Best iteration: 2231
2026-09-19 09:29:31 | DEBUG | Trial 7 | Fold 2/5 | AUC: 0.94137 | Best iteration: 2231
2026-09-19 09:30:41 | DEBUG | Trial 7 | Fold 3/5 | AUC: 0.94140 | Best iteration: 2248
2026-09-19 09:31:55 | DEBUG | Trial 7 | Fold 4/5 | AUC: 0.94160 | Best iteration: 2218
2026-09-19 09:33:06 | DEBUG | Trial 7 | Fold 5/5 | AUC: 0.94010 | Best iteration: 2247
2026-09-19 09:33:06 | INFO | Trial 7 finished | Mean AUC: 0.94133 | Std: 0.00068
2026-09-19 09:33:35 | DEBUG | Trial 8 | Fold 1/5 | AUC: 0.94208 | Best iteration: 933
2026-09-19 09:34:04 | DEBUG | Trial 8 | Fold 2/5 | AUC: 0.94124 | Best iteration: 923
2026-09-19 09:34:34 | DEBUG | Trial 8 | Fold 3/5 | AUC: 0.94129 | Best iteration: 930
2026-09-19 09:35:03 | DEBUG | Trial 8 | Fold 4/5 | AUC: 0.94157 | Best iteration: 902
2026-09-19 09:35:32 | DEBUG | Trial 8 | Fold 5/5 | AUC: 0.93999 | Best iteration: 933
2026-09-19 09:35:32 | INFO | Trial 8 finished | Mean AUC: 0.94123 | Std: 0.00069
2026-09-19 09:35:56 | DEBUG | Trial 9 | Fold 1/5 | AUC: 0.94167 | Best iteration: 245
2026-09-19 09:36:20 | DEBUG | Trial 9 | Fold 2/5 | AUC: 0.94079 | Best iteration: 245
2026-09-19 09:36:48 | DEBUG | Trial 9 | Fold 3/5 | AUC: 0.94095 | Best iteration: 346
```

2026-09-19 09:37:11	DEBUG	Trial 9	Fold 4/5	AUC: 0.94122	Best iteration: 234
2026-09-19 09:37:36	DEBUG	Trial 9	Fold 5/5	AUC: 0.93950	Best iteration: 245
2026-09-19 09:37:36	INFO	Trial 9 finished Mean AUC: 0.94083 Std: 0.00073			
2026-09-19 09:38:18	DEBUG	Trial 10	Fold 1/5	AUC: 0.94179	Best iteration: 1526
2026-09-19 09:39:00	DEBUG	Trial 10	Fold 2/5	AUC: 0.94103	Best iteration: 1526
2026-09-19 09:39:41	DEBUG	Trial 10	Fold 3/5	AUC: 0.94109	Best iteration: 1526
2026-09-19 09:40:24	DEBUG	Trial 10	Fold 4/5	AUC: 0.94125	Best iteration: 1526
2026-09-19 09:41:06	DEBUG	Trial 10	Fold 5/5	AUC: 0.93970	Best iteration: 1526
2026-09-19 09:41:06	INFO	Trial 10 finished Mean AUC: 0.94097 Std: 0.00069			
2026-09-19 09:41:50	DEBUG	Trial 11	Fold 1/5	AUC: 0.94212	Best iteration: 1182
2026-09-19 09:42:27	DEBUG	Trial 11	Fold 2/5	AUC: 0.94130	Best iteration: 966
2026-09-19 09:43:00	DEBUG	Trial 11	Fold 3/5	AUC: 0.94137	Best iteration: 854
2026-09-19 09:43:40	DEBUG	Trial 11	Fold 4/5	AUC: 0.94165	Best iteration: 1066
2026-09-19 09:44:23	DEBUG	Trial 11	Fold 5/5	AUC: 0.94019	Best iteration: 1162
2026-09-19 09:44:23	INFO	Trial 11 finished Mean AUC: 0.94133 Std: 0.00064			
2026-09-19 09:45:22	DEBUG	Trial 12	Fold 1/5	AUC: 0.94210	Best iteration: 1477
2026-09-19 09:46:17	DEBUG	Trial 12	Fold 2/5	AUC: 0.94125	Best iteration: 1369
2026-09-19 09:47:14	DEBUG	Trial 12	Fold 3/5	AUC: 0.94135	Best iteration: 1399
2026-09-19 09:48:10	DEBUG	Trial 12	Fold 4/5	AUC: 0.94158	Best iteration: 1385
2026-09-19 09:49:12	DEBUG	Trial 12	Fold 5/5	AUC: 0.94010	Best iteration: 1592
2026-09-19 09:49:12	INFO	Trial 12 finished Mean AUC: 0.94127 Std: 0.00066			
2026-09-19 09:49:43	DEBUG	Trial 13	Fold 1/5	AUC: 0.94200	Best iteration: 656
2026-09-19 09:50:16	DEBUG	Trial 13	Fold 2/5	AUC: 0.94124	Best iteration: 709
2026-09-19 09:50:45	DEBUG	Trial 13	Fold 3/5	AUC: 0.94135	Best iteration: 562
2026-09-19 09:51:14	DEBUG	Trial 13	Fold 4/5	AUC: 0.94155	Best iteration: 612
2026-09-19 09:51:54	DEBUG	Trial 13	Fold 5/5	AUC: 0.94002	Best iteration: 935
2026-09-19 09:51:54	INFO	Trial 13 finished Mean AUC: 0.94123 Std: 0.00066			
2026-09-19 09:52:54	DEBUG	Trial 14	Fold 1/5	AUC: 0.94123	Best iteration: 2102
2026-09-19 09:53:52	DEBUG	Trial 14	Fold 2/5	AUC: 0.94035	Best iteration: 2102
2026-09-19 09:54:52	DEBUG	Trial 14	Fold 3/5	AUC: 0.94048	Best iteration: 2102
2026-09-19 09:55:51	DEBUG	Trial 14	Fold 4/5	AUC: 0.94070	Best iteration: 2102
2026-09-19 09:56:50	DEBUG	Trial 14	Fold 5/5	AUC: 0.93898	Best iteration: 2102
2026-09-19 09:56:50	INFO	Trial 14 finished Mean AUC: 0.94035 Std: 0.00075			
2026-09-19 09:57:50	DEBUG	Trial 15	Fold 1/5	AUC: 0.94211	Best iteration: 1715
2026-09-19 09:58:48	DEBUG	Trial 15	Fold 2/5	AUC: 0.94131	Best iteration: 1641
2026-09-19 09:59:47	DEBUG	Trial 15	Fold 3/5	AUC: 0.94142	Best iteration: 1724
2026-09-19 10:00:47	DEBUG	Trial 15	Fold 4/5	AUC: 0.94162	Best iteration: 1675
2026-09-19 10:01:47	DEBUG	Trial 15	Fold 5/5	AUC: 0.94012	Best iteration: 1722
2026-09-19 10:01:47	INFO	Trial 15 finished Mean AUC: 0.94132 Std: 0.00066			
2026-09-19 10:02:41	DEBUG	Trial 16	Fold 1/5	AUC: 0.94213	Best iteration: 1504
2026-09-19 10:03:30	DEBUG	Trial 16	Fold 2/5	AUC: 0.94134	Best iteration: 1317
2026-09-19 10:04:22	DEBUG	Trial 16	Fold 3/5	AUC: 0.94138	Best iteration: 1466
2026-09-19 10:05:12	DEBUG	Trial 16	Fold 4/5	AUC: 0.94160	Best iteration: 1366
2026-09-19 10:06:13	DEBUG	Trial 16	Fold 5/5	AUC: 0.94015	Best iteration: 1765
2026-09-19 10:06:14	INFO	Trial 16 finished Mean AUC: 0.94132 Std: 0.00065			
2026-09-19 10:07:42	DEBUG	Trial 17	Fold 1/5	AUC: 0.94194	Best iteration: 2134
2026-09-19 10:09:11	DEBUG	Trial 17	Fold 2/5	AUC: 0.94098	Best iteration: 2262
2026-09-19 10:10:40	DEBUG	Trial 17	Fold 3/5	AUC: 0.94116	Best iteration: 2075
2026-09-19 10:12:03	DEBUG	Trial 17	Fold 4/5	AUC: 0.94147	Best iteration: 1897
2026-09-19 10:13:31	DEBUG	Trial 17	Fold 5/5	AUC: 0.93986	Best iteration: 2027
2026-09-19 10:13:31	INFO	Trial 17 finished Mean AUC: 0.94108 Std: 0.00069			
2026-09-19 10:14:25	DEBUG	Trial 18	Fold 1/5	AUC: 0.94216	Best iteration: 1515
2026-09-19 10:15:16	DEBUG	Trial 18	Fold 2/5	AUC: 0.94138	Best iteration: 1451
2026-09-19 10:16:11	DEBUG	Trial 18	Fold 3/5	AUC: 0.94138	Best iteration: 1597
2026-09-19 10:17:01	DEBUG	Trial 18	Fold 4/5	AUC: 0.94163	Best iteration: 1367
2026-09-19 10:17:55	DEBUG	Trial 18	Fold 5/5	AUC: 0.94015	Best iteration: 1572
2026-09-19 10:17:55	INFO	Trial 18 finished Mean AUC: 0.94134 Std: 0.00066			
2026-09-19 10:18:37	DEBUG	Trial 19	Fold 1/5	AUC: 0.94208	Best iteration: 1507
2026-09-19 10:19:18	DEBUG	Trial 19	Fold 2/5	AUC: 0.94133	Best iteration: 1507
2026-09-19 10:19:59	DEBUG	Trial 19	Fold 3/5	AUC: 0.94134	Best iteration: 1506

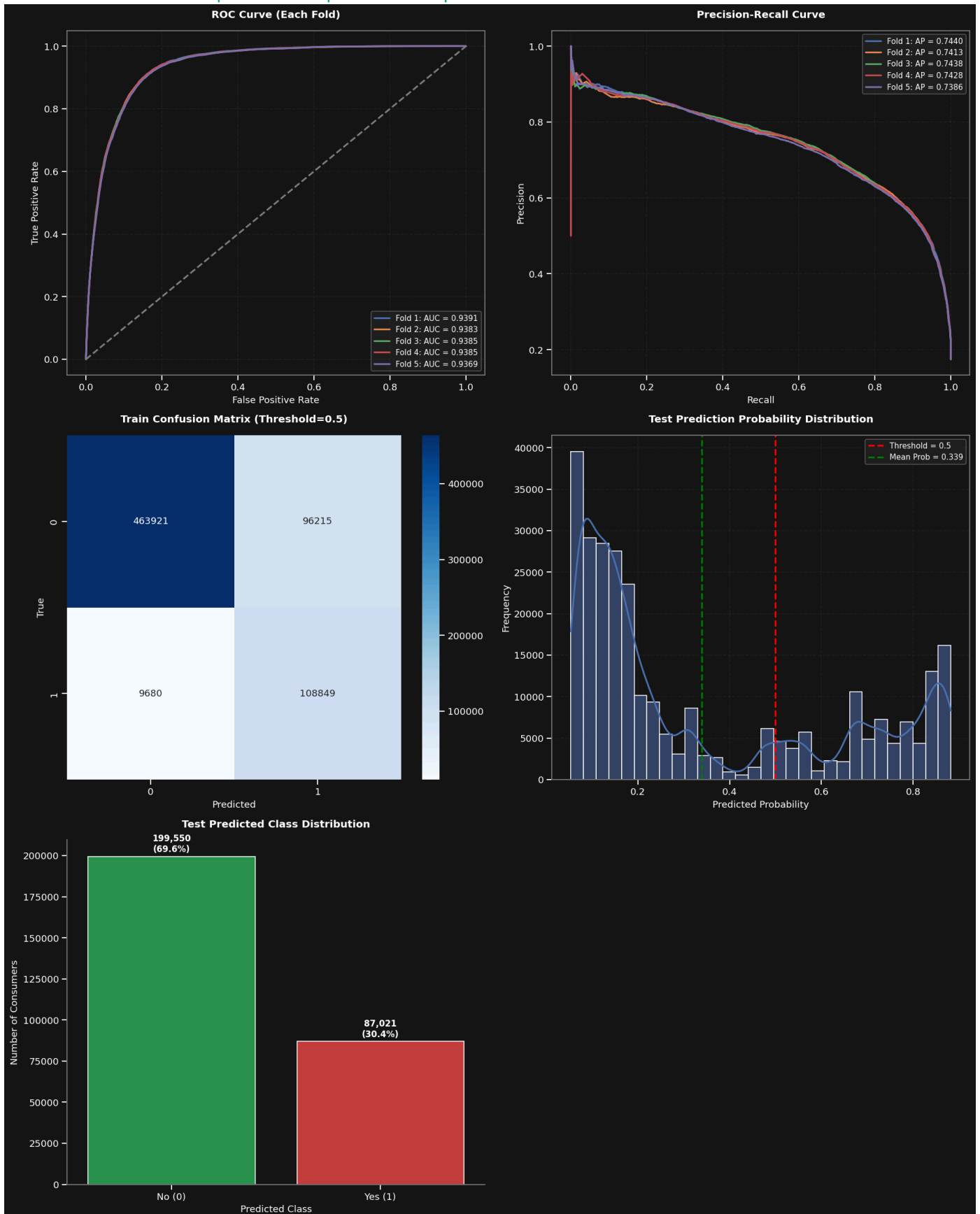
2026-09-19 10:20:40	DEBUG	Trial 19	Fold 4/5	AUC: 0.94153	Best iteration: 1496
2026-09-19 10:21:21	DEBUG	Trial 19	Fold 5/5	AUC: 0.93999	Best iteration: 1506
2026-09-19 10:21:21	INFO	Trial 19 finished Mean AUC: 0.94125 Std: 0.00069			
2026-09-19 10:22:23	DEBUG	Trial 20	Fold 1/5	AUC: 0.94204	Best iteration: 1371
2026-09-19 10:23:24	DEBUG	Trial 20	Fold 2/5	AUC: 0.94112	Best iteration: 1337
2026-09-19 10:24:27	DEBUG	Trial 20	Fold 3/5	AUC: 0.94128	Best iteration: 1385
2026-09-19 10:25:27	DEBUG	Trial 20	Fold 4/5	AUC: 0.94156	Best iteration: 1298
2026-09-19 10:26:32	DEBUG	Trial 20	Fold 5/5	AUC: 0.93997	Best iteration: 1416
2026-09-19 10:26:32	INFO	Trial 20 finished Mean AUC: 0.94119 Std: 0.00069			
2026-09-19 10:27:25	DEBUG	Trial 21	Fold 1/5	AUC: 0.94217	Best iteration: 1515
2026-09-19 10:28:12	DEBUG	Trial 21	Fold 2/5	AUC: 0.94141	Best iteration: 1287
2026-09-19 10:29:07	DEBUG	Trial 21	Fold 3/5	AUC: 0.94144	Best iteration: 1537
2026-09-19 10:29:59	DEBUG	Trial 21	Fold 4/5	AUC: 0.94167	Best iteration: 1450
2026-09-19 10:30:57	DEBUG	Trial 21	Fold 5/5	AUC: 0.94021	Best iteration: 1627
2026-09-19 10:30:57	INFO	Trial 21 finished Mean AUC: 0.94138 Std: 0.00065			
2026-09-19 10:31:57	DEBUG	Trial 22	Fold 1/5	AUC: 0.94223	Best iteration: 1939
2026-09-19 10:32:57	DEBUG	Trial 22	Fold 2/5	AUC: 0.94144	Best iteration: 1847
2026-09-19 10:33:58	DEBUG	Trial 22	Fold 3/5	AUC: 0.94145	Best iteration: 1742
2026-09-19 10:34:57	DEBUG	Trial 22	Fold 4/5	AUC: 0.94163	Best iteration: 1646
2026-09-19 10:35:58	DEBUG	Trial 22	Fold 5/5	AUC: 0.94019	Best iteration: 1956
2026-09-19 10:35:58	INFO	Trial 22 finished Mean AUC: 0.94139 Std: 0.00066			
2026-09-19 10:36:46	DEBUG	Trial 23	Fold 1/5	AUC: 0.94209	Best iteration: 1095
2026-09-19 10:37:37	DEBUG	Trial 23	Fold 2/5	AUC: 0.94128	Best iteration: 1213
2026-09-19 10:38:27	DEBUG	Trial 23	Fold 3/5	AUC: 0.94141	Best iteration: 1188
2026-09-19 10:39:16	DEBUG	Trial 23	Fold 4/5	AUC: 0.94154	Best iteration: 1130
2026-09-19 10:40:10	DEBUG	Trial 23	Fold 5/5	AUC: 0.94014	Best iteration: 1362
2026-09-19 10:40:10	INFO	Trial 23 finished Mean AUC: 0.94129 Std: 0.00064			
2026-09-19 10:41:01	DEBUG	Trial 24	Fold 1/5	AUC: 0.94217	Best iteration: 1411
2026-09-19 10:41:44	DEBUG	Trial 24	Fold 2/5	AUC: 0.94134	Best iteration: 1134
2026-09-19 10:42:26	DEBUG	Trial 24	Fold 3/5	AUC: 0.94140	Best iteration: 1089
2026-09-19 10:43:07	DEBUG	Trial 24	Fold 4/5	AUC: 0.94158	Best iteration: 1084
2026-09-19 10:43:57	DEBUG	Trial 24	Fold 5/5	AUC: 0.94011	Best iteration: 1339
2026-09-19 10:43:57	INFO	Trial 24 finished Mean AUC: 0.94132 Std: 0.00067			
2026-09-19 10:44:45	DEBUG	Trial 25	Fold 1/5	AUC: 0.94207	Best iteration: 1738
2026-09-19 10:45:33	DEBUG	Trial 25	Fold 2/5	AUC: 0.94134	Best iteration: 1738
2026-09-19 10:46:21	DEBUG	Trial 25	Fold 3/5	AUC: 0.94133	Best iteration: 1738
2026-09-19 10:47:09	DEBUG	Trial 25	Fold 4/5	AUC: 0.94152	Best iteration: 1736
2026-09-19 10:47:57	DEBUG	Trial 25	Fold 5/5	AUC: 0.94000	Best iteration: 1738
2026-09-19 10:47:57	INFO	Trial 25 finished Mean AUC: 0.94125 Std: 0.00068			
2026-09-19 10:48:27	DEBUG	Trial 26	Fold 1/5	AUC: 0.94195	Best iteration: 524
2026-09-19 10:48:57	DEBUG	Trial 26	Fold 2/5	AUC: 0.94107	Best iteration: 517
2026-09-19 10:49:30	DEBUG	Trial 26	Fold 3/5	AUC: 0.94122	Best iteration: 587
2026-09-19 10:50:03	DEBUG	Trial 26	Fold 4/5	AUC: 0.94143	Best iteration: 599
2026-09-19 10:50:38	DEBUG	Trial 26	Fold 5/5	AUC: 0.93992	Best iteration: 635
2026-09-19 10:50:38	INFO	Trial 26 finished Mean AUC: 0.94112 Std: 0.00067			
2026-09-19 10:51:39	DEBUG	Trial 27	Fold 1/5	AUC: 0.94211	Best iteration: 1522
2026-09-19 10:52:43	DEBUG	Trial 27	Fold 2/5	AUC: 0.94130	Best iteration: 1584
2026-09-19 10:53:50	DEBUG	Trial 27	Fold 3/5	AUC: 0.94144	Best iteration: 1776
2026-09-19 10:54:54	DEBUG	Trial 27	Fold 4/5	AUC: 0.94161	Best iteration: 1572
2026-09-19 10:56:02	DEBUG	Trial 27	Fold 5/5	AUC: 0.94017	Best iteration: 1828
2026-09-19 10:56:02	INFO	Trial 27 finished Mean AUC: 0.94133 Std: 0.00064			
2026-09-19 10:56:44	DEBUG	Trial 28	Fold 1/5	AUC: 0.94215	Best iteration: 1344
2026-09-19 10:57:27	DEBUG	Trial 28	Fold 2/5	AUC: 0.94135	Best iteration: 1332
2026-09-19 10:58:09	DEBUG	Trial 28	Fold 3/5	AUC: 0.94138	Best iteration: 1349
2026-09-19 10:58:51	DEBUG	Trial 28	Fold 4/5	AUC: 0.94160	Best iteration: 1344
2026-09-19 10:59:34	DEBUG	Trial 28	Fold 5/5	AUC: 0.94007	Best iteration: 1344
2026-09-19 10:59:34	INFO	Trial 28 finished Mean AUC: 0.94131 Std: 0.00068			
2026-09-19 11:00:24	DEBUG	Trial 29	Fold 1/5	AUC: 0.94220	Best iteration: 1850
2026-09-19 11:01:14	DEBUG	Trial 29	Fold 2/5	AUC: 0.94140	Best iteration: 1848
2026-09-19 11:02:06	DEBUG	Trial 29	Fold 3/5	AUC: 0.94143	Best iteration: 1762

2026-09-19 11:02:56	DEBUG	Trial 29	Fold 4/5	AUC: 0.94167	Best iteration: 1813
2026-09-19 11:03:47	DEBUG	Trial 29	Fold 5/5	AUC: 0.94015	Best iteration: 1840
2026-09-19 11:03:47	INFO	Trial 29 finished Mean AUC: 0.94137 Std: 0.00068			
2026-09-19 11:04:33	DEBUG	Trial 30	Fold 1/5	AUC: 0.94216	Best iteration: 1659
2026-09-19 11:05:17	DEBUG	Trial 30	Fold 2/5	AUC: 0.94138	Best iteration: 1570
2026-09-19 11:06:03	DEBUG	Trial 30	Fold 3/5	AUC: 0.94142	Best iteration: 1662
2026-09-19 11:06:48	DEBUG	Trial 30	Fold 4/5	AUC: 0.94160	Best iteration: 1661
2026-09-19 11:07:34	DEBUG	Trial 30	Fold 5/5	AUC: 0.94009	Best iteration: 1656
2026-09-19 11:07:34	INFO	Trial 30 finished Mean AUC: 0.94133 Std: 0.00068			
2026-09-19 11:08:25	DEBUG	Trial 31	Fold 1/5	AUC: 0.94216	Best iteration: 1825
2026-09-19 11:09:16	DEBUG	Trial 31	Fold 2/5	AUC: 0.94134	Best iteration: 1812
2026-09-19 11:10:07	DEBUG	Trial 31	Fold 3/5	AUC: 0.94134	Best iteration: 1821
2026-09-19 11:10:58	DEBUG	Trial 31	Fold 4/5	AUC: 0.94155	Best iteration: 1827
2026-09-19 11:11:48	DEBUG	Trial 31	Fold 5/5	AUC: 0.94006	Best iteration: 1830
2026-09-19 11:11:48	INFO	Trial 31 finished Mean AUC: 0.94129 Std: 0.00068			
2026-09-19 11:12:36	DEBUG	Trial 32	Fold 1/5	AUC: 0.94218	Best iteration: 1332
2026-09-19 11:13:20	DEBUG	Trial 32	Fold 2/5	AUC: 0.94140	Best iteration: 1180
2026-09-19 11:14:02	DEBUG	Trial 32	Fold 3/5	AUC: 0.94142	Best iteration: 1066
2026-09-19 11:14:46	DEBUG	Trial 32	Fold 4/5	AUC: 0.94163	Best iteration: 1187
2026-09-19 11:15:36	DEBUG	Trial 32	Fold 5/5	AUC: 0.94016	Best iteration: 1355
2026-09-19 11:15:36	INFO	Trial 32 finished Mean AUC: 0.94136 Std: 0.00066			
2026-09-19 11:16:28	DEBUG	Trial 33	Fold 1/5	AUC: 0.94218	Best iteration: 1447
2026-09-19 11:17:12	DEBUG	Trial 33	Fold 2/5	AUC: 0.94140	Best iteration: 1172
2026-09-19 11:17:55	DEBUG	Trial 33	Fold 3/5	AUC: 0.94140	Best iteration: 1114
2026-09-19 11:18:33	DEBUG	Trial 33	Fold 4/5	AUC: 0.94167	Best iteration: 1023
2026-09-19 11:19:23	DEBUG	Trial 33	Fold 5/5	AUC: 0.94016	Best iteration: 1357
2026-09-19 11:19:23	INFO	Trial 33 finished Mean AUC: 0.94136 Std: 0.00066			
2026-09-19 11:20:21	DEBUG	Trial 34	Fold 1/5	AUC: 0.94215	Best iteration: 2074
2026-09-19 11:21:18	DEBUG	Trial 34	Fold 2/5	AUC: 0.94139	Best iteration: 2079
2026-09-19 11:22:15	DEBUG	Trial 34	Fold 3/5	AUC: 0.94138	Best iteration: 2079
2026-09-19 11:23:12	DEBUG	Trial 34	Fold 4/5	AUC: 0.94157	Best iteration: 2075
2026-09-19 11:24:12	DEBUG	Trial 34	Fold 5/5	AUC: 0.94008	Best iteration: 2079
2026-09-19 11:24:12	INFO	Trial 34 finished Mean AUC: 0.94131 Std: 0.00068			
2026-09-19 11:24:44	DEBUG	Trial 35	Fold 1/5	AUC: 0.94203	Best iteration: 647
2026-09-19 11:25:16	DEBUG	Trial 35	Fold 2/5	AUC: 0.94118	Best iteration: 674
2026-09-19 11:25:48	DEBUG	Trial 35	Fold 3/5	AUC: 0.94129	Best iteration: 654
2026-09-19 11:26:18	DEBUG	Trial 35	Fold 4/5	AUC: 0.94151	Best iteration: 625
2026-09-19 11:26:54	DEBUG	Trial 35	Fold 5/5	AUC: 0.94003	Best iteration: 785
2026-09-19 11:26:55	INFO	Trial 35 finished Mean AUC: 0.94121 Std: 0.00066			
2026-09-19 11:28:03	DEBUG	Trial 36	Fold 1/5	AUC: 0.94226	Best iteration: 2174
2026-09-19 11:29:10	DEBUG	Trial 36	Fold 2/5	AUC: 0.94144	Best iteration: 2044
2026-09-19 11:30:20	DEBUG	Trial 36	Fold 3/5	AUC: 0.94148	Best iteration: 2168
2026-09-19 11:31:29	DEBUG	Trial 36	Fold 4/5	AUC: 0.94169	Best iteration: 2065
2026-09-19 11:32:39	DEBUG	Trial 36	Fold 5/5	AUC: 0.94023	Best iteration: 2128
2026-09-19 11:32:39	INFO	Trial 36 finished Mean AUC: 0.94142 Std: 0.00066			
2026-09-19 11:33:42	DEBUG	Trial 37	Fold 1/5	AUC: 0.94210	Best iteration: 2277
2026-09-19 11:34:44	DEBUG	Trial 37	Fold 2/5	AUC: 0.94136	Best iteration: 2278
2026-09-19 11:35:46	DEBUG	Trial 37	Fold 3/5	AUC: 0.94134	Best iteration: 2272
2026-09-19 11:36:48	DEBUG	Trial 37	Fold 4/5	AUC: 0.94152	Best iteration: 2278
2026-09-19 11:37:51	DEBUG	Trial 37	Fold 5/5	AUC: 0.94007	Best iteration: 2277
2026-09-19 11:37:51	INFO	Trial 37 finished Mean AUC: 0.94128 Std: 0.00066			
2026-09-19 11:39:20	DEBUG	Trial 38	Fold 1/5	AUC: 0.94149	Best iteration: 2353
2026-09-19 11:40:49	DEBUG	Trial 38	Fold 2/5	AUC: 0.94064	Best iteration: 2353
2026-09-19 11:42:19	DEBUG	Trial 38	Fold 3/5	AUC: 0.94081	Best iteration: 2353
2026-09-19 11:43:48	DEBUG	Trial 38	Fold 4/5	AUC: 0.94099	Best iteration: 2353
2026-09-19 11:45:18	DEBUG	Trial 38	Fold 5/5	AUC: 0.93930	Best iteration: 2353
2026-09-19 11:45:18	INFO	Trial 38 finished Mean AUC: 0.94065 Std: 0.00073			
2026-09-19 11:46:01	DEBUG	Trial 39	Fold 1/5	AUC: 0.94171	Best iteration: 522
2026-09-19 11:46:41	DEBUG	Trial 39	Fold 2/5	AUC: 0.94086	Best iteration: 457
2026-09-19 11:47:27	DEBUG	Trial 39	Fold 3/5	AUC: 0.94107	Best iteration: 584

2026-09-19 11:48:12	DEBUG	Trial 39	Fold 4/5	AUC: 0.94129	Best iteration: 564
2026-09-19 11:48:59	DEBUG	Trial 39	Fold 5/5	AUC: 0.93966	Best iteration: 583
2026-09-19 11:48:59	INFO	Trial 39 finished Mean AUC: 0.94092 Std: 0.00069			
2026-09-19 11:49:58	DEBUG	Trial 40	Fold 1/5	AUC: 0.94189	Best iteration: 1209
2026-09-19 11:51:05	DEBUG	Trial 40	Fold 2/5	AUC: 0.94103	Best iteration: 1431
2026-09-19 11:52:06	DEBUG	Trial 40	Fold 3/5	AUC: 0.94115	Best iteration: 1256
2026-09-19 11:53:01	DEBUG	Trial 40	Fold 4/5	AUC: 0.94144	Best iteration: 1124
2026-09-19 11:54:06	DEBUG	Trial 40	Fold 5/5	AUC: 0.93989	Best iteration: 1390
2026-09-19 11:54:06	INFO	Trial 40 finished Mean AUC: 0.94108 Std: 0.00067			
2026-09-19 11:55:10	DEBUG	Trial 41	Fold 1/5	AUC: 0.94222	Best iteration: 1883
2026-09-19 11:56:10	DEBUG	Trial 41	Fold 2/5	AUC: 0.94142	Best iteration: 1685
2026-09-19 11:57:15	DEBUG	Trial 41	Fold 3/5	AUC: 0.94150	Best iteration: 1997
2026-09-19 11:58:22	DEBUG	Trial 41	Fold 4/5	AUC: 0.94167	Best iteration: 2108
2026-09-19 11:59:26	DEBUG	Trial 41	Fold 5/5	AUC: 0.94023	Best iteration: 2070
2026-09-19 11:59:26	INFO	Trial 41 finished Mean AUC: 0.94141 Std: 0.00065			
2026-09-19 12:00:32	DEBUG	Trial 42	Fold 1/5	AUC: 0.94225	Best iteration: 2096
2026-09-19 12:01:39	DEBUG	Trial 42	Fold 2/5	AUC: 0.94145	Best iteration: 2022
2026-09-19 12:02:45	DEBUG	Trial 42	Fold 3/5	AUC: 0.94146	Best iteration: 2053
2026-09-19 12:03:51	DEBUG	Trial 42	Fold 4/5	AUC: 0.94170	Best iteration: 2102
2026-09-19 12:04:57	DEBUG	Trial 42	Fold 5/5	AUC: 0.94023	Best iteration: 2121
2026-09-19 12:04:57	INFO	Trial 42 finished Mean AUC: 0.94142 Std: 0.00066			
2026-09-19 12:06:03	DEBUG	Trial 43	Fold 1/5	AUC: 0.94223	Best iteration: 2045
2026-09-19 12:07:09	DEBUG	Trial 43	Fold 2/5	AUC: 0.94147	Best iteration: 2034
2026-09-19 12:08:15	DEBUG	Trial 43	Fold 3/5	AUC: 0.94144	Best iteration: 2119
2026-09-19 12:09:23	DEBUG	Trial 43	Fold 4/5	AUC: 0.94169	Best iteration: 2158
2026-09-19 12:10:31	DEBUG	Trial 43	Fold 5/5	AUC: 0.94024	Best iteration: 2102
2026-09-19 12:10:32	INFO	Trial 43 finished Mean AUC: 0.94141 Std: 0.00065			
2026-09-19 12:11:41	DEBUG	Trial 44	Fold 1/5	AUC: 0.94215	Best iteration: 2173
2026-09-19 12:12:50	DEBUG	Trial 44	Fold 2/5	AUC: 0.94132	Best iteration: 2169
2026-09-19 12:14:00	DEBUG	Trial 44	Fold 3/5	AUC: 0.94138	Best iteration: 2173
2026-09-19 12:15:09	DEBUG	Trial 44	Fold 4/5	AUC: 0.94158	Best iteration: 2173
2026-09-19 12:16:19	DEBUG	Trial 44	Fold 5/5	AUC: 0.94007	Best iteration: 2170
2026-09-19 12:16:19	INFO	Trial 44 finished Mean AUC: 0.94130 Std: 0.00068			
2026-09-19 12:17:21	DEBUG	Trial 45	Fold 1/5	AUC: 0.94212	Best iteration: 1516
2026-09-19 12:18:27	DEBUG	Trial 45	Fold 2/5	AUC: 0.94129	Best iteration: 1612
2026-09-19 12:19:30	DEBUG	Trial 45	Fold 3/5	AUC: 0.94142	Best iteration: 1546
2026-09-19 12:20:29	DEBUG	Trial 45	Fold 4/5	AUC: 0.94160	Best iteration: 1445
2026-09-19 12:21:32	DEBUG	Trial 45	Fold 5/5	AUC: 0.94016	Best iteration: 1525
2026-09-19 12:21:32	INFO	Trial 45 finished Mean AUC: 0.94132 Std: 0.00064			
2026-09-19 12:22:47	DEBUG	Trial 46	Fold 1/5	AUC: 0.94211	Best iteration: 2344
2026-09-19 12:24:00	DEBUG	Trial 46	Fold 2/5	AUC: 0.94130	Best iteration: 2345
2026-09-19 12:25:15	DEBUG	Trial 46	Fold 3/5	AUC: 0.94134	Best iteration: 2349
2026-09-19 12:26:29	DEBUG	Trial 46	Fold 4/5	AUC: 0.94156	Best iteration: 2349
2026-09-19 12:27:43	DEBUG	Trial 46	Fold 5/5	AUC: 0.94002	Best iteration: 2350
2026-09-19 12:27:43	INFO	Trial 46 finished Mean AUC: 0.94127 Std: 0.00069			
2026-09-19 12:28:48	DEBUG	Trial 47	Fold 1/5	AUC: 0.94222	Best iteration: 2020
2026-09-19 12:29:53	DEBUG	Trial 47	Fold 2/5	AUC: 0.94145	Best iteration: 1980
2026-09-19 12:30:59	DEBUG	Trial 47	Fold 3/5	AUC: 0.94149	Best iteration: 2143
2026-09-19 12:32:04	DEBUG	Trial 47	Fold 4/5	AUC: 0.94172	Best iteration: 1937
2026-09-19 12:33:10	DEBUG	Trial 47	Fold 5/5	AUC: 0.94022	Best iteration: 2100
2026-09-19 12:33:10	INFO	Trial 47 finished Mean AUC: 0.94142 Std: 0.00066			
2026-09-19 12:34:20	DEBUG	Trial 48	Fold 1/5	AUC: 0.94213	Best iteration: 1792
2026-09-19 12:35:21	DEBUG	Trial 48	Fold 2/5	AUC: 0.94136	Best iteration: 1541
2026-09-19 12:36:32	DEBUG	Trial 48	Fold 3/5	AUC: 0.94147	Best iteration: 1968
2026-09-19 12:37:40	DEBUG	Trial 48	Fold 4/5	AUC: 0.94163	Best iteration: 1707
2026-09-19 12:38:53	DEBUG	Trial 48	Fold 5/5	AUC: 0.94018	Best iteration: 2087
2026-09-19 12:38:53	INFO	Trial 48 finished Mean AUC: 0.94135 Std: 0.00064			
2026-09-19 12:39:59	DEBUG	Trial 49	Fold 1/5	AUC: 0.94219	Best iteration: 1914
2026-09-19 12:41:03	DEBUG	Trial 49	Fold 2/5	AUC: 0.94142	Best iteration: 1847
2026-09-19 12:42:07	DEBUG	Trial 49	Fold 3/5	AUC: 0.94145	Best iteration: 1880

```
2026-09-19 12:43:10 | DEBUG | Trial 49 | Fold 4/5 | AUC: 0.94165 | Best iteration: 1808
2026-09-19 12:44:21 | DEBUG | Trial 49 | Fold 5/5 | AUC: 0.94020 | Best iteration: 2262
2026-09-19 12:44:21 | INFO | Trial 49 finished | Mean AUC: 0.94138 | Std: 0.00065
2026-09-19 12:44:21 | INFO | LightGBM hyperparameter tuning completed
2026-09-19 12:44:21 | INFO | Best CV AUC: 0.94142
2026-09-19 12:44:21 | INFO | Best trial: 36
2026-09-19 12:44:21 | INFO | Best parameters: {'learning_rate': 0.022544626824392143,
'num_leaves': 144, 'max_depth': 4, 'min_child_samples': 13, 'subsample': 0.8306290992873027,
'colsample_bytree': 0.6644225858102292, 'reg_alpha': 9.64546653135789, 'reg_lambda':
0.030892034431607027, 'iterations': 2176}
2026-09-19 12:44:21 | INFO | -----
-----
2026-09-19 12:44:21 | INFO | Initializing LightGBM model with best hyperparameters
2026-09-19 12:44:21 | INFO | -----
-----
2026-09-19 12:44:21 | INFO | Starting LightGBM OOF evaluation
2026-09-19 12:44:21 | INFO | Training samples: 678,665 | Features: 18 | Categorical
features: 8 | CV Folds: 5
2026-09-19 12:44:21 | INFO | Starting OOF training with LGBMClassifier
2026-09-19 12:44:21 | INFO | -----
-----
2026-09-19 12:44:21 | INFO | Starting Fold 1/5
2026-09-19 12:44:21 | DEBUG | Fold 1 | Train samples: 542,932 | Validation samples: 135,733
2026-09-19 12:44:24 | INFO | Fold 1 training completed | Best iteration: 100
2026-09-19 12:44:25 | INFO | Fold 1 AUC: 0.93915
2026-09-19 12:44:25 | DEBUG | Fold 1 feature importance collected successfully
2026-09-19 12:44:25 | DEBUG | Fold 1 temporary memory released
2026-09-19 12:44:25 | INFO | -----
-----
2026-09-19 12:44:25 | INFO | Starting Fold 2/5
2026-09-19 12:44:25 | DEBUG | Fold 2 | Train samples: 542,932 | Validation samples: 135,733
2026-09-19 12:44:29 | INFO | Fold 2 training completed | Best iteration: 99
2026-09-19 12:44:30 | INFO | Fold 2 AUC: 0.93834
2026-09-19 12:44:30 | DEBUG | Fold 2 feature importance collected successfully
2026-09-19 12:44:30 | DEBUG | Fold 2 temporary memory released
2026-09-19 12:44:30 | INFO | -----
-----
2026-09-19 12:44:30 | INFO | Starting Fold 3/5
2026-09-19 12:44:30 | DEBUG | Fold 3 | Train samples: 542,932 | Validation samples: 135,733
2026-09-19 12:44:33 | INFO | Fold 3 training completed | Best iteration: 98
2026-09-19 12:44:34 | INFO | Fold 3 AUC: 0.93852
2026-09-19 12:44:34 | DEBUG | Fold 3 feature importance collected successfully
2026-09-19 12:44:34 | DEBUG | Fold 3 temporary memory released
2026-09-19 12:44:34 | INFO | -----
-----
2026-09-19 12:44:34 | INFO | Starting Fold 4/5
2026-09-19 12:44:34 | DEBUG | Fold 4 | Train samples: 542,932 | Validation samples: 135,733
2026-09-19 12:44:38 | INFO | Fold 4 training completed | Best iteration: 100
2026-09-19 12:44:39 | INFO | Fold 4 AUC: 0.93853
2026-09-19 12:44:39 | DEBUG | Fold 4 feature importance collected successfully
2026-09-19 12:44:39 | DEBUG | Fold 4 temporary memory released
2026-09-19 12:44:39 | INFO | -----
-----
2026-09-19 12:44:39 | INFO | Starting Fold 5/5
2026-09-19 12:44:39 | DEBUG | Fold 5 | Train samples: 542,932 | Validation samples: 135,733
2026-09-19 12:44:43 | INFO | Fold 5 training completed | Best iteration: 97
2026-09-19 12:44:44 | INFO | Fold 5 AUC: 0.93690
2026-09-19 12:44:44 | DEBUG | Fold 5 feature importance collected successfully
2026-09-19 12:44:44 | DEBUG | Fold 5 temporary memory released
2026-09-19 12:44:44 | INFO | OOF training completed
```

2026-09-19 12:44:44 | INFO | OOF AUC: 0.93819
2026-09-19 12:44:44 | INFO | Fold AUCs: [np.float64(0.9391456856007772),
np.float64(0.9383427257258132), np.float64(0.9385153833744195), np.float64(0.9385311972122977),
np.float64(0.9369005722051806)]
2026-09-19 12:44:44 | INFO | Mean Fold AUC: 0.93829 +/- 0.00074
2026-09-19 12:44:45 | INFO | Predicted positive ratio on test set: 0.3037



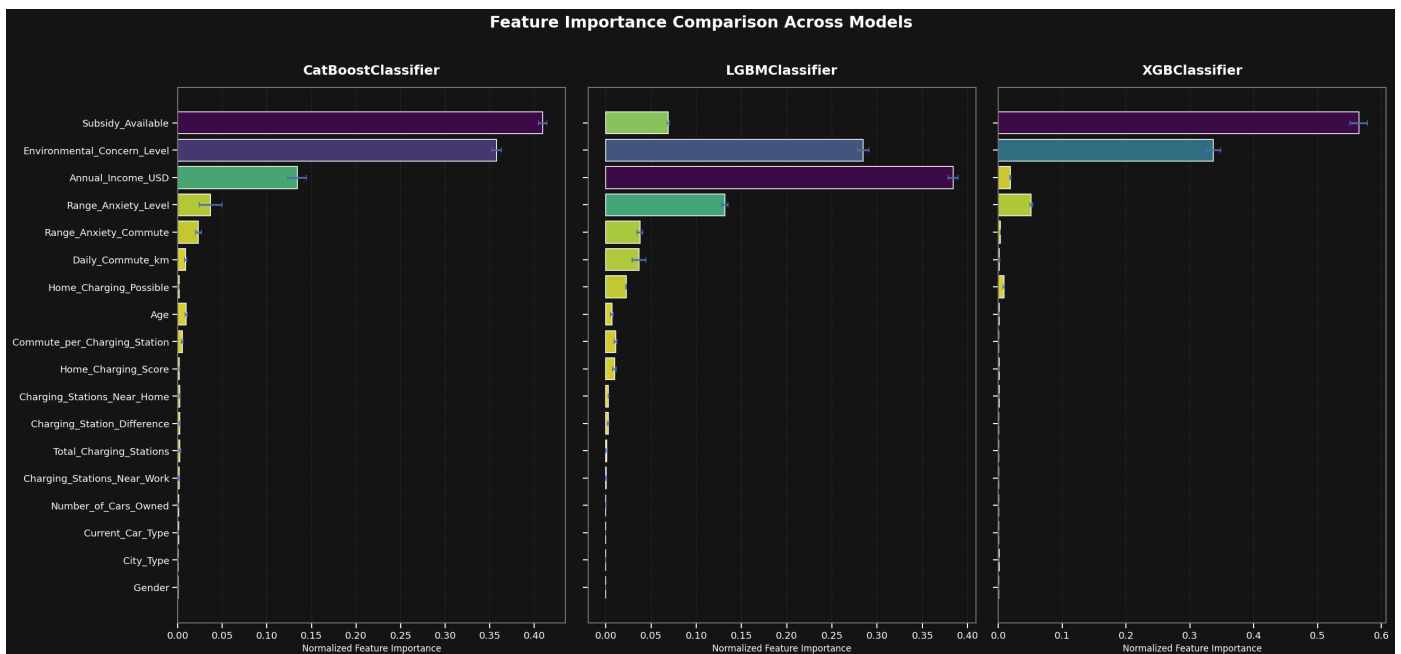
2026-09-19 12:44:47 | INFO | LightGBM OOF evaluation completed

2026-09-19 12:44:47 | INFO | -----

Evaluation

```
feature_importances = {
    "CatBoostClassifier": results_cb["feature_importance"],
    "LGBMClassifier": results_lgbm["feature_importance"],
    "XGBClassifier": results_xgb["feature_importance"]
}

fig = plot_feature_importance_comparison(
    feature_importances=feature_importances,
    top_n=20,
    figsize=(22, 10),
    show_std=True
)
```



```
# Extract OOF and validation predictions
oof_pred_cb = results_cb["oof_pred"]
oof_pred_lgbm = results_lgbm["oof_pred"]
oof_pred_xgb = results_xgb["oof_pred"]

xgb_auc = roc_auc_score(y, oof_pred_xgb)
lgb_auc = roc_auc_score(y, oof_pred_lgbm)
cat_auc = roc_auc_score(y, oof_pred_cb)

# AUC-weighted blend
total_auc = xgb_auc + lgb_auc + cat_auc
w_xgb = xgb_auc / total_auc
```

```

w_lgb = lgb_auc / total_auc
w_cat = cat_auc / total_auc

blended_oof = w_xgb * oof_pred_xgb + w_lgb * oof_pred_lgbm + w_cat * oof_pred_cb
blended_test = w_xgb * results_xgb["test_pred"] + w_lgb * results_lgbm["test_pred"] + \
               w_cat * results_cb["test_pred"]

blend_auc = roc_auc_score(y, blended_oof)

logger.info("=*80)
logger.info("ENSEMBLE BLENDING")
logger.info("=*80)
logger.info(f"XGB OOF AUC: {xgb_auc:.5f} (Weight: {w_xgb:.5f})")
logger.info(f"LGBM OOF AUC: {lgb_auc:.5f} (Weight: {w_lgb:.5f})")
logger.info(f"CATBOOST OOF AUC: {cat_auc:.5f} (Weight: {w_cat:.5f})")
logger.info(f"Blended OOF AUC: {blend_auc:.5f}")
logger.info("=*80)

# Prepare submission file
submission = pd.DataFrame({
    "id": list_test_id,
    "Will_Buy_EV": blended_test
})

submission.to_csv("submission.csv", index=False)
print("\nSubmission file saved!")
submission.head()

```

```

2026-09-19 12:44:49 | INFO      |
=====
2026-09-19 12:44:49 | INFO      | ENSEMBLE BLENDING
2026-09-19 12:44:49 | INFO      |
=====
2026-09-19 12:44:49 | INFO      | XGB OOF AUC: 0.94141 (Weight: 0.33374)
2026-09-19 12:44:49 | INFO      | LGBM OOF AUC: 0.93819 (Weight: 0.33260)
2026-09-19 12:44:49 | INFO      | CATBOOST OOF AUC: 0.94121 (Weight: 0.33367)
2026-09-19 12:44:49 | INFO      | Blended OOF AUC: 0.94105
2026-09-19 12:44:49 | INFO      |
=====

```

Submission file saved!

	id	Will_Buy_EV
0	668665	0.086158
1	668666	0.144917
2	668667	0.070736
3	668668	0.045350
4	668669	0.129709

```

# Prediction Distribution Visualization
fig, axes = plt.subplots(1, 4, figsize=(22, 5), sharey=True)
prediction_data = [
    ("CatBoost", oof_pred_cb, cat_auc),
    ("LightGBM", oof_pred_lgbm, lgb_auc),
    ("XGBoost", oof_pred_xgb, xgb_auc),
    ("Blended Ensemble", blended_oof, blend_auc)
]

```

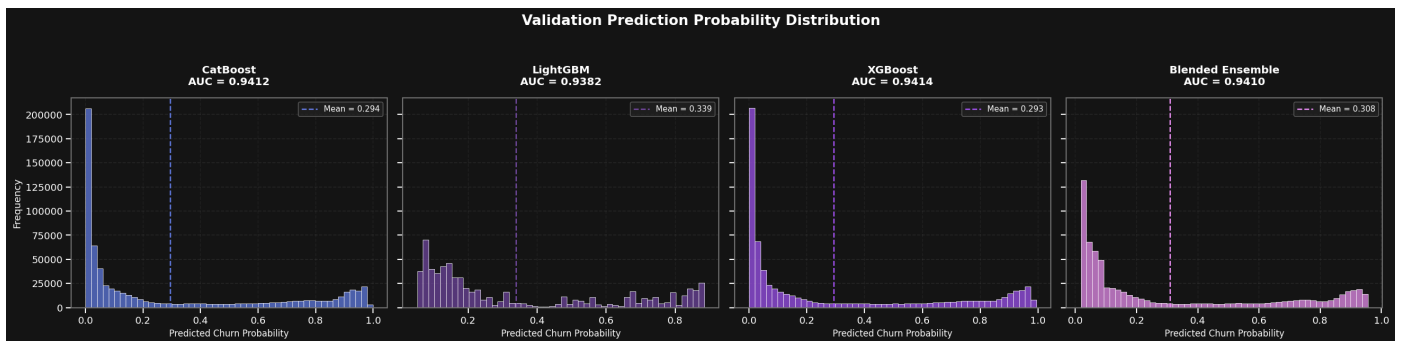
```

]

# Plot distributions
colors = list(config["colors"])
for i, (ax, (model_name, predictions, auc)) in enumerate(zip(axes, prediction_data)):
    sns.histplot(predictions, color=colors[i], alpha=0.7, ax=ax, bins=50)
    ax.axvline(predictions.mean(), color=colors[i], linestyle="--", linewidth=1.5, label=f"Mean = {predictions.mean():.3f}")
    ax.set_title(f"{model_name}\nAUC = {auc:.4f}", fontsize=12, fontweight="bold", pad=15)
    ax.set_xlabel("Predicted Churn Probability", fontsize=10)
    ax.set_ylabel("Frequency")
    ax.grid(axis="y", alpha=0.3)
    ax.legend()

# Final layout
fig.suptitle("Validation Prediction Probability Distribution", fontsize=16, fontweight="bold",
y=1.05)
plt.tight_layout()
plt.show()

```



Recommendation



Recommendations

Area	Recommendation
EV Subsidies	Expand and target EV subsidy programs toward consumer segments where financial incentives may have the greatest potential to support adoption. Subsidy_Available is consistently one of the most important features across all three models.
Environmental Awareness	Strengthen environmental awareness and EV education campaigns. Environmental_Concern_Level remains highly important across CatBoost, LightGBM, and XGBoost, while the EDA also shows a strong relationship between environmental concern and EV purchase behavior.
Affordability & Income	Consider affordability-focused strategies such as competitive pricing, financing options, and incentives. Annual_Income_USD is among the most important predictors, although its relative importance varies across models.
Range Anxiety	Reduce range anxiety through better communication of vehicle range, charging availability, and practical charging solutions. Range_Anxiety_Level shows meaningful importance in the models and a strong relationship with purchase behavior in the EDA.

Charging Infrastructure

Continue improving charging accessibility, particularly around consumers' homes and workplaces. However, the relatively low model importance of the individual charging-station features suggests that infrastructure alone may not explain EV purchase decisions in this dataset.

Customer Segmentation

Use income, environmental concern, subsidy availability, and range anxiety to develop more targeted EV marketing and customer segmentation rather than relying heavily on demographic characteristics alone.

Note: Feature importance indicates how much each feature contributes to model predictions; it does not by itself establish a causal relationship. Recommendations should therefore be interpreted as data-driven business directions rather than causal conclusions.